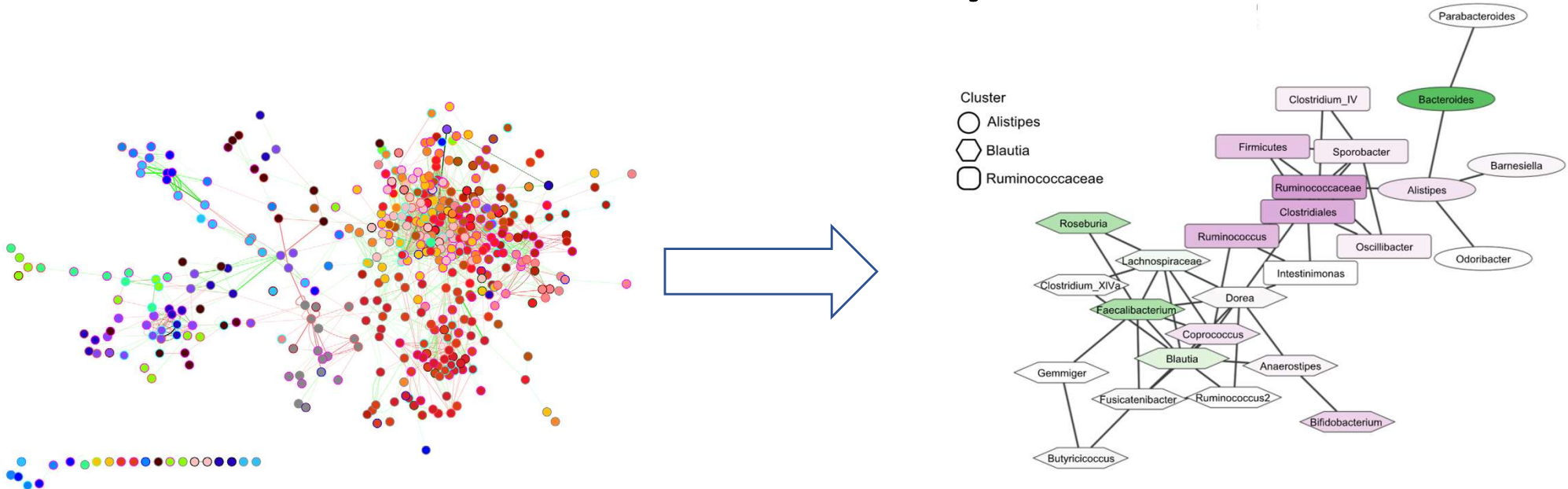
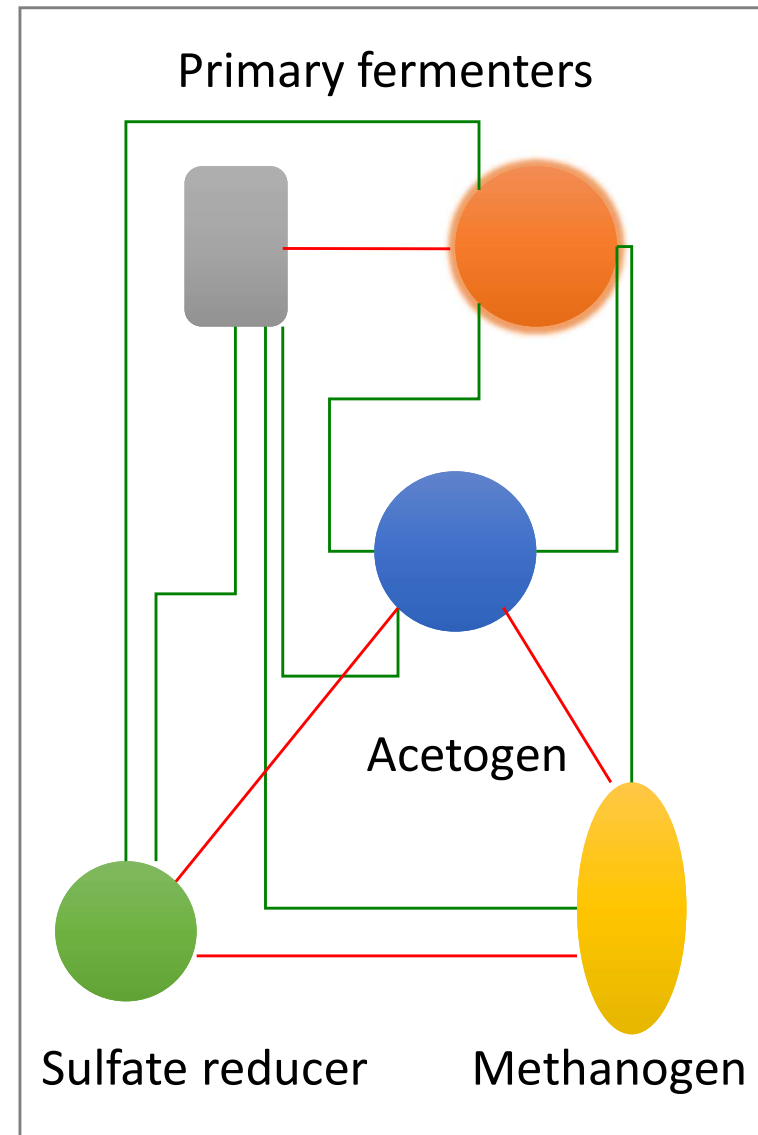
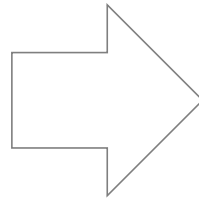


Microbial network inference and analysis



(with some metabolic network analysis sprinkled in)

1. Introduction: microbial networks



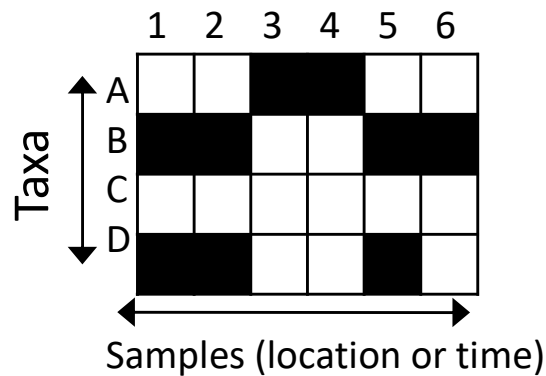
Who is there and with which abundance?

Who interacts with whom?

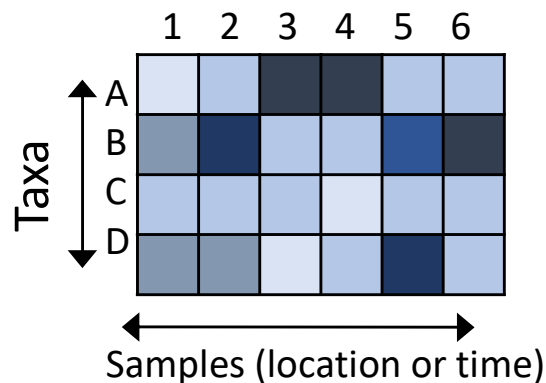
Microbial network inference

INPUT

Presences/absences



Abundances



NETWORK INFERENCE (dozens of tools)

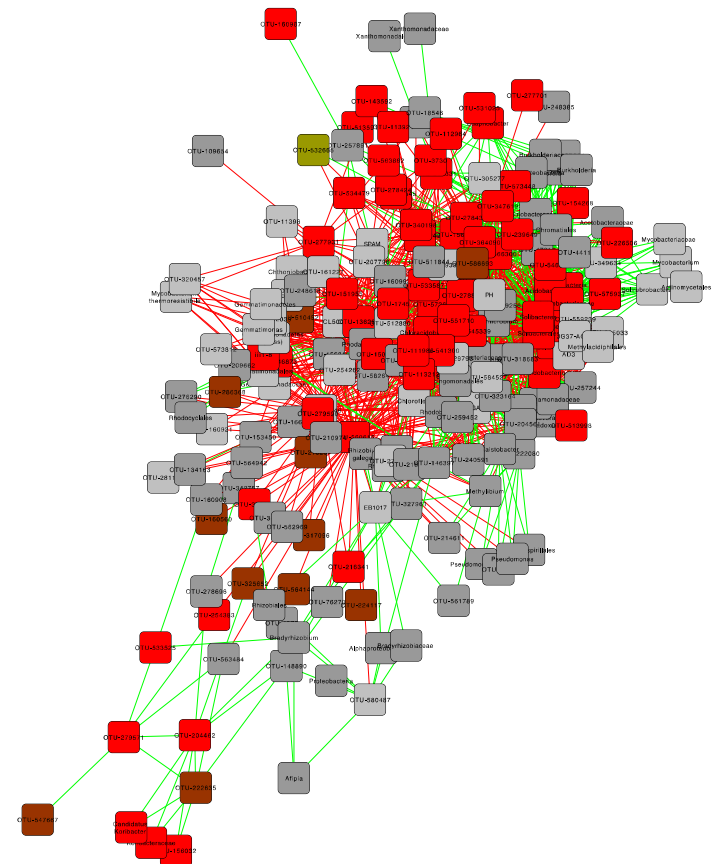
Similarity, Dissimilarity
Correlation, ...



OUTPUT

Microbial association network
Nodes: taxa (OTUs, genera, ...) or
metadata (pH, temperature, ...)

Edges: significant associations



2. Challenges of microbial network inference

- Varying sequencing depth
- Compositional data
- Zeros
- Indirect edges/heterogeneity
- Sampling scale



Varying sequencing depth needs to be treated...

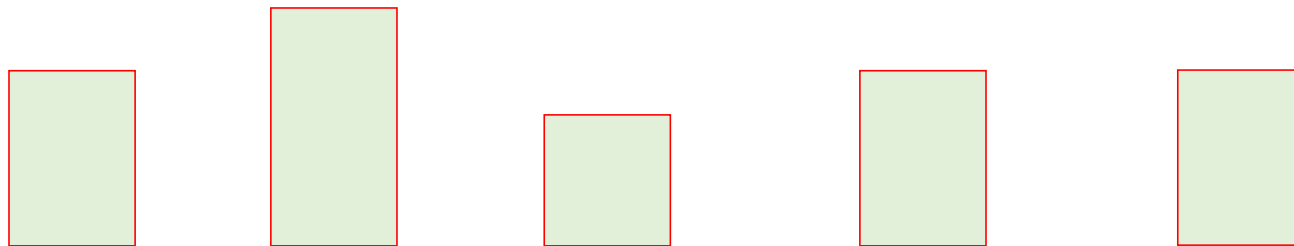
Read count $\neq$ cell count



Blue taxon



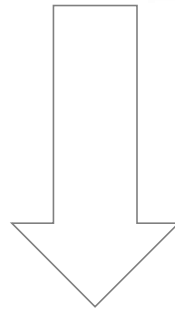
Red taxon



...by removal of sequencing depth differences



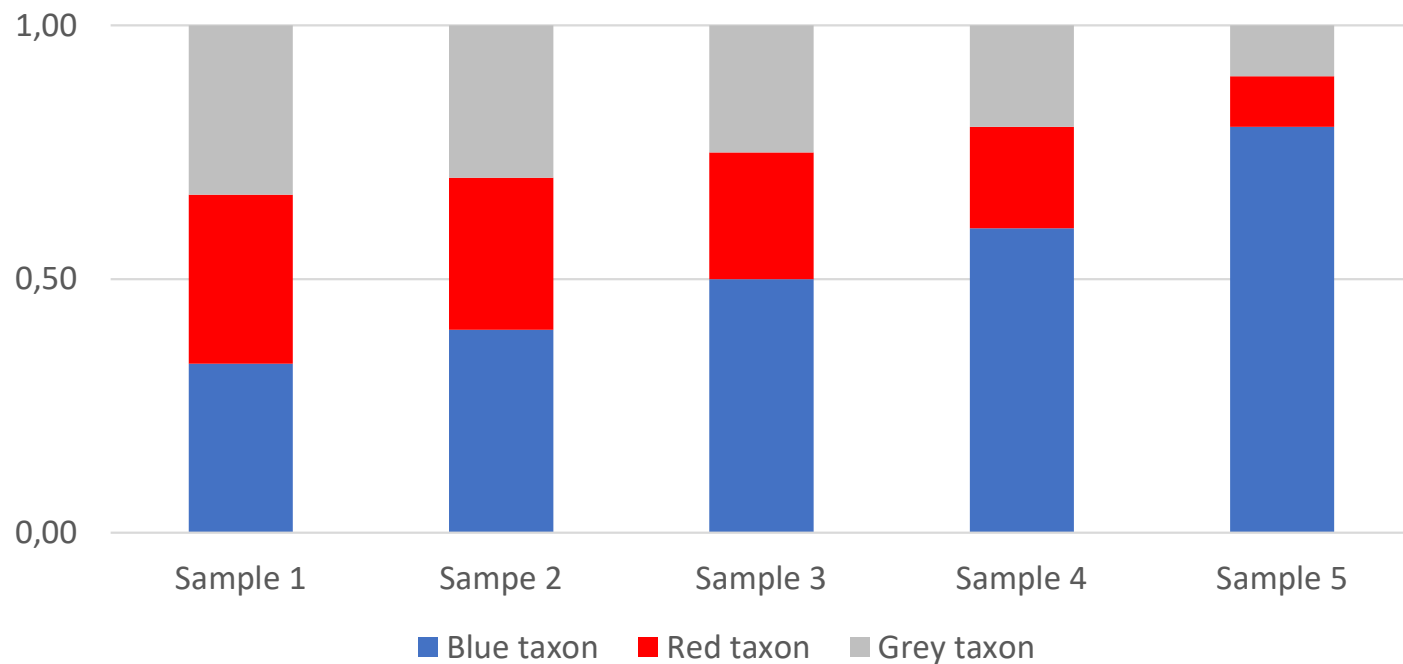
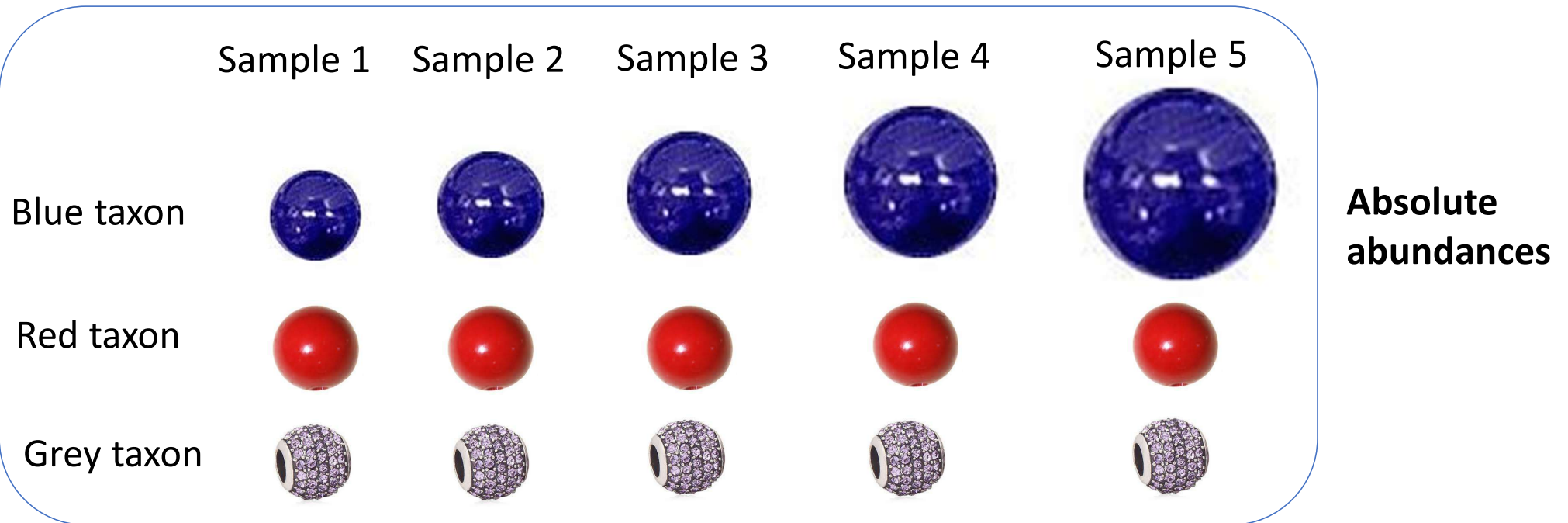
Rarefaction: Select beads from the big bucket with a probability equal to their proportion, until selected bead number is the same as in the small bucket
=> Additional zeros can be introduced, counts are preserved



Normalization: Convert counts into relative abundances (proportions) => counts (integers) are lost, no additional zeros

+ several other, less frequently used techniques

...which leads to compositionality



How to deal with compositionality

Algorithmically

- Transform data e.g. using centered log ratio (clr) transformation (ratio trick: ratios are not changed by normalization)
- Use compositional-robust measures such as Aitchison distance or Bray Curtis dissimilarity

OR

Experimentally

- Quantify total cell counts, e.g. using flow cytometry

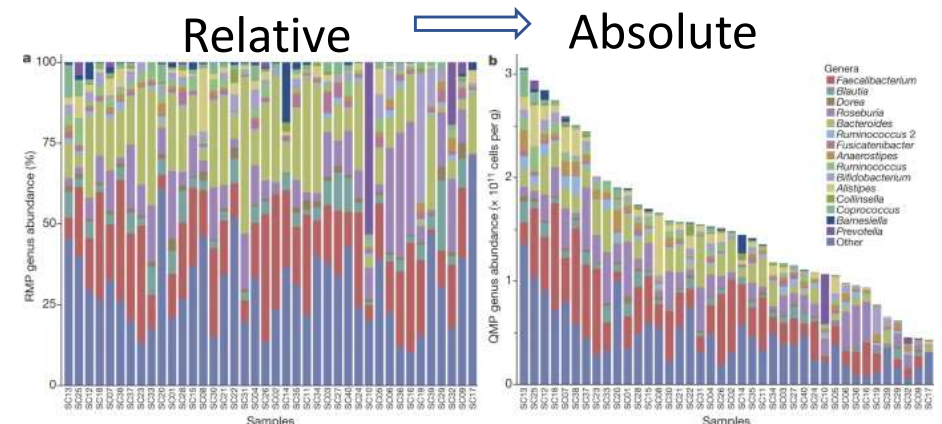


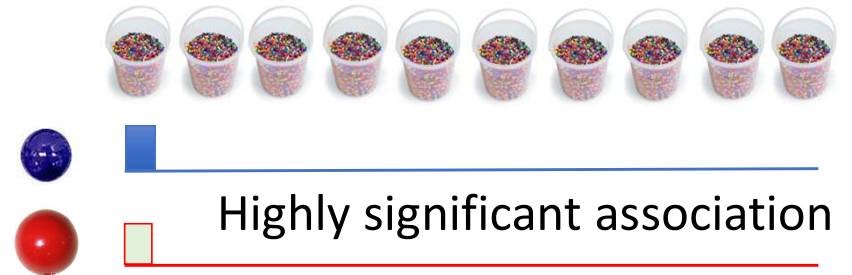
Image source: Vandeputte et al. Nature 551, 507-511 (2017).

- $\Rightarrow$ Working with log ratios poses a zero treatment problem
- $\Rightarrow$ Relative abundances can be multiplied with cell counts, but if cell counts do not depend on microbial interactions, they act as a confounder

Zero treatment as a challenge

- **Structural zeros** (true absences) versus **sampling zeros** (present but not observed)
- Co-absences in rare taxa lead to strong associations, which may be misleading (counts may vary randomly below detection limit)
- Log ratio transformations do not deal well with zeros; they either lose data (by ignoring zeros) or modify data (by adding pseudocounts)
- Rare taxa are filtered ad-hoc (but lower bounds on prevalence have been proposed, see Cougoul et al.)

The problem of co-absences



Ad-hoc solution: Prevalence filter

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
	0	5	2	0	3	0
	0	0	0	2	0	0
	58	56	45	129	81	18
	0	2	0	175	0	0

Presence in at least 50% of samples

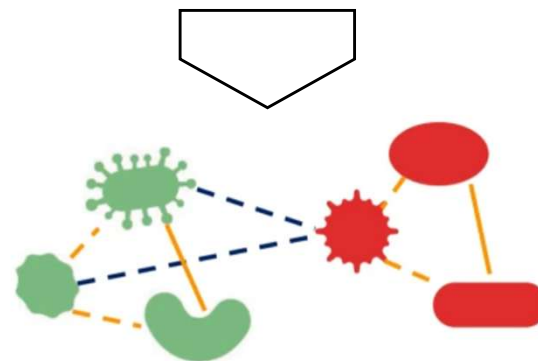
Garbage taxon

	S1	S2	S3	S4	S5	S6
	0	5	2	0	3	0
	58	56	45	129	81	18
	0	2	0	177	0	0

Cougoul et al. (2019) "Rarity of microbial species: in search of reliable associations."
PLoS ONE 14, e0200458.

Indirect edges and heterogeneity

	Low pH	Low pH	Low pH	Low pH	High pH	High pH	High pH	High pH
	54	32	68	49	215	203	193	201
	1	0	0	1	245	226	201	243
	39	30	42	35	271	258	223	249
	231	252	268	235	21	20	13	21
	189	217	195	202	9	5	12	7
	195	201	219	198	15	13	10	12
pH	6.5	6.3	6.2	6.4	7.5	7.2	7.1	7.2

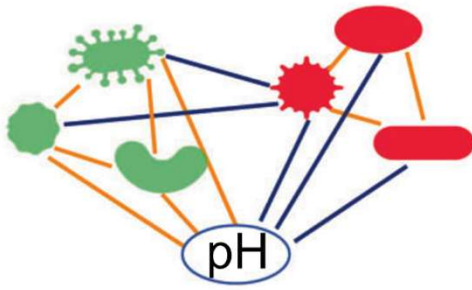


Correlation != Causation

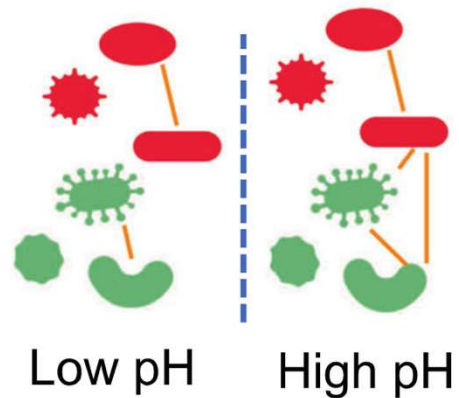
- Positive edge
- Negative edge
- Indirect edge

And what to do about it

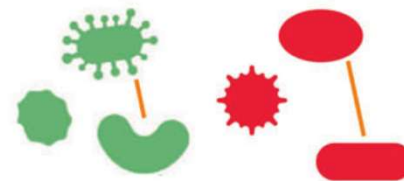
1) Include environmental factors in network



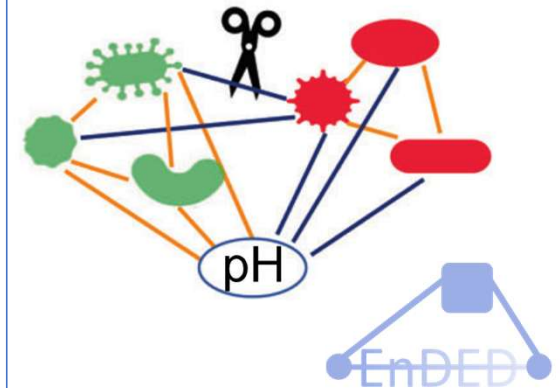
2) Stratify samples and compute a network per sample group



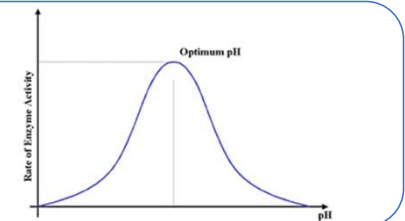
3) Regress out environmental factors before network construction



4) Filter network



Problem with third solution: non-linear dependencies of taxon abundance on environmental factors (optima)



EnDED: Deutschmann et al. (2021) “Disentangling environmental effects in microbial association networks” Microbiome 9, 232.

Zero treatment in FlashWeave

- Heterogeneous and homogeneous mode
- **Heterogeneous mode: zeros are** interpreted as structural (true absences) and are **excluded from computations** (treated as missing values)
- Reduces indirect edges due to environmental heterogeneity
- But leads to loss of information in case zeros are present for other reasons
- The user informs the inference method about which zero type is more likely (data from the same or from different environments)

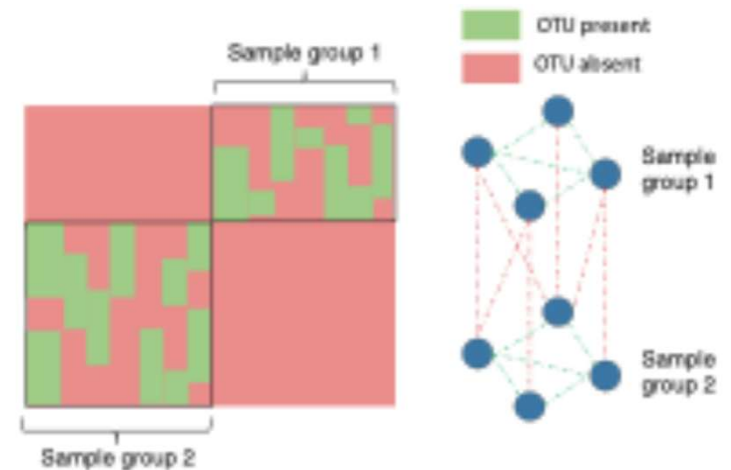
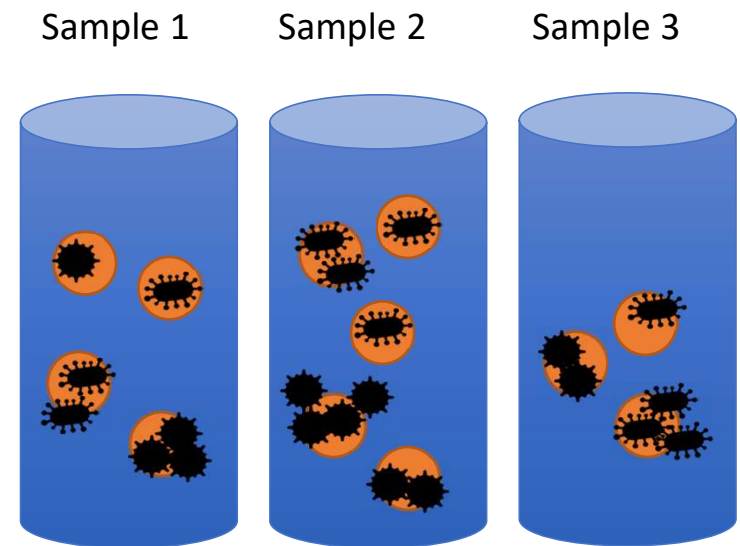


Image taken from FlashWeave paper

Sampling resolution

- Edge sign may differ depending on sampling resolution
- Variability at the microscale allows survival of competing species (one or the other dominates locally, but both co-occur globally)
- Problem of experimental design



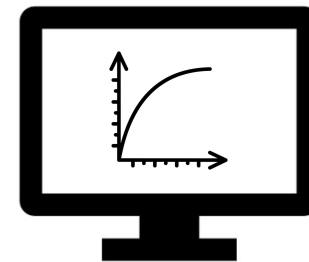
Inferred edge is negative within a sample but positive across the samples

This was one of the motivations for 3D'omics 😊

Chesson (2000) "Mechanisms of Maintenance of Species Diversity" 31, 343-366.
Armitage & Jones (2019) "How sample heterogeneity can obscure the signal of microbial interactions" The ISME Journal 13, 2639-2646.

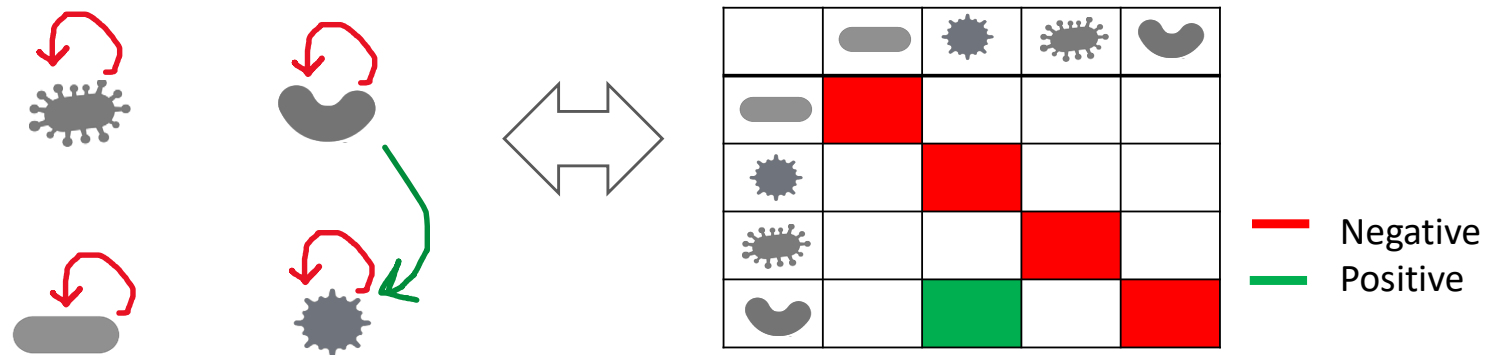
3. Evaluation and improvement of microbial network inference

- How accurate is microbial network inference?
- Only very few microbial interaction networks have been resolved experimentally
- Evaluation of microbial network inference is largely carried out on in silico data



Data simulation with the generalised Lotka Volterra model

- The species network can be represented by the **directed** interaction matrix A ; entries represent interaction strengths



- Species abundance vector X changes as a function of species initial abundance, growth rates B and interaction matrix A

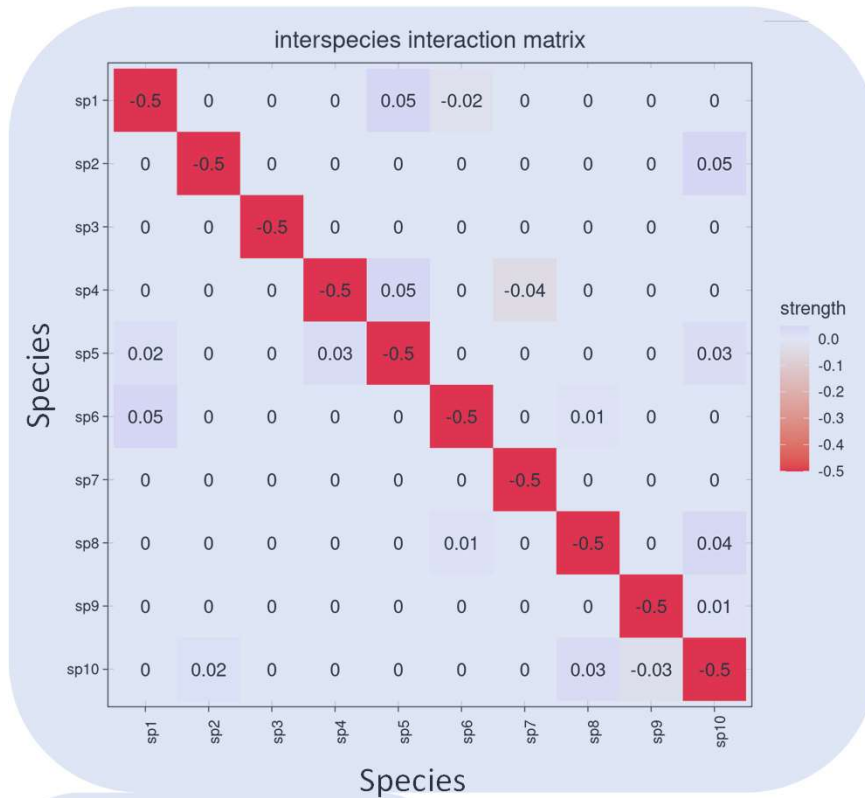
$$\frac{dX(t)}{dt} = X(t)(B + AX(t))$$

generalized Lotka
Volterra (gLTV)

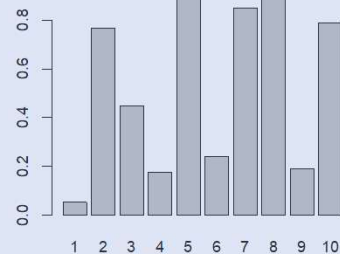
We'll cover this in-depth in the Leuven retreat

Example simulation with generalized Lotka-Volterra

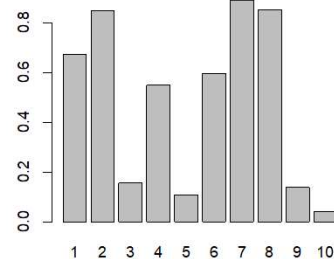
Parameters



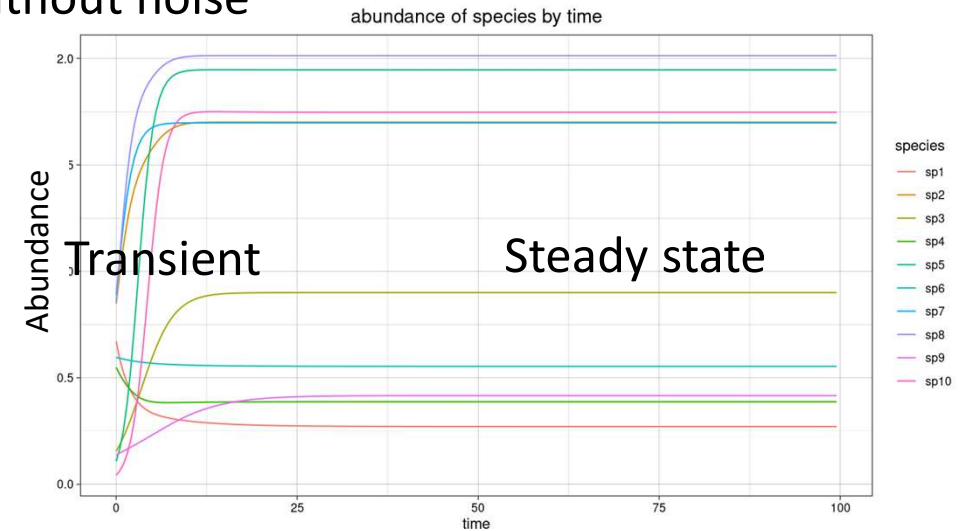
Growth rates



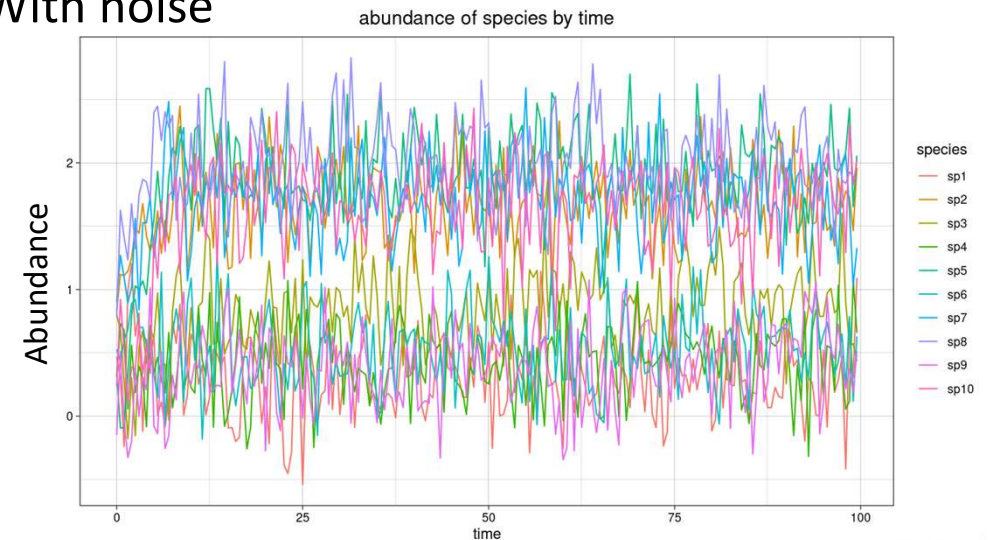
Initial abundances



Without noise



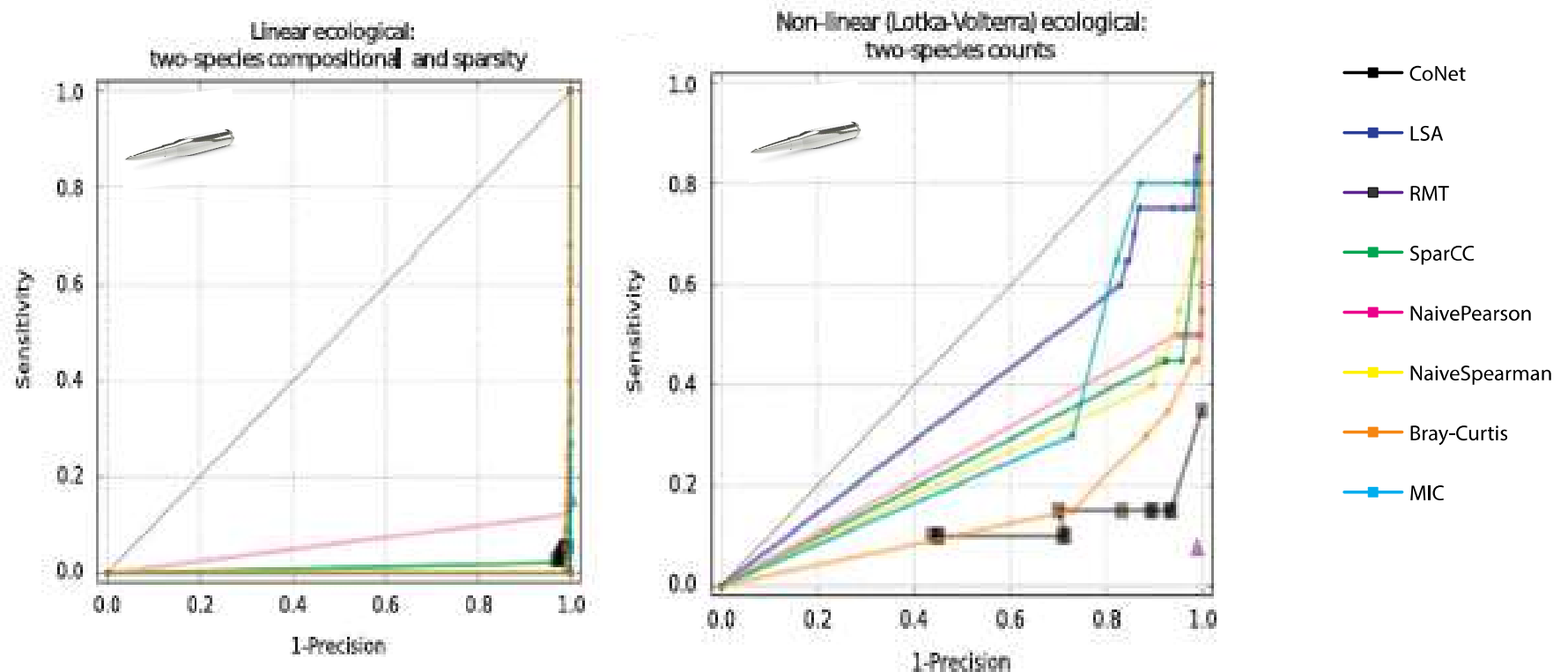
With noise



Microbial network inference has low accuracy on synthetic data

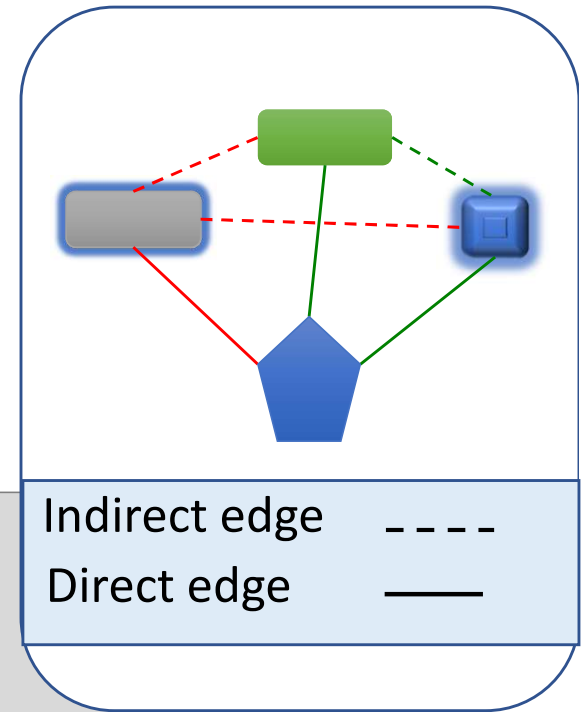


Position of optimal tool (silver bullet)

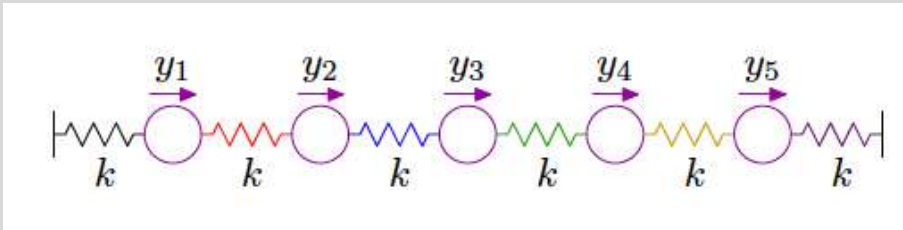


Inverse covariance to the rescue?

- Idea: remove indirect edges
- Zero in the inverse covariance matrix: conditional independence
- **Assumption: all relevant variables are taken into consideration**



Intuitive example by David MacKay:



Weights (nodes)
connected by springs
(edges)

Covariance matrix

$$\mathbf{K} = \frac{T}{k} \begin{bmatrix} 0.83 & 0.67 & 0.50 & 0.33 & 0.17 \\ 0.67 & 1.33 & 1.00 & 0.67 & 0.33 \\ 0.50 & 1.00 & 1.50 & 1.00 & 0.50 \\ 0.33 & 0.67 & 1.00 & 1.33 & 0.67 \\ 0.17 & 0.33 & 0.50 & 0.67 & 0.83 \end{bmatrix}$$

Inverse covariance matrix

$$\mathbf{K}^{-1} = \frac{k}{T} \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}$$

Tools implementing inverse covariance in some form

- SPIEC-EASI (GL and MB)
- SPRING
- gCoda
- HARMONIES
- COZINE
- FlashWeave
- ...

(graphical Gaussian model, precision matrix or partial correlations: same principle)

SPIEC-EASI: Kurtz et al. (2015) “Sparse and Compositionally Robust Inference of Microbial Ecological Networks” PLoS Computational Biology 11, e1004226.

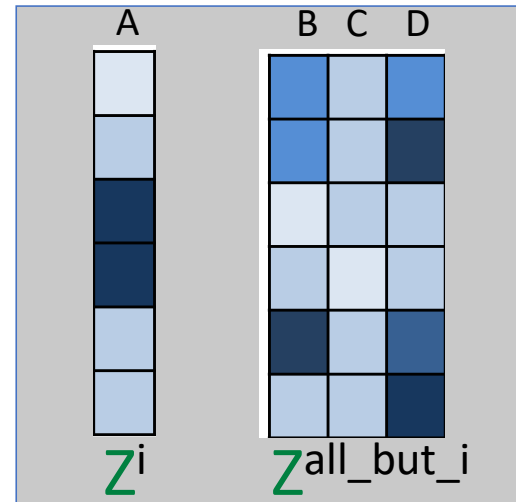
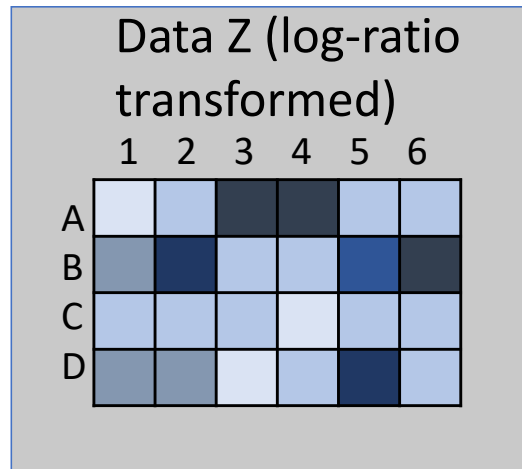
SPRING: Yoon et al. (2019) “Microbial Networks in SPRING - Semi-parametric Rank-Based Correlation and Partial Correlation Estimation for Quantitative Microbiome Data” Frontiers in Genetics 10, 00516.

Fang et al. (2017) “gCoda: Conditional Dependence Network Inference for Compositional Data” Journal of Computational Biology 24, 699-708

COZINE: Ha et al. (2020) “Compositional zero-inflated network estimation for microbiome data”, BMC Bioinformatics 21, 581.

Example: SPIEC-EASI MB

Among SPIEC-EASI's methods to infer the inverse covariance matrix:
Meinshausen & Bühlmann method (neighborhood selection)



$$\hat{\beta}^i, \lambda = \arg \min_{\beta \in \mathbb{R}^{p-1}} \left(\frac{1}{n} \left\| Z^i - Z^{-i} \beta \right\|^2 + \lambda \left\| \beta \right\|_1 \right)$$

Penalty parameter
 λ n β $\left\| \beta \right\|_1$
Species number Sample number Regression coefficients

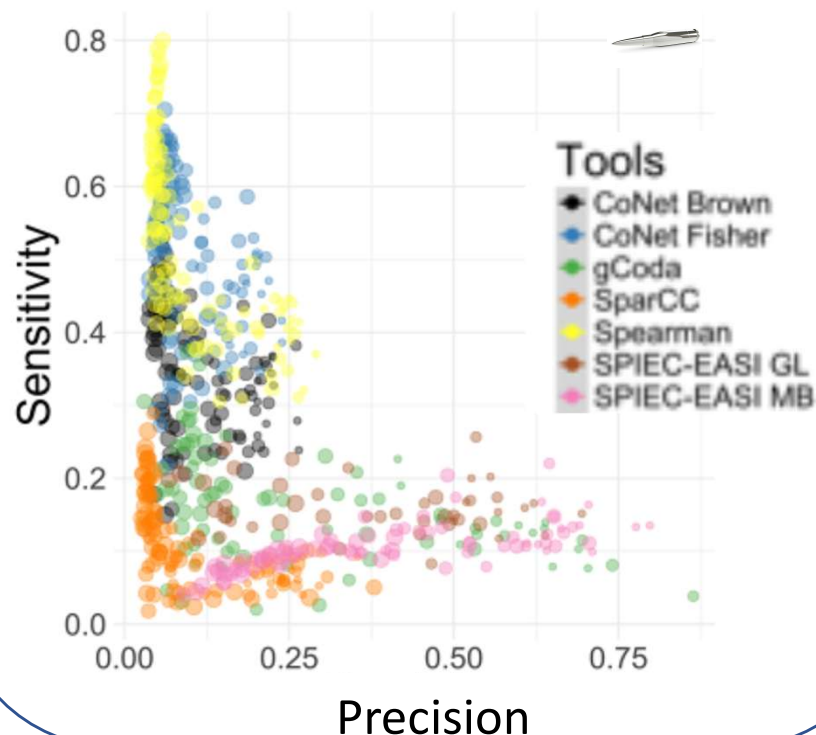
In words:

For all species abundance vectors i : minimize the least square error between the abundance vector of species i and the abundance vector predicted from selected other species. Selection is achieved through a penalty term. We build the network one node at a time.

Does not work that well on simulated data

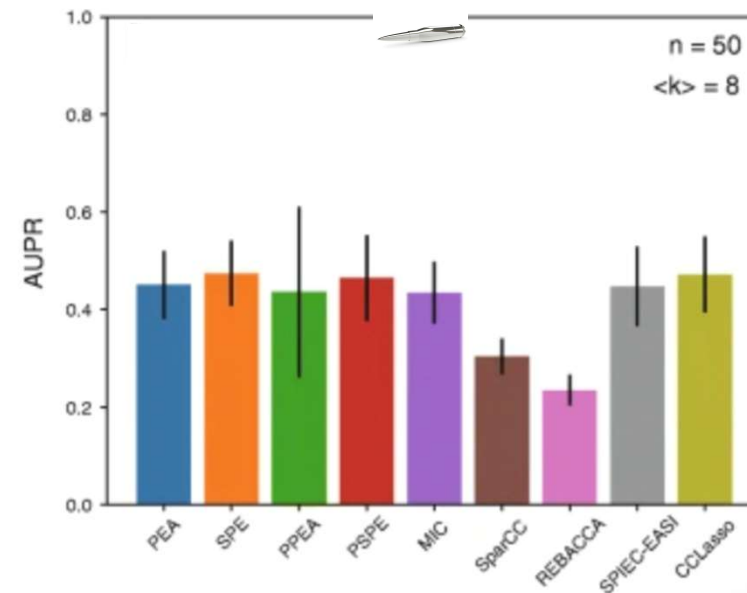
Röttjers & Faust (2018) FEMS
Microbiology Reviews 42, 761-780.

- gLV with environmental influence (encoded by node size)
- Increasing environmental influence lowers precision of inverse covariance tools



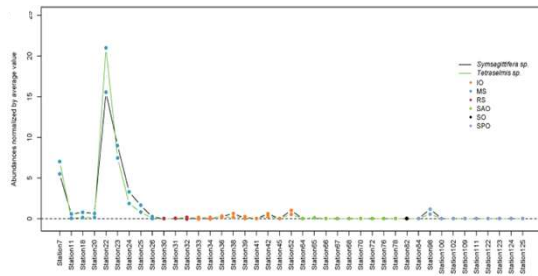
Hirano & Takemoto (2019)
BMC Bioinformatics 20, 329.

- gLV simulations
- Correlation methods (PEA/SPE) perform as well as partial correlation (PPEA, PSPE) and inverse-covariance based methods (SPIEC-EASI)

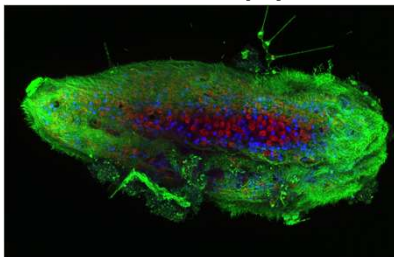


Microbial networks can be useful despite low accuracy

Prediction



Microscopy

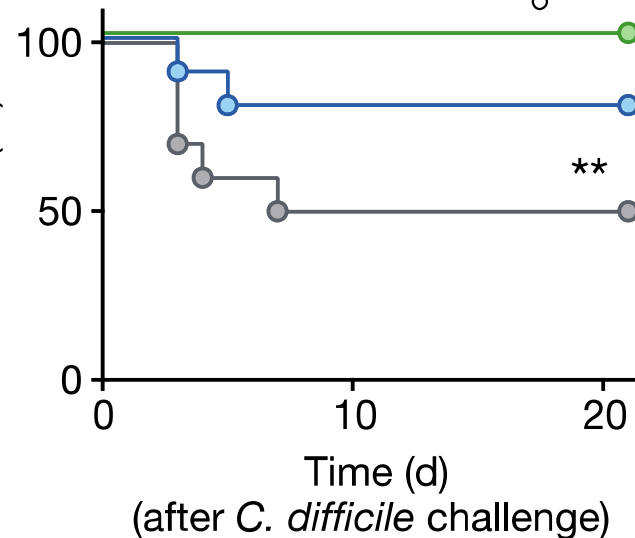


TARA Oceans
Lima-Mendez et al. Science
348, 1262073 (2015).

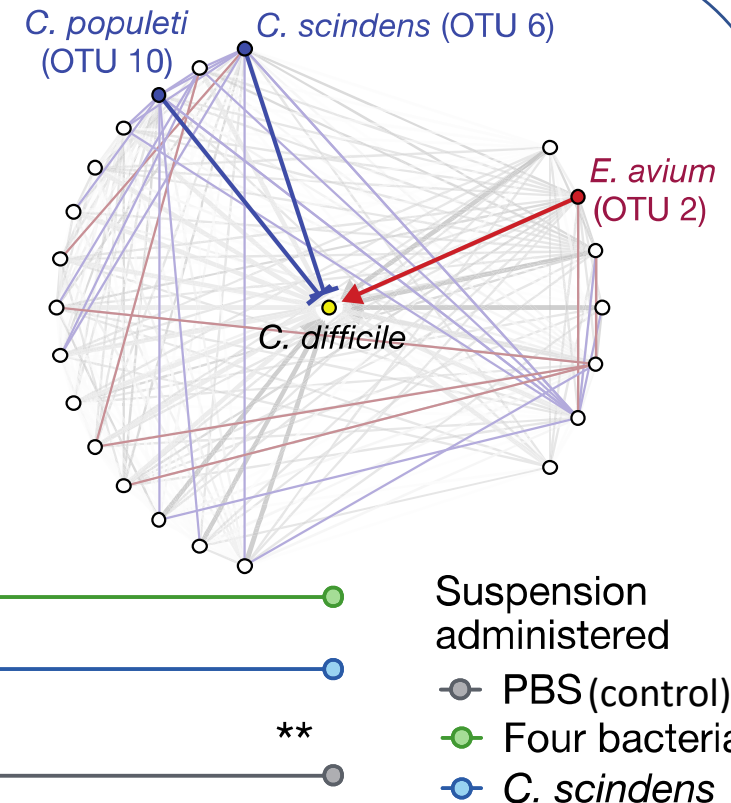
Prediction of pathogen inhibitors led to a probiotics-based *C. difficile* treatment that worked in mice



Survival (%)

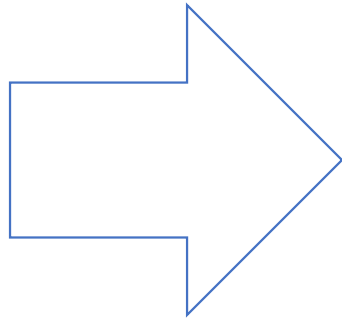
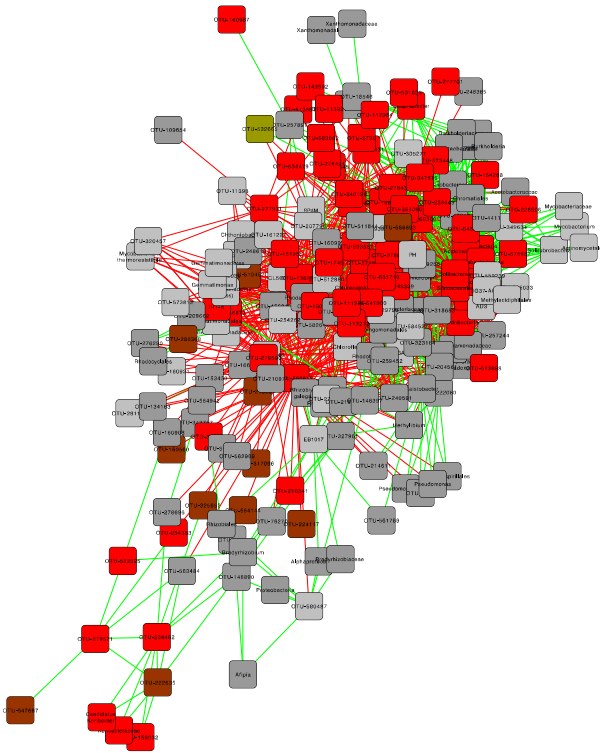


Buffie et al. Nature 517, 205-208 (2014).



Chapters 4 and 5: How to analyze microbial networks?

Hairball



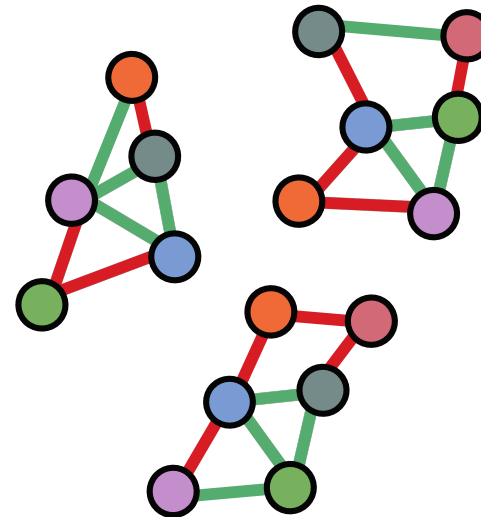
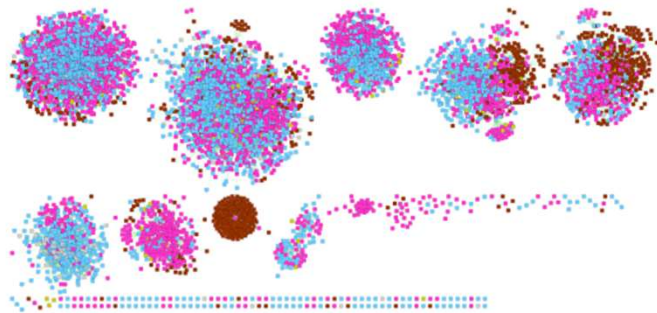
Hypotheses

- Who interacts with whom
- Which taxa respond to which environmental factors
- Which taxa cluster together and why
-

Knowing that edges are not very reliable...

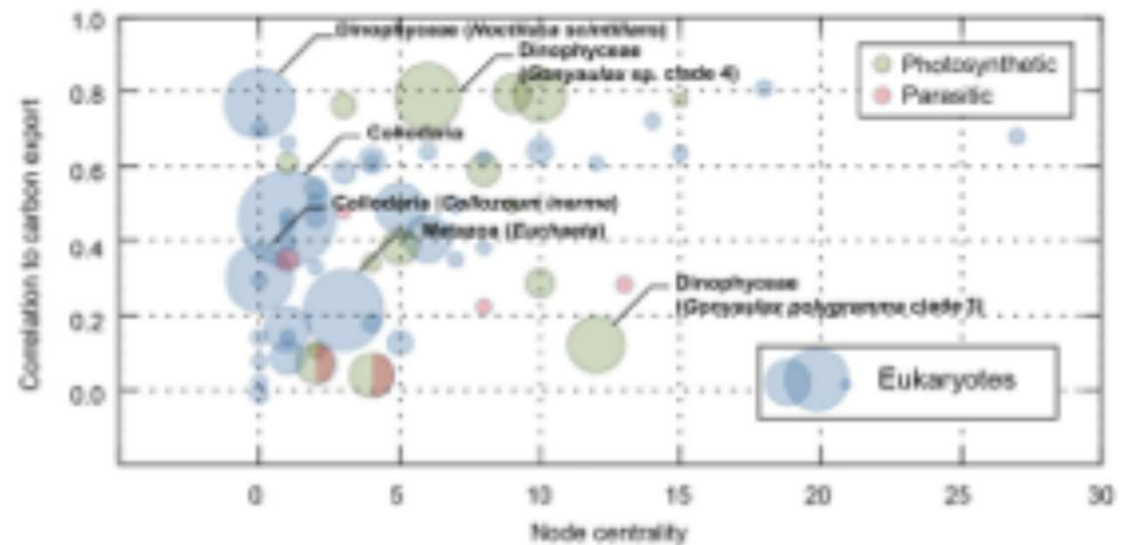
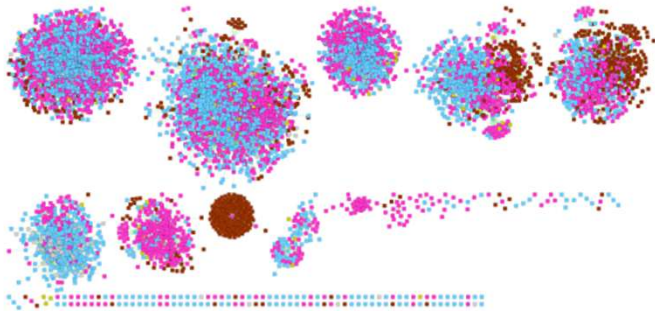
4. Topological analysis of microbial networks

- What can we learn from the network structure?



Microbial network clustering

- Break the hairball into manageable (and hopefully biologically informative) bits
- Example: clustering of marine microbial network and correlation of representative cluster members to carbon export



Guidi et al. (2016) "Plankton networks driving carbon export in the oligotrophic ocean" Nature 532, 465-470.

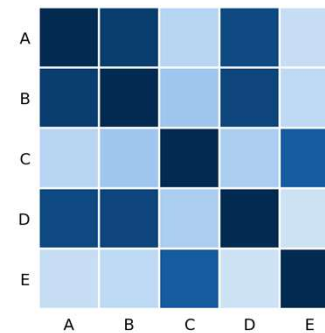
Microbial network clustering: WGCNA

Weighted (Gene) Correlation Network Analysis – Steps:

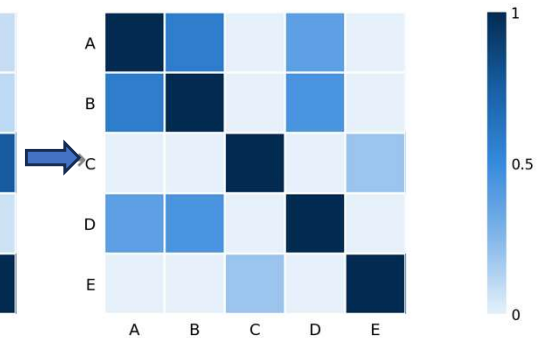


- Compute a pairwise correlation (or similarity) matrix between taxa
- Soft-threshold with: $\alpha_{ij} = |cor(x_i, x_j)|^\beta$
- Convert resulting network into Topological Overlap Matrix (TOM) by computing for each node pair to what extent neighbor nodes are shared
- Apply average-linkage hierarchical clustering to TOM to identify modules
- Eigen-taxon: first principal component of the abundance profiles in a module (this is what was correlated to carbon export in the example)

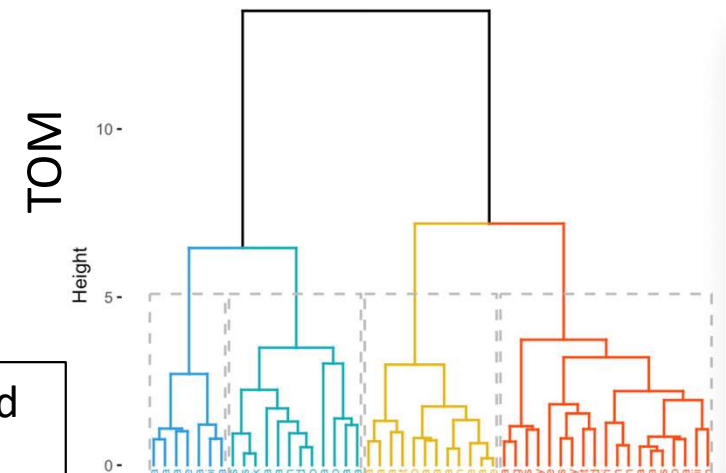
Correlation matrix



Taking to the power β
→ modules emerge



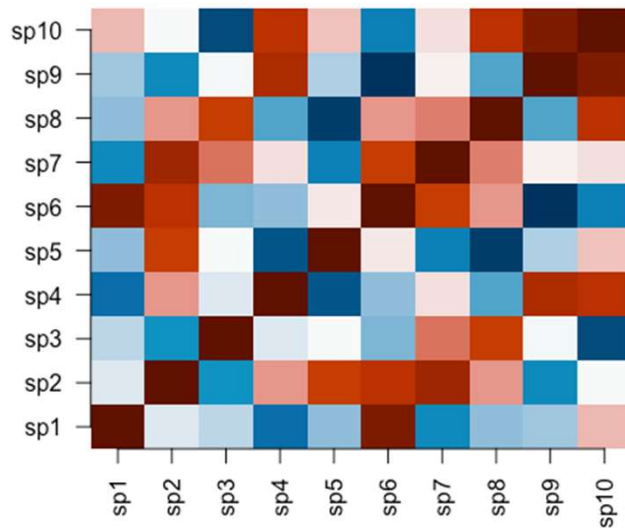
WGCNA identifies β by assuming that the network is scale-free.



Langfelder & Horvath (2008) “WGCNA: an R package for weighted correlation network analysis”, BMC Bioinformatics 9, 559.

WGCNA – Simplified

Correlations



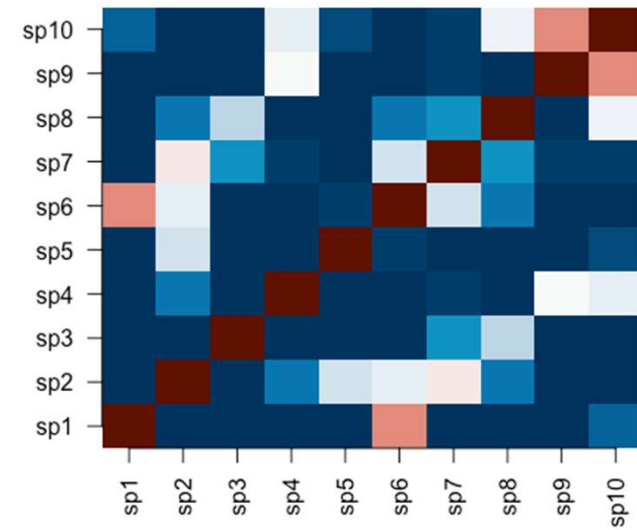
(absolute values)

Take each entry in correlation matrix to given power

$$> A = C^6$$



Soft-thresholded correlations

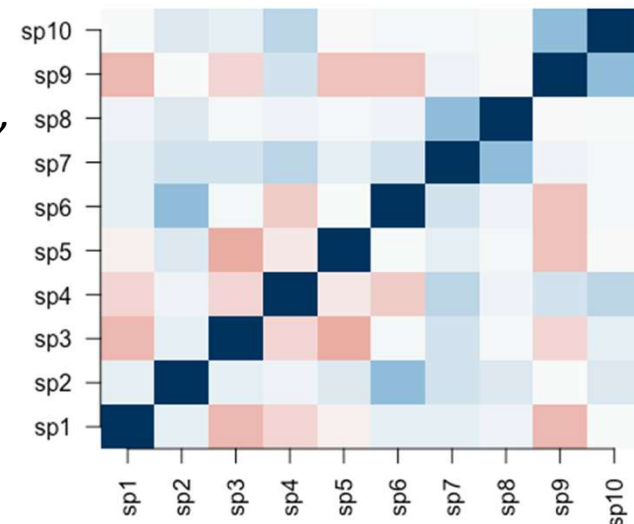


Compute TOM

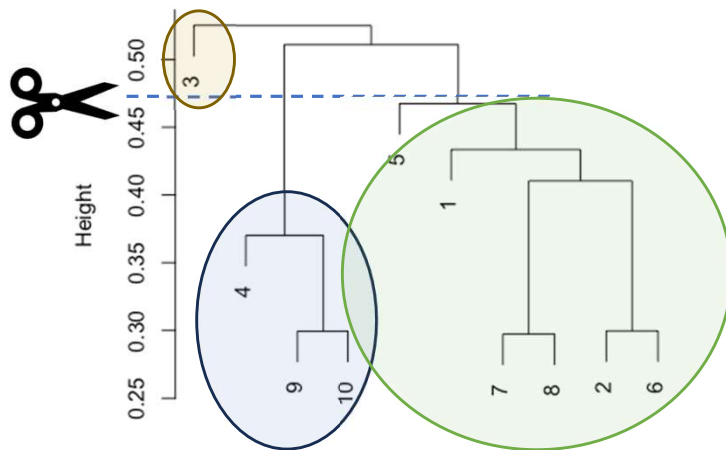
$$> tomMat = TOMdist(A)$$



TOM dissimilarities



TOM dendrogram



Identify modules

```
> hc = hclust(as.dist(tomMat),
method="average")
```

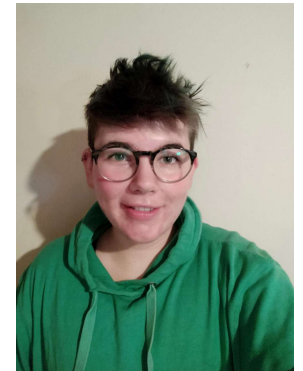


```
> modules = cutree(hc, k=3)
```

(WGCNA identifies k automatically through dynamic tree-cutting)

Microbial network clustering with manta

- Challenges:
 - Microbial networks have low accuracy
 - Several existing network cluster algorithms do not exploit information present in negative edges (e.g. MCL) or make assumptions about the network structure (e.g. WGCNA)
- Manta addresses these challenges



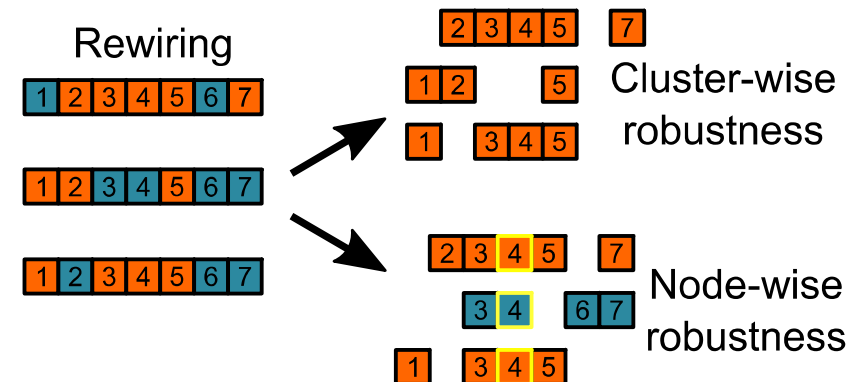
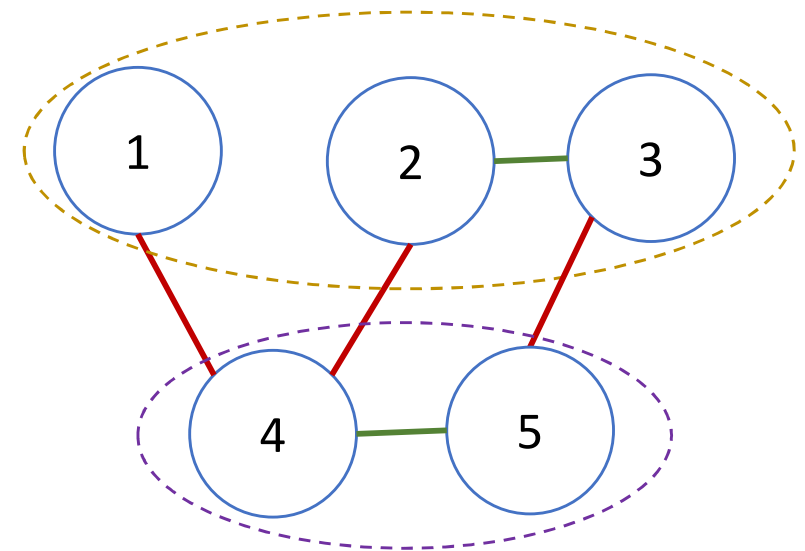
Sam Röttjers



Manta – key ideas



- Matrix operations do not lose sign; nodes can be grouped based on shared negative edges (“the enemy of my enemy is my friend”)
- Hierarchical clustering; cluster number determined by optimizing sparsity score (sum of edge numbers with negative weights between clusters and positive weights within clusters)
- Weak node assignments (nodes that cannot be clustered)
- Repeat clustering on partly rewired networks to assess robustness of clusters and cluster memberships



Röttjers & Faust (2020) “manta-a clustering algorithm for weighted ecological networks” mSystems 5 e00903-19.

How to evaluate a network cluster algorithm?



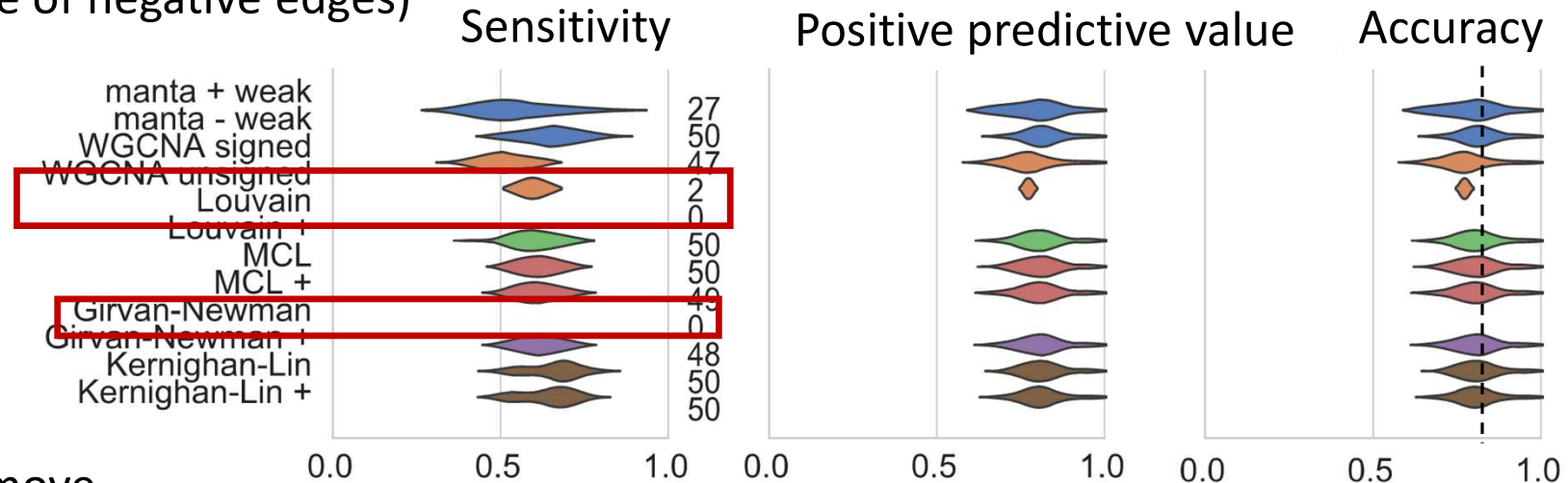
- We need a data set with known clusters to check whether the tool finds them back
- Microbiome data with known clusters are hard to find
- Generate synthetic microbial abundances with known clusters for manta:
 - 1. Population model (gLV) simulating different environmental effects on predefined groups of taxa
 - 2. Biclust generation with FABIA
- Choice of data generation process often biases tool evaluation (two processes better than one)

Hochreiter et al. (2010) "FABIA: factor analysis for bicluster acquisition."
Bioinformatics 26, 1520-1527.

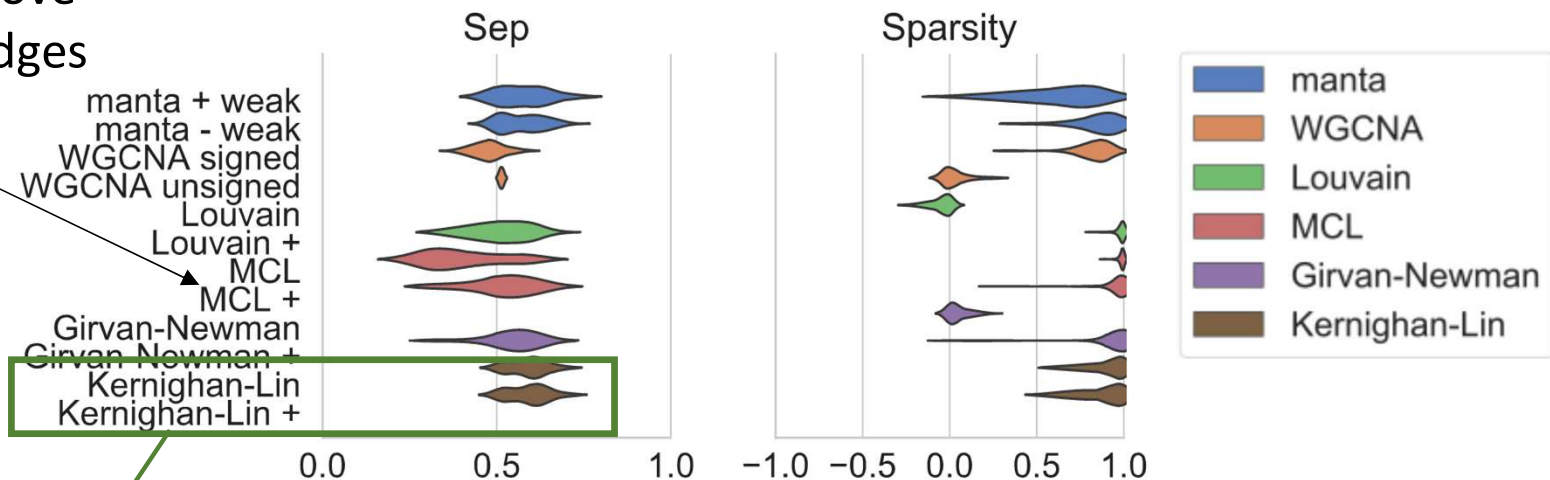
Evaluation with population model



Louvain, GN and WGCNA unsigned failed
(because of negative edges)



Better remove
negative edges
for MCL



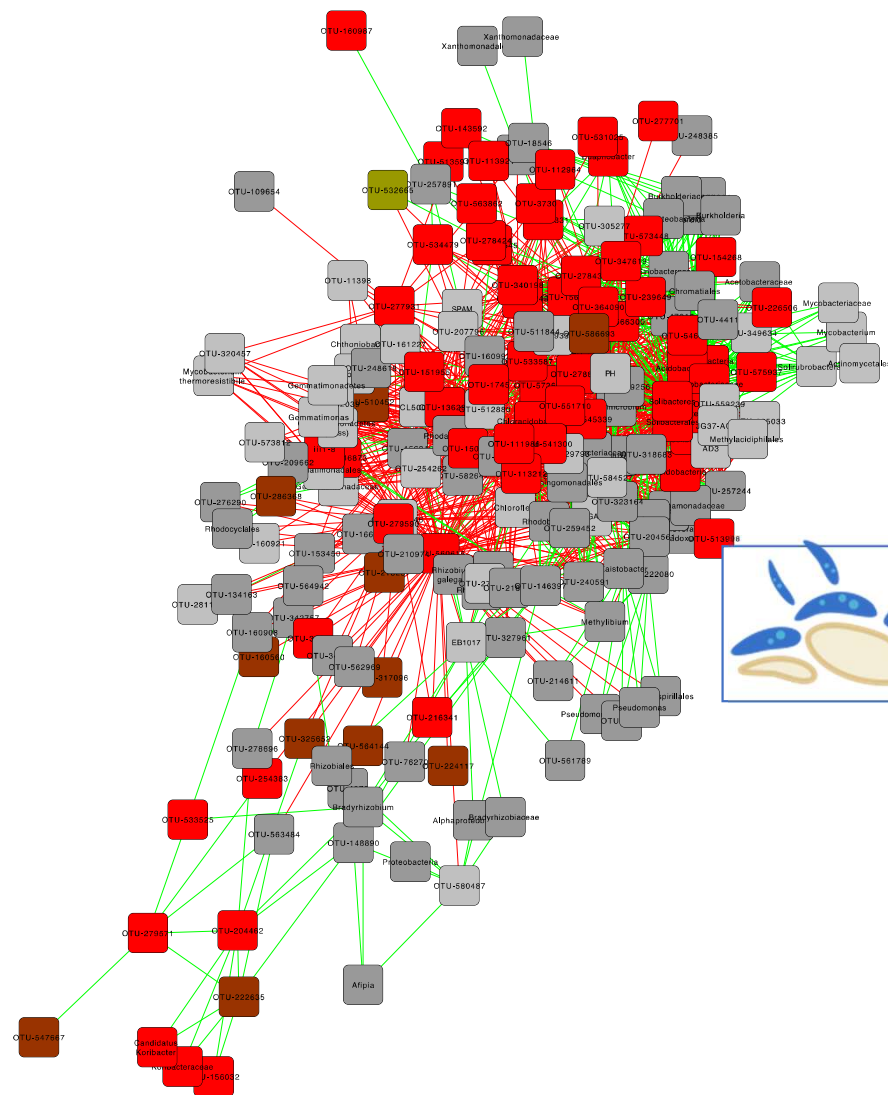
KL performs well but only allows 2 clusters.

Sep (cluster separation) &
sparsity: cluster quality scores

Manta in action: Tundra soil



Hairball

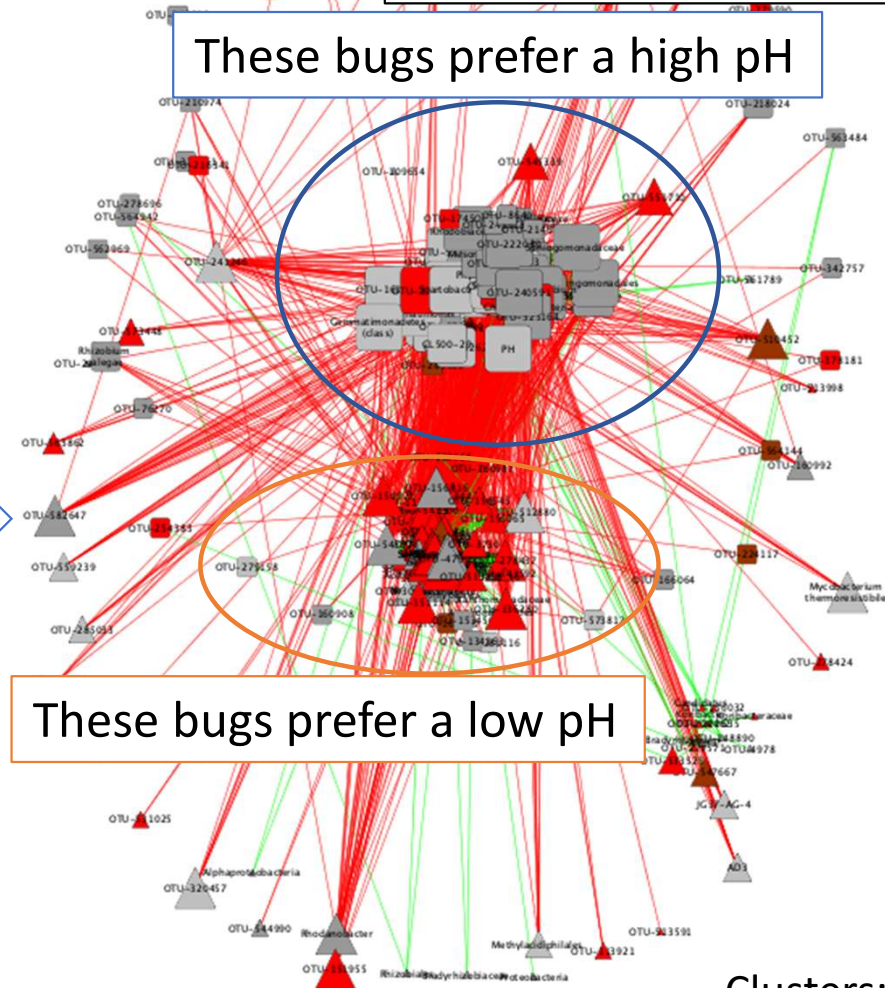


Tundra soil network
Layout: organic

Data: Chu et al. (2010)
Environ Microbiol 12(11),
2998-3006

Informative network

Node size is proportional
to robustness of cluster
assignment



Clusters:

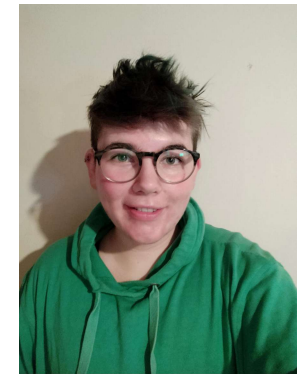
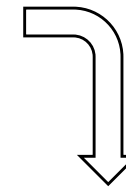
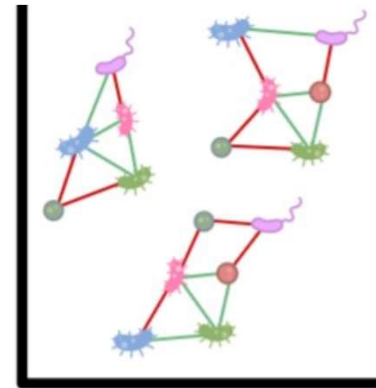


Tundra soil network after
clustering & layout with manta

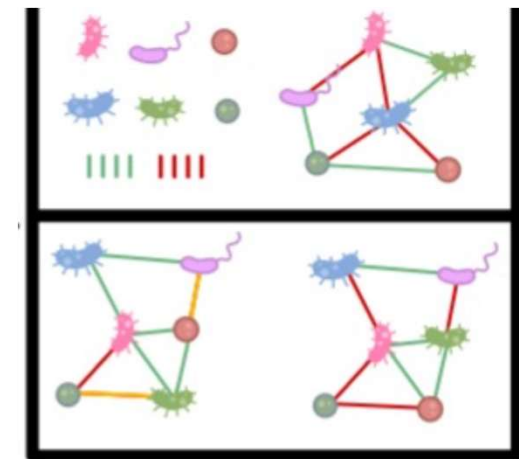
Anuran: a toolbox for comparing noisy microbial networks



- Network set: is a core network present or do networks overlap by chance?
- Anuran tests whether a network property or a core network is significant given randomized networks
- Anuran implements 2 types of null models: network randomization with and without preserving node degree distribution



Sam Röttjers

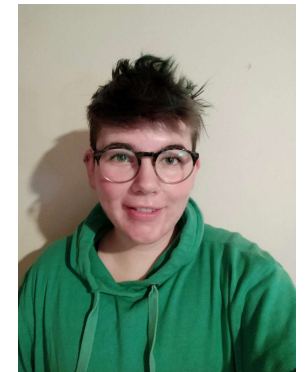
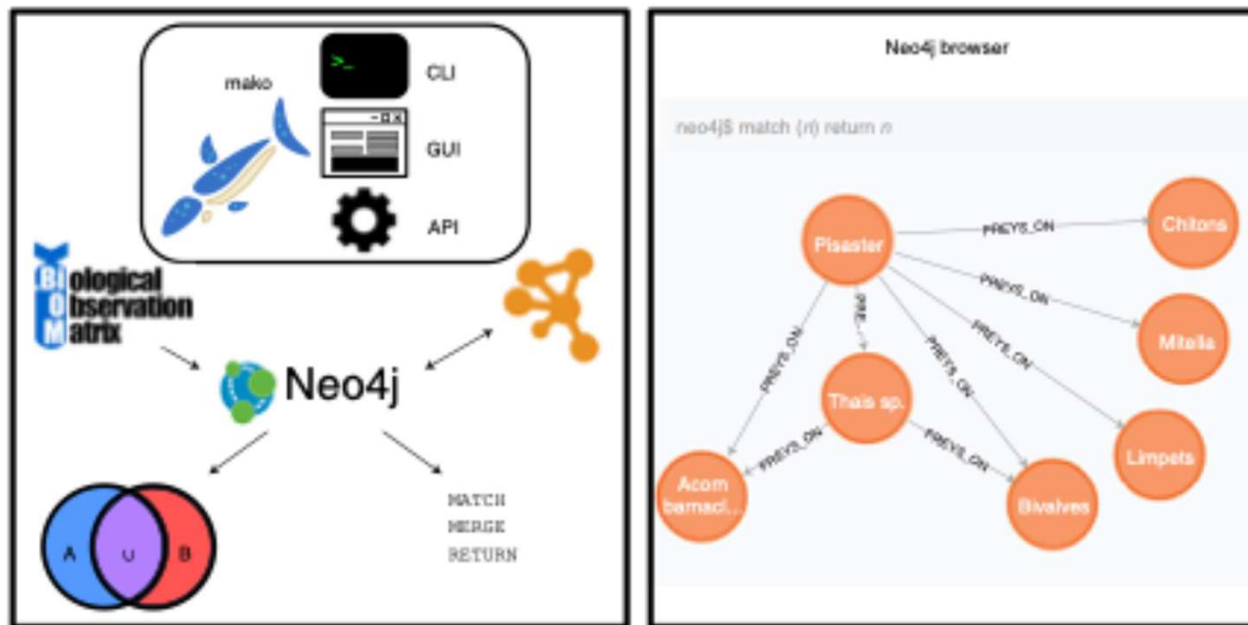


Röttjers S, Vandeputte D, Raes J and Faust K (2021). “Null-model-based network comparison reveals core associations” ISME Communications 1, 36

Mako: Querying microbial networks



- Microbial network database enables queries across networks
- With dozens of network inference tools and their settings, flexible and local solution is preferable over a centralized database
- Mako creates network database on the fly from user networks
- Uses neo4j network database and CYPHER network query language

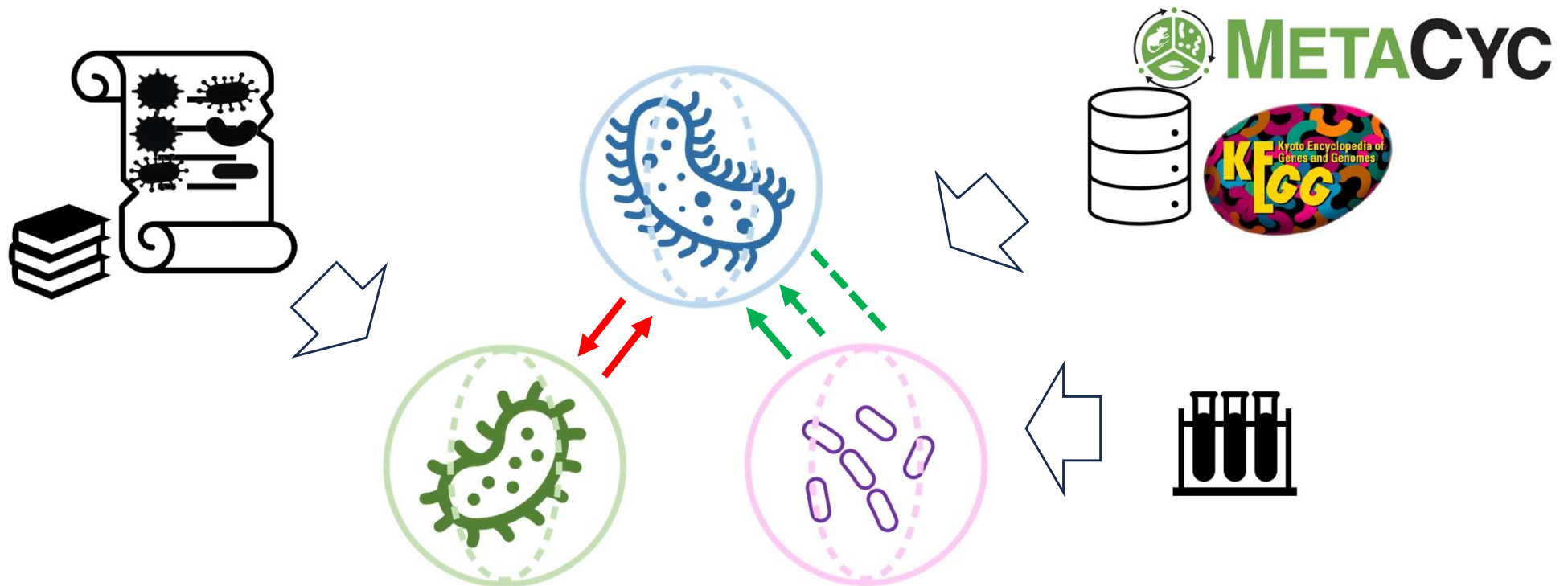


Sam Röttjers

Röttjers & Faust “Fast and flexible analysis of linked microbiome data with mako” (2022) Nature Methods 19, 51-54.

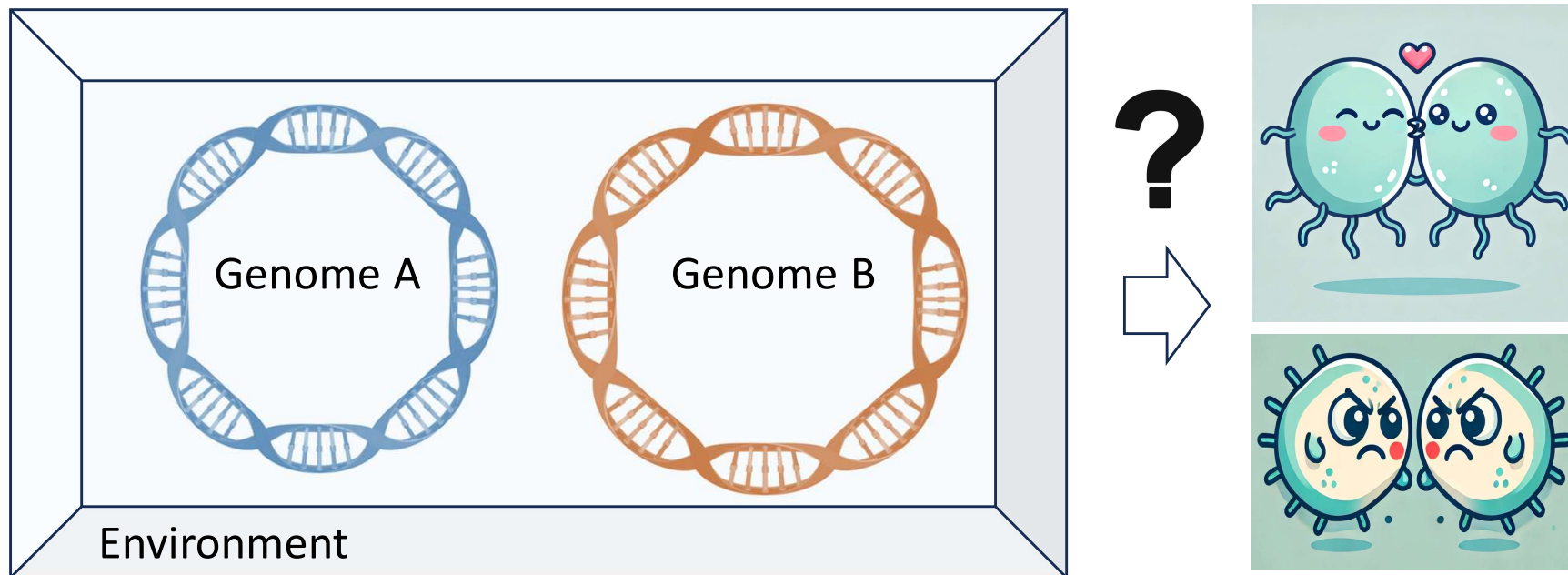
5. Context-based analysis of microbial networks

- Combine microbial networks with information from other sources to help interpret associations
- Microbial networks as data integration tools



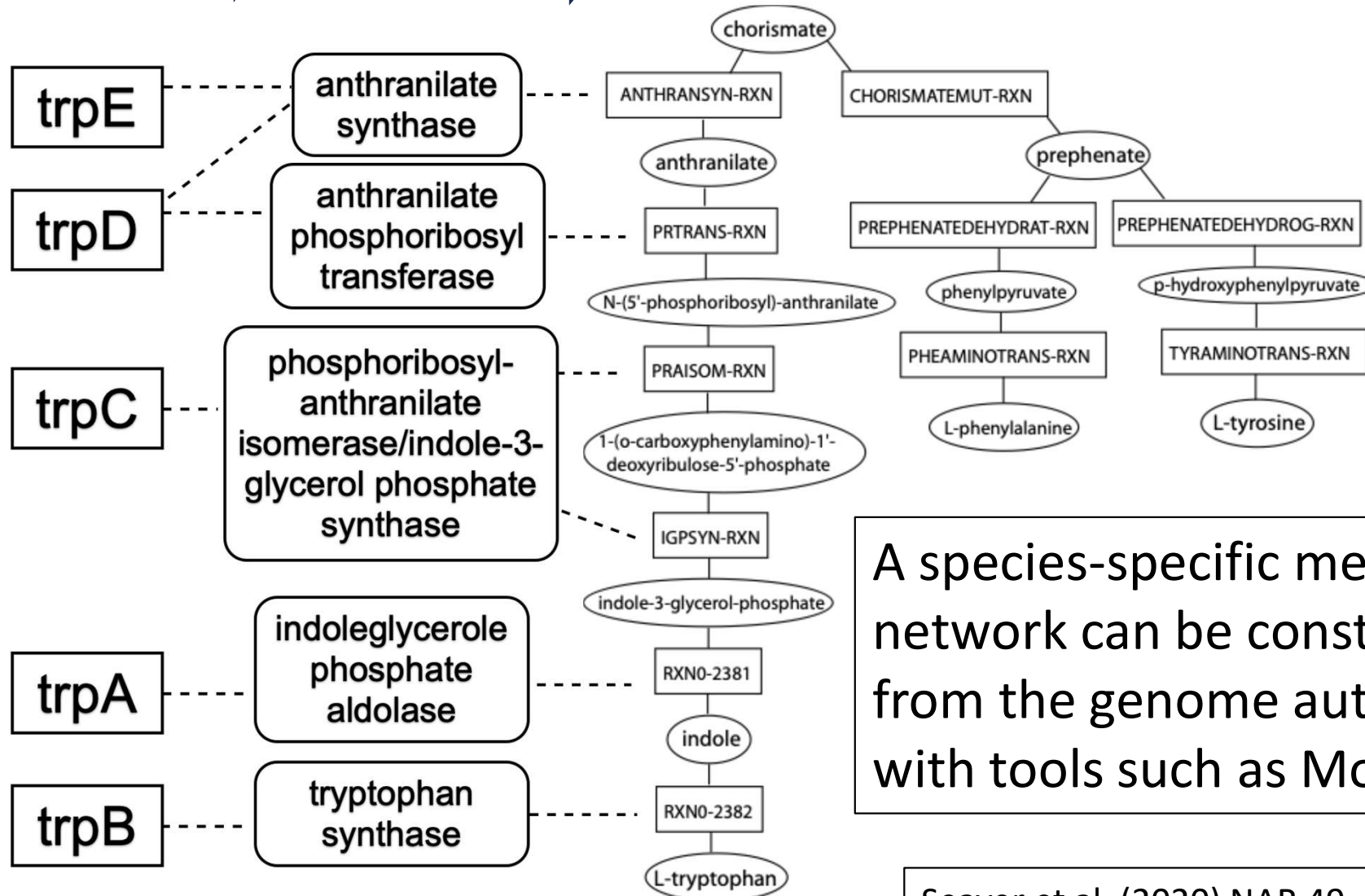
Reverse ecology

- Reverse ecology: learn about a species' lifestyle and its interactions from the genome
- Several traits (e.g. presence of flagella, metabolite production) can be predicted quite reliably from the genome
- Interactions are context-dependent, but context (environment) can be defined



Metabolic reconstruction: from genes to metabolic networks

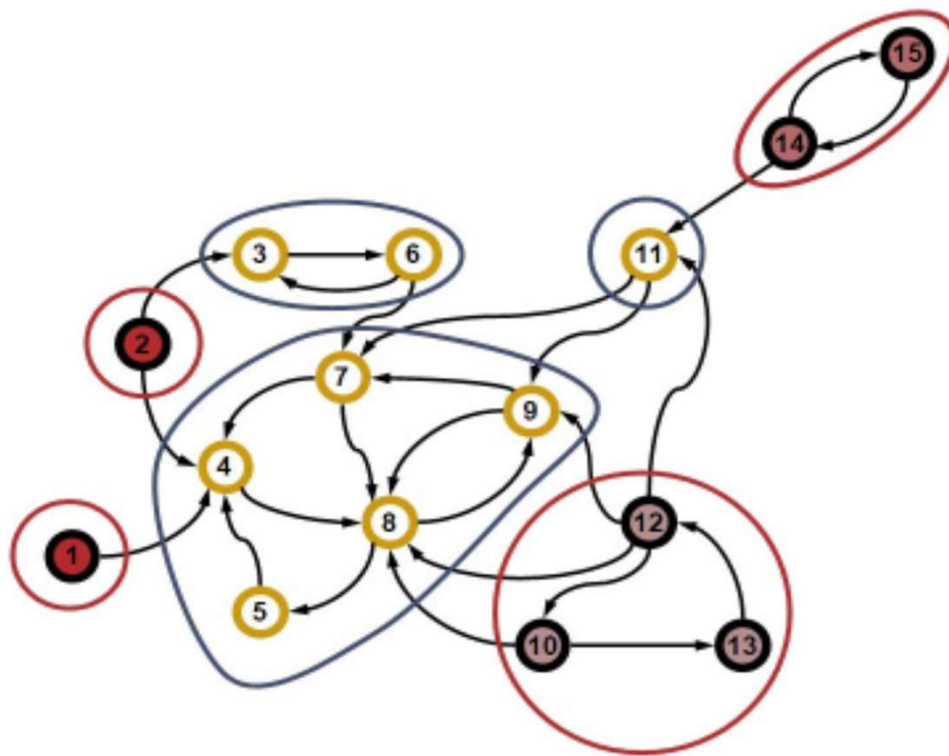
genes $\Rightarrow$ enzymes $\Rightarrow$ metabolic network



A species-specific metabolic network can be constructed from the genome automatically with tools such as ModelSEED

Metabolic networks and seeds

- **Seeds:** metabolite nodes without incoming arcs
- they cannot be produced by the metabolic network and therefore **represent essential nutrients**



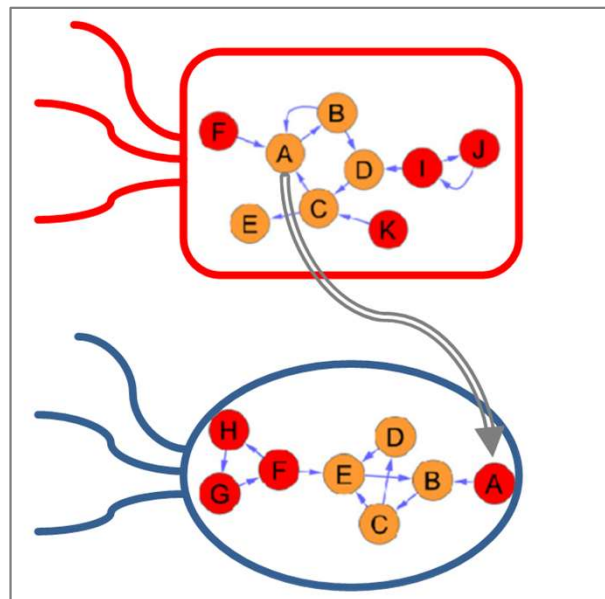
Detection of **seeds** involves decomposition of network into strongly connected components (circles)

Borenstein et al. "Large-scale reconstruction and phylogenetic analysis of metabolic environments" PNAS 2008.

Reverse ecology: genome-based interaction prediction

- **Metabolic complementarity:** based on the number of metabolites produced by organism A that are seeds of B's metabolic network
- **Metabolic overlap:** based on the number of overlapping seeds

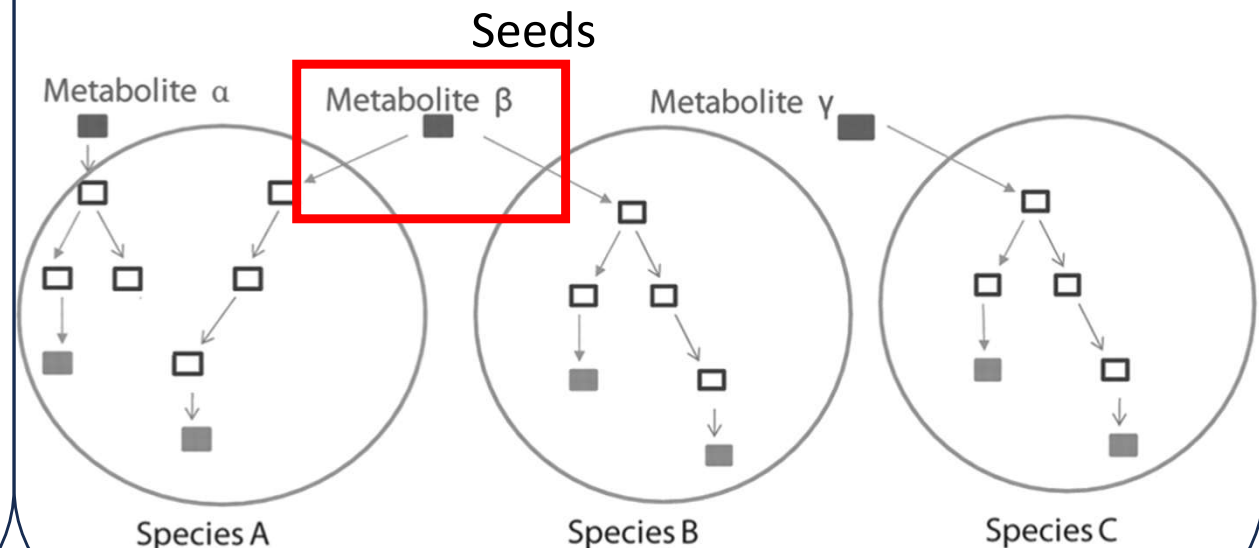
Complementarity (Cross-feeding)



Seed compounds

Intermediate compounds

Overlap (Competition)

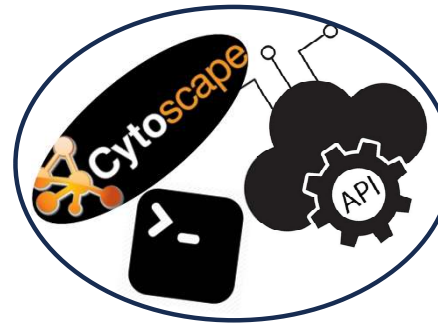


NetCooperate (complementarity): Levy et al. (2015) BMC Bioinformatics 16, 164
NetCmpt (overlap): Kreimer et al. (2012) Bioinformatics 28, 2195-2197

Microbial network annotation

<https://microbetag.readthedocs.io>

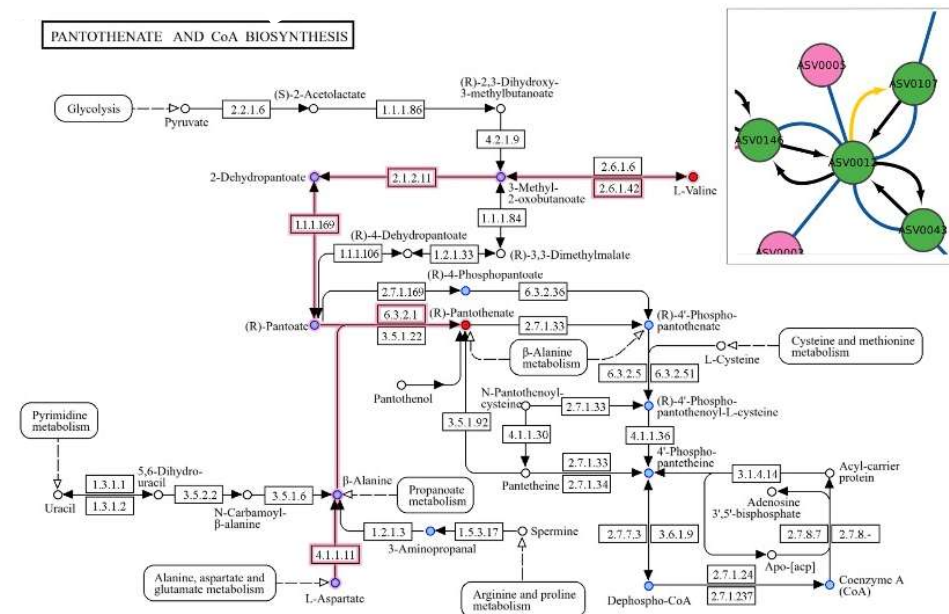
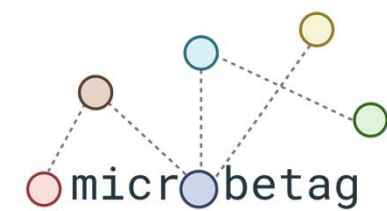
- microbial network annotation tool with GUI (Cytoscape app), server (API) and CLI access
- Annotates taxon nodes with predicted phenotypic traits from phenDB and FAPROTAX
- Annotates edges with predicted cross-feeding/competition (reverse ecology approach)



Ermis Delopoulos



Haris Zafeiropoulos

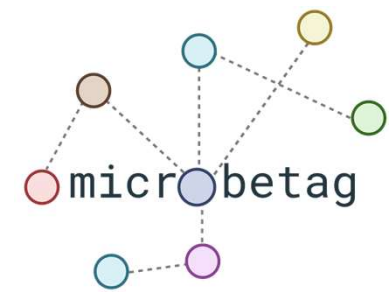


Zafeiropoulos et al. "microbetag: simplifying microbial network interpretation through annotation, enrichment tests and metabolic complementarity analysis" Genome Biology, 26, 292.

PhenDB: Feldbauer et al. (2015) BMC Bioinformatics 16(14):1-8

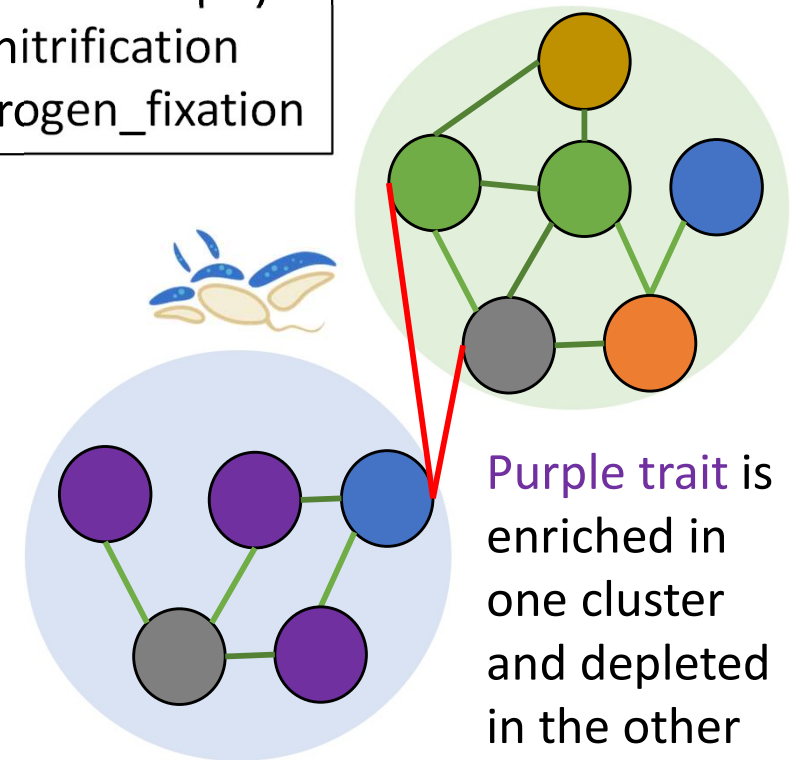
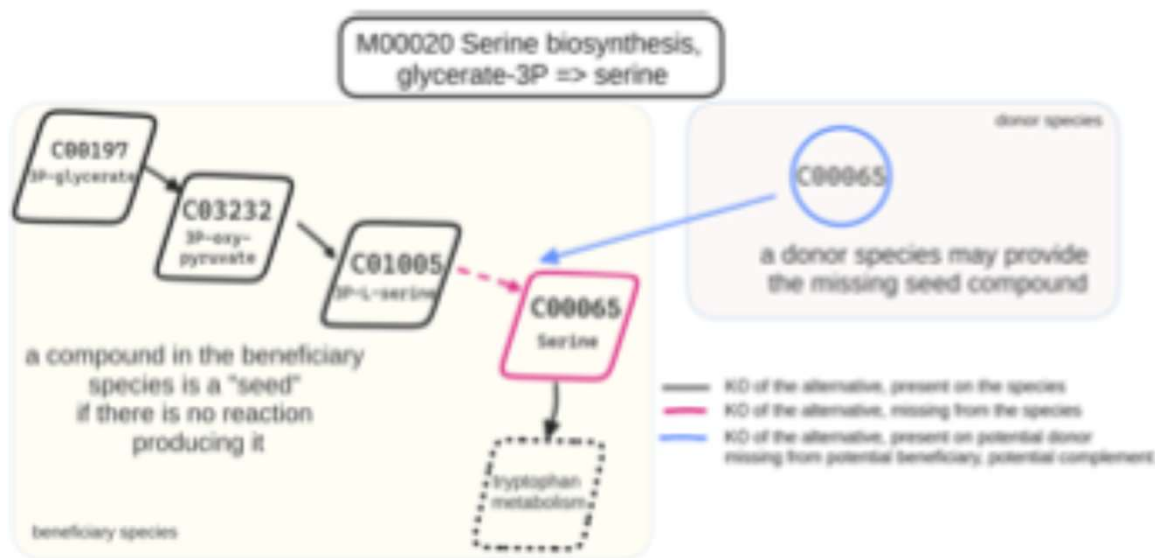
FAPROTAX: Louca et al. (2016) Science 353(6305):1272-1277

Microbial network annotation

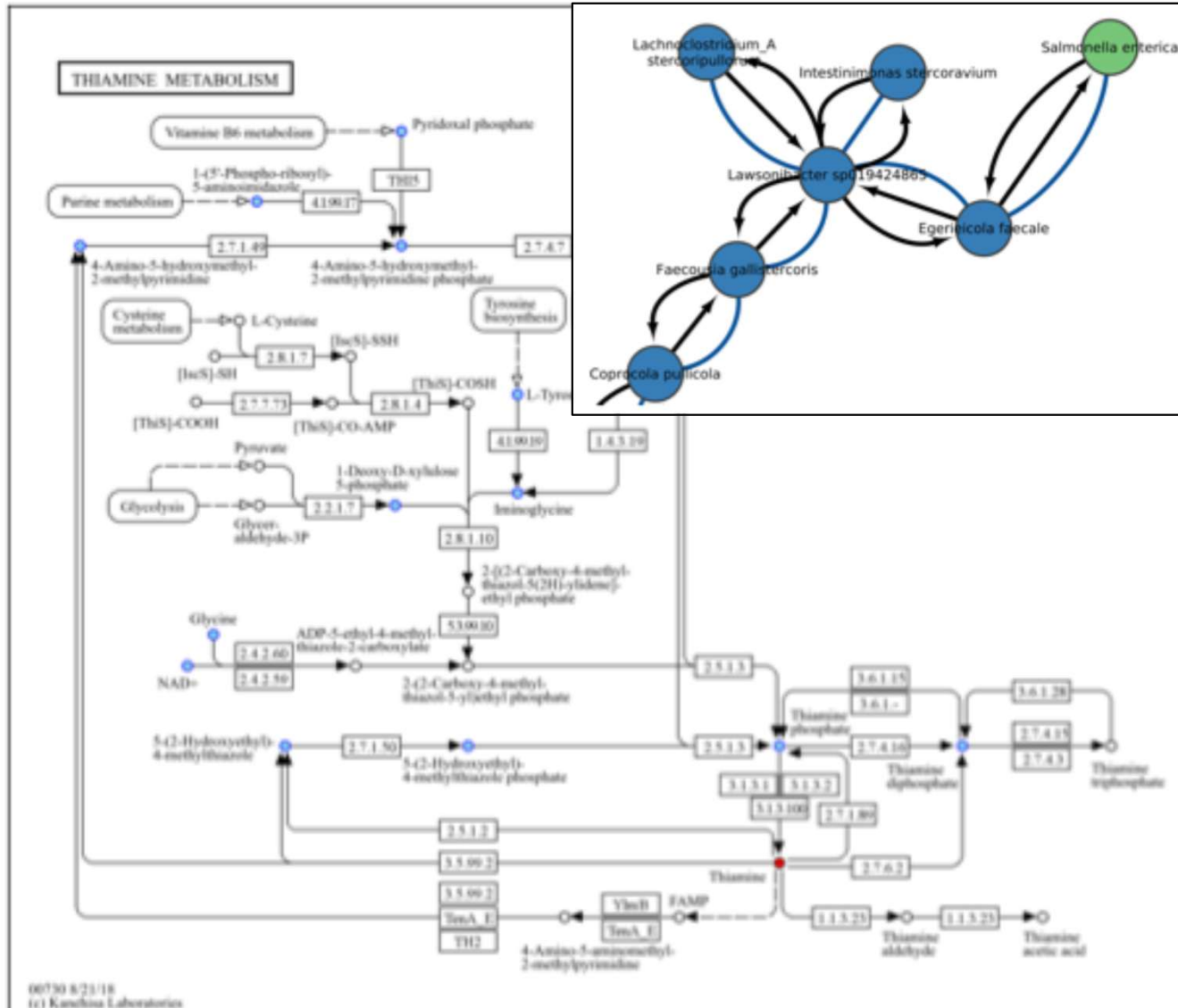
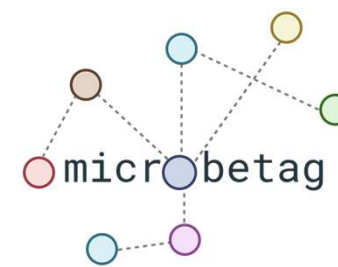


- *microbetag* wraps FlashWeave for network construction and manta for network clustering
- *microbetag* optionally tests for trait enrichment/depletion in clusters
- *microbetag* reports donor metabolites that complement recipient metabolism at the network and pathway level

Trait examples:
plant_pathogen
methanotrophy
denitrification
nitrogen_fixation



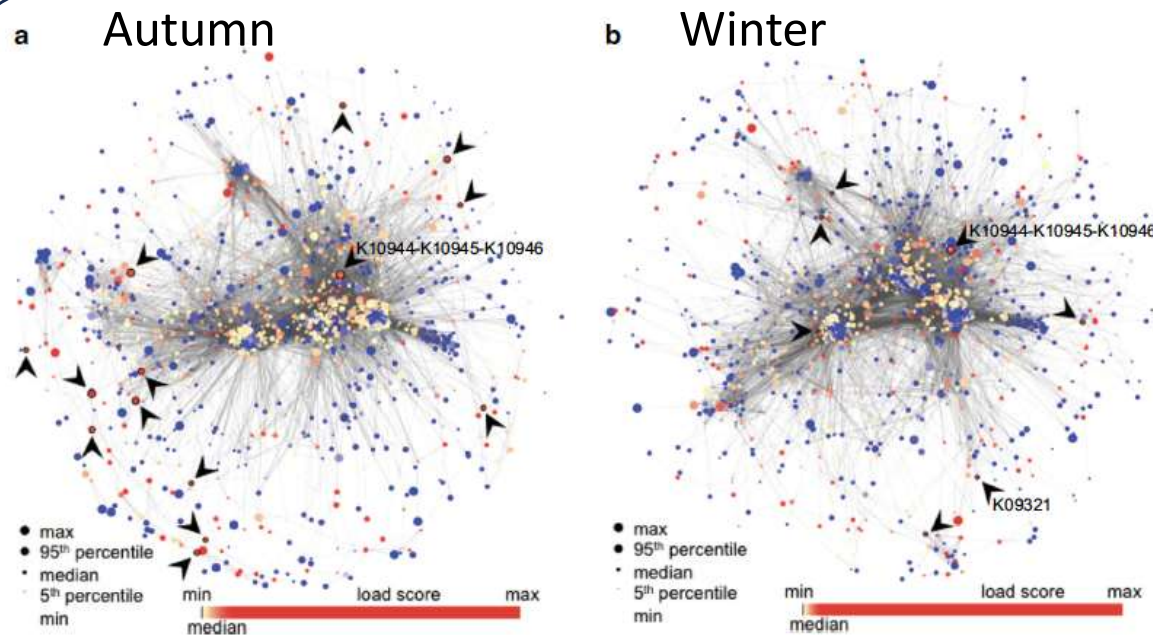
Application example



- Chicken infected with *Salmonella enterica* and treated with five probiotic species plus inulin (PoultryStar®)
- *microbetag* run locally (computing cluster) on custom MAG catalogue
- Networks constructed from MAG abundances with FlashWeave
- *S. enterica* predicted to be a thiamine auxotroph
- *Egeriella faecale* can provide thiamine to *S. enterica*

Metabolic analysis of microbial meta-omics data

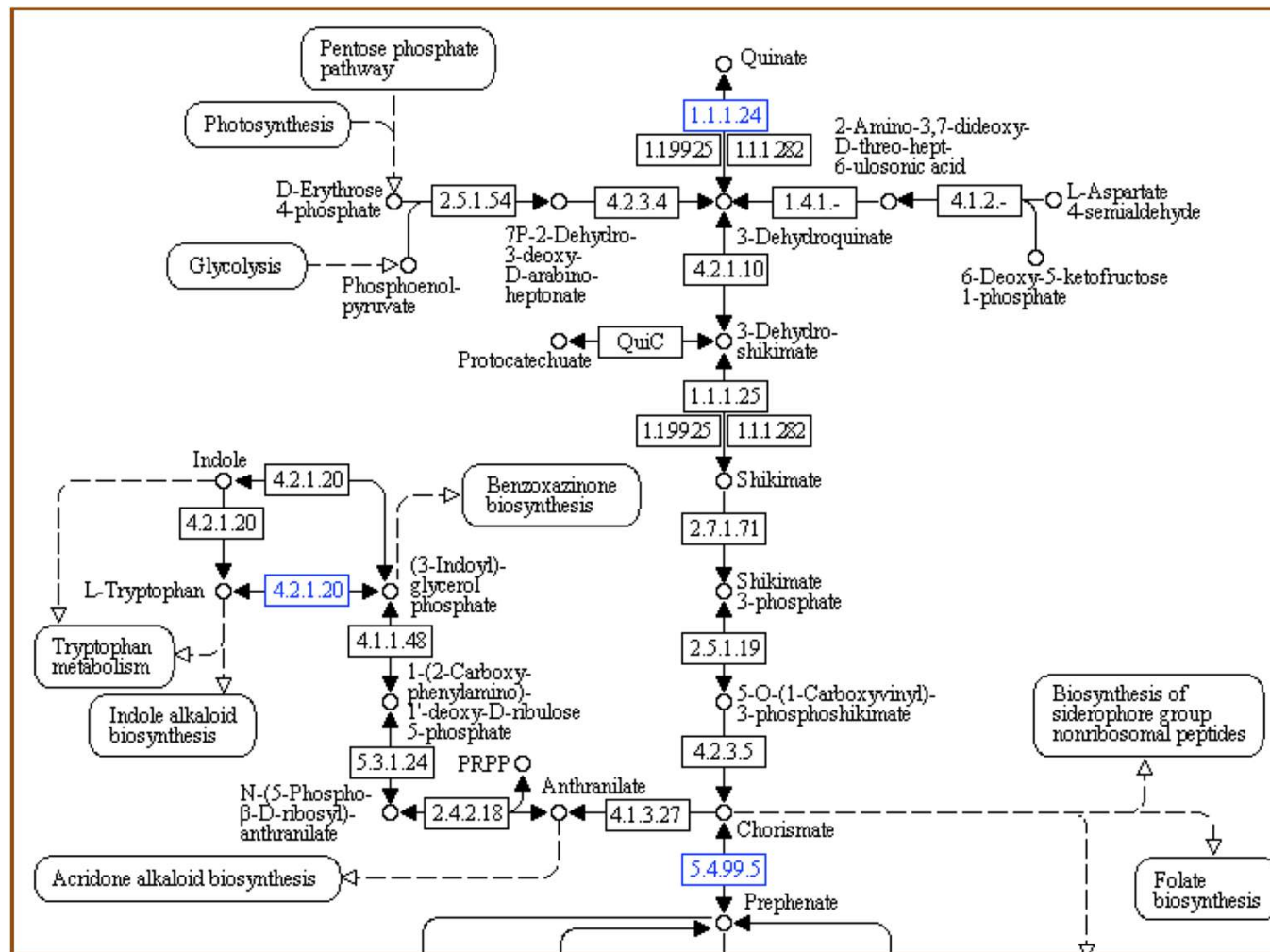
- microbetag annotates inferred networks between microbes
- Next step: find out whether and which part of the community metabolism changes across conditions and who is responsible for the change
- Without computing correlations



Roume et al. “Comparative integrated omics: identification of key functionalities in microbial community-wide metabolic networks” npj Biofilms & Microbes 2015.

Metabolic sub-network extraction

Given **nodes of interest** (metabolites and/or reactions and/or enzymes and/or enzyme-coding transcripts/genes) and a **metabolic network**...

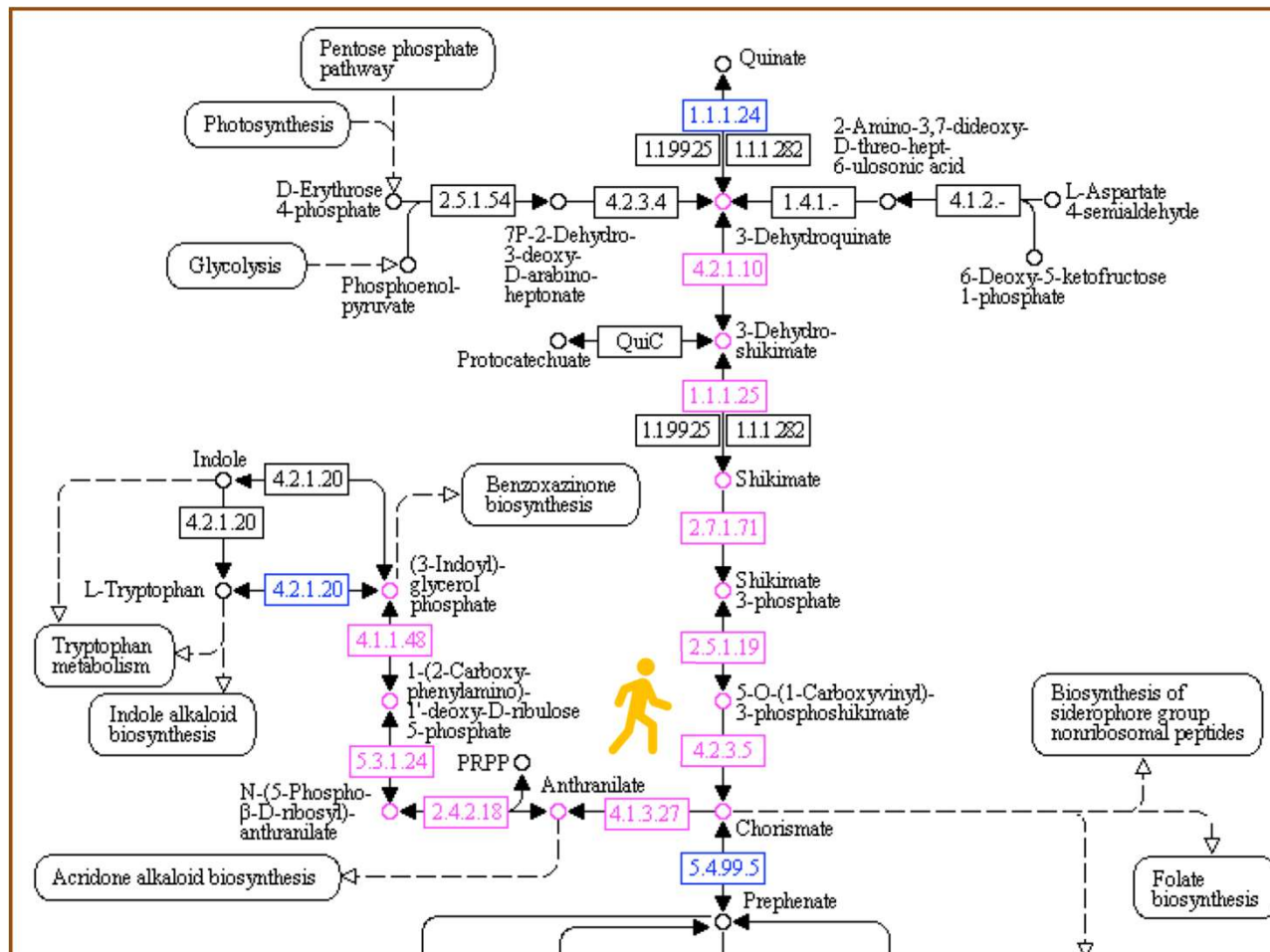


Metabolic sub-network extraction

...extract a **metabolic pathway** connecting the **nodes of interest** using a **sub-network extraction algorithm**



Biologically meaningless shortcuts via highly connected metabolites (e.g. ADP) can be avoided using weights or KEGG reactant pairs

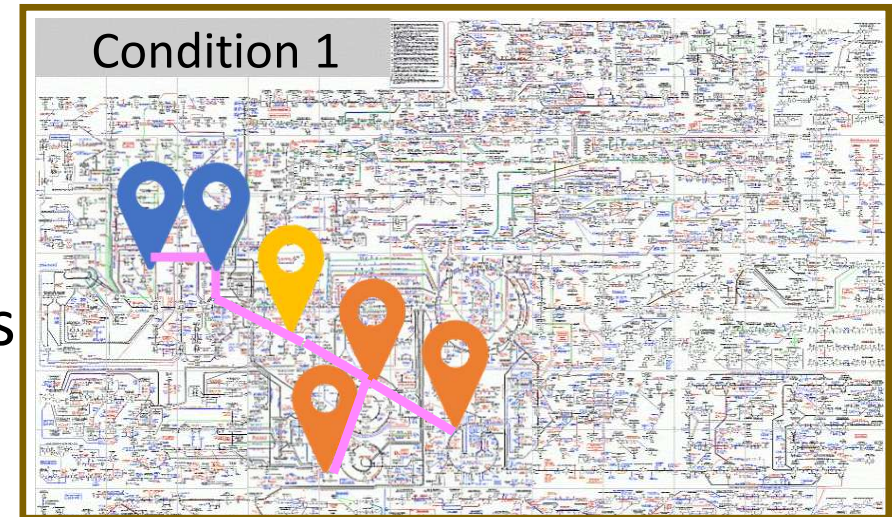


Tool development for application to meta-omics data in progress (Anna Greffe)

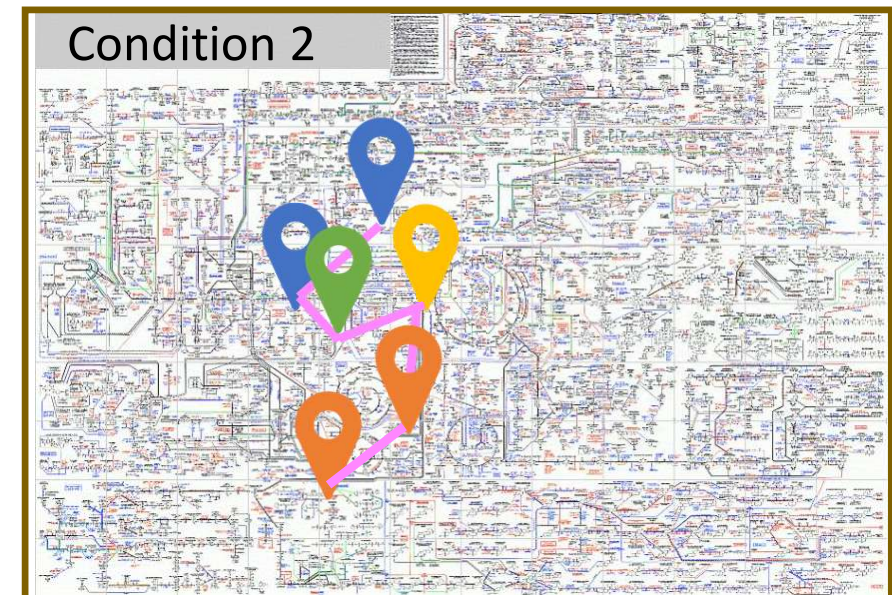
Faust et al. (2010) "Pathway discovery in metabolic networks by subgraph extraction" Bioinformatics 26, 1211-1218 (command line tools available)

Application of metabolic sub-network extraction to meta-omics

- Identify **genes/transcripts/proteins** (optionally linked to different species) **and/or metabolites** significantly differentially abundant across conditions
- Map them onto the **metabolic network** of a microbial community and extract and compare **sub-networks**
- Generate mechanistic hypotheses for meta-omic data sets using metabolic knowledge without relying on potentially biased pathway definitions
- Principle can be extended beyond metabolic networks to known gene regulation or protein-protein interactions

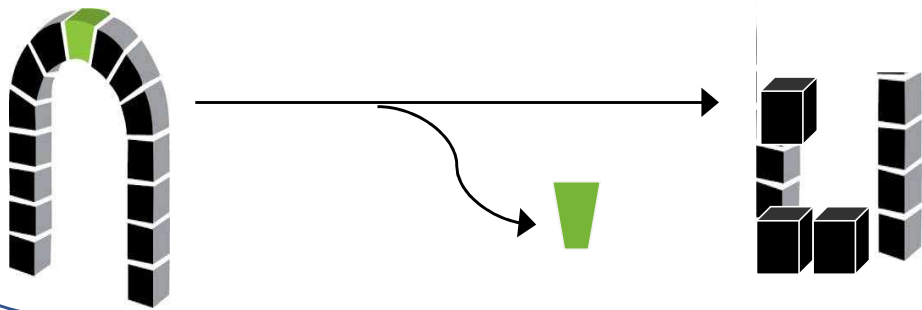


Enzymes of interest in species **A**, **B**, **C**
Metabolites of interest

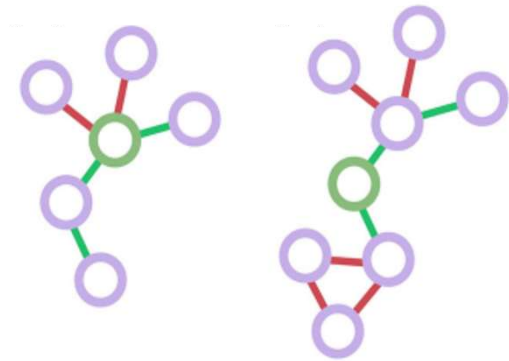


6. What about network properties?

Paine on **keystone species**: “These individual populations are the keystone of the community’s structure, and the integrity of the community and its unaltered persistence through time, that is, stability are determined by their activities and abundances”



Can we identify keystone species with networks?

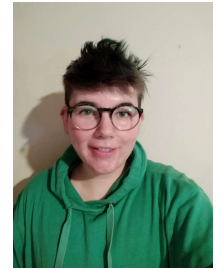


Hub taxon:
high degree

Connector
taxon: high
betweenness

R.T. Paine (1969). A Note on Trophic Complexity and Community Stability. *The American Naturalist* 103, 91-93.

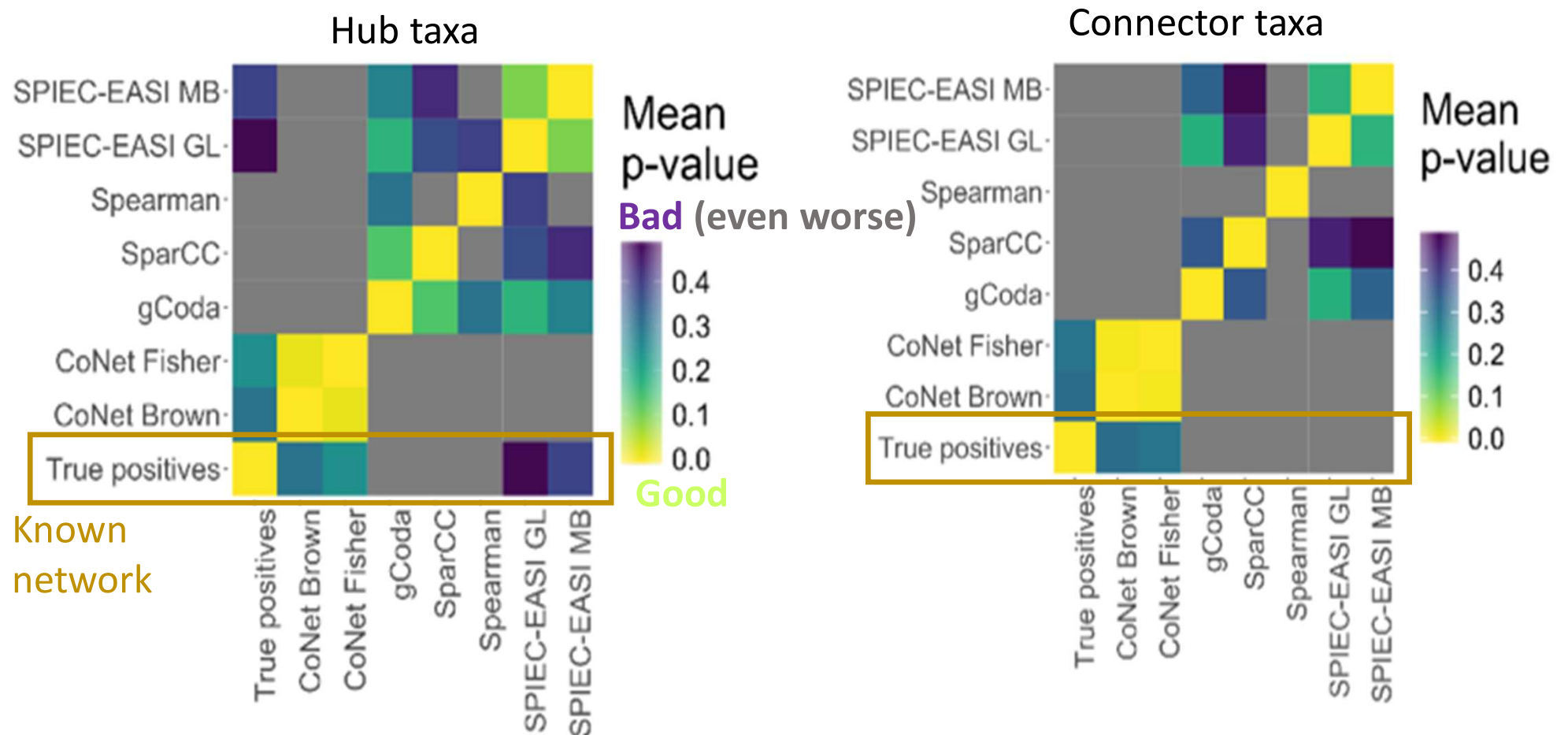
Can microbial network inference tools predict hub taxa?



Sam Röttjers

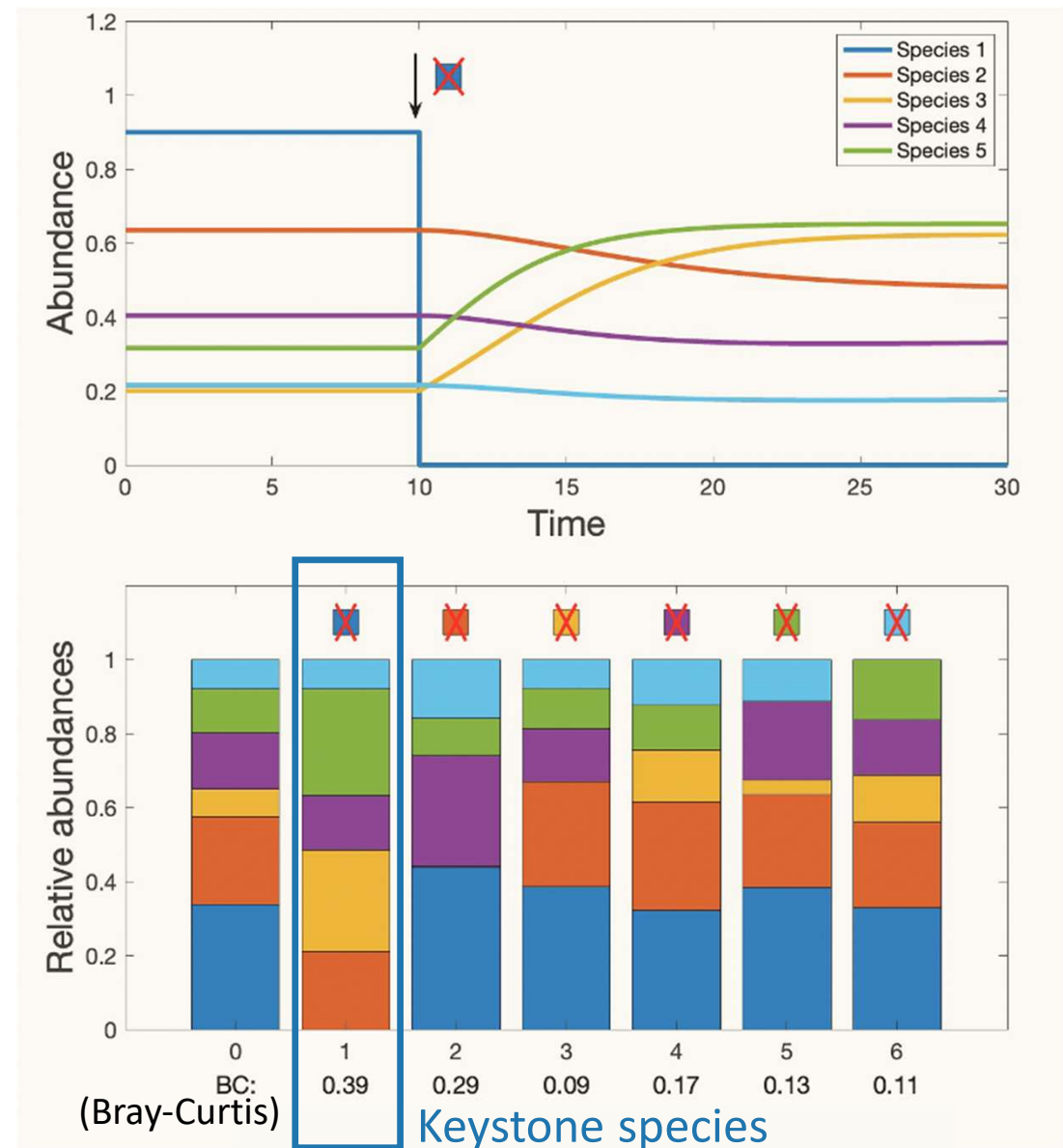
- Not well (in synthetic data with known hub taxa)

Intersection between known and inferred hub/connector taxa:



How do we detect keystones?

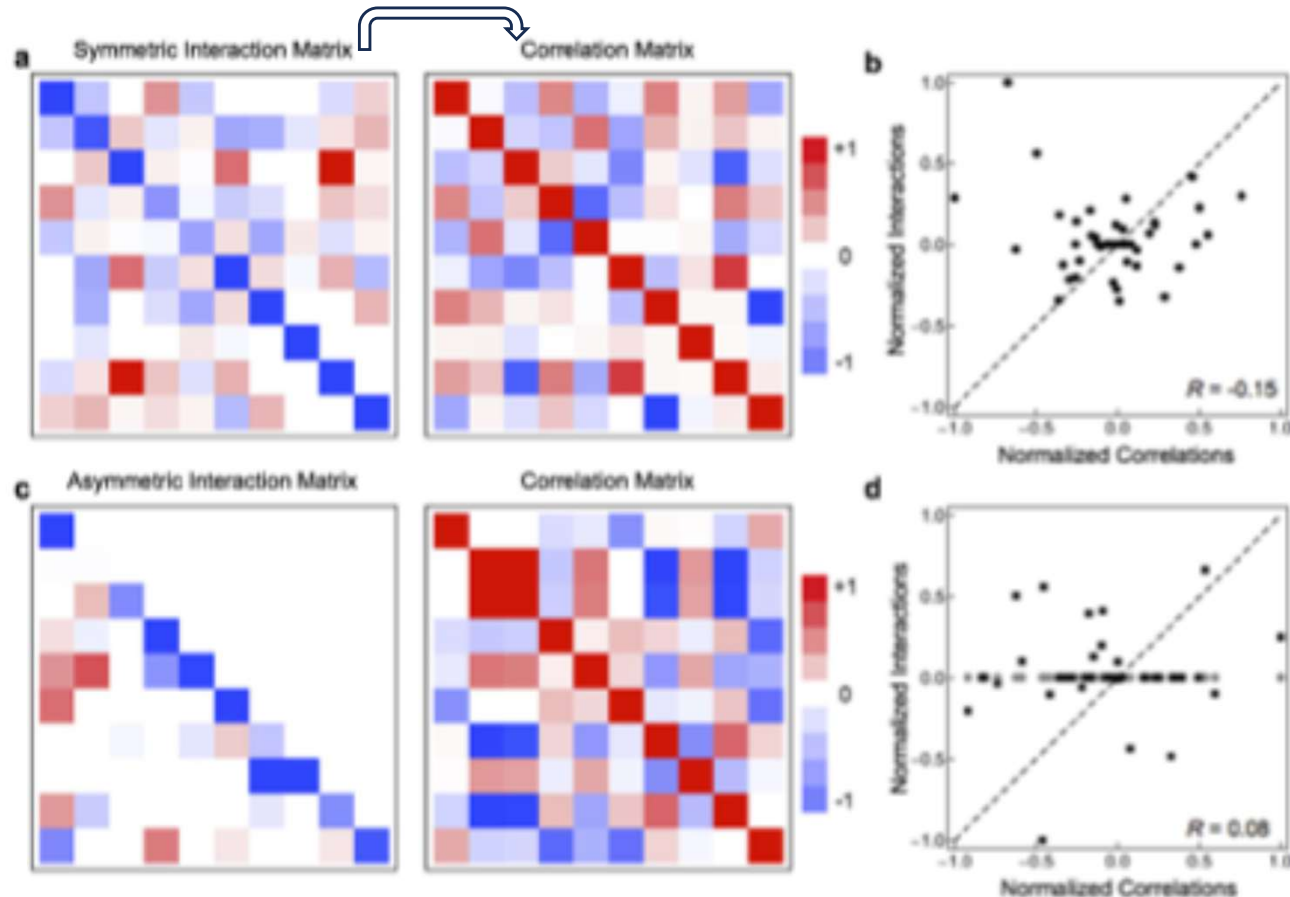
- Operational definition: taxa (or functions) whose change in abundance alters the taxonomic (and/or functional) composition of the community **more strongly than the change of any other taxa**
- Can be experimentally tested by systematic leave-one-out experiments (for N species, assemble the community N times without 1 species)



Garza, Gonze & Faust (2026) "Keystone concept revisited: insights into microbial community dynamics and control" Nature Review Microbiology 24, 359-371

...but interaction networks are not association networks

Simulations with generalized Lotka-Volterra

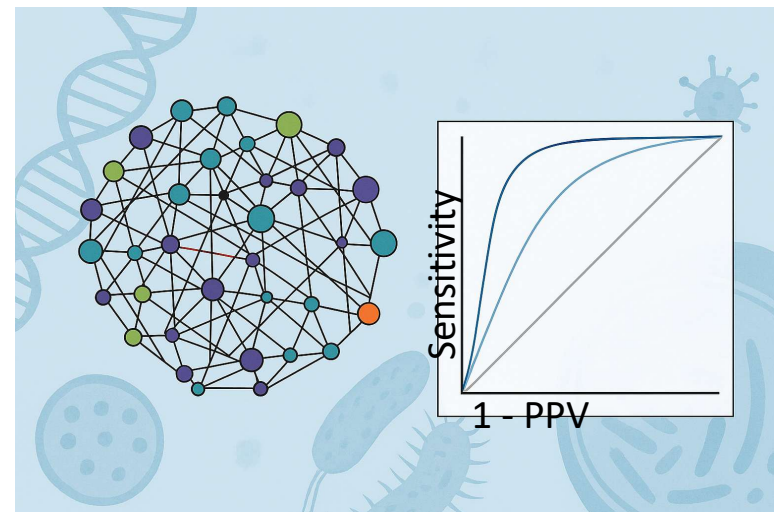


- Systematic study of known keystones and their properties still missing

Fisher & Mehta (2014) "Identifying Keystone Species in the Human Gut Microbiome from Metagenomic Timeseries Using Sparse Linear Regression" PLoS ONE 9, e102451.

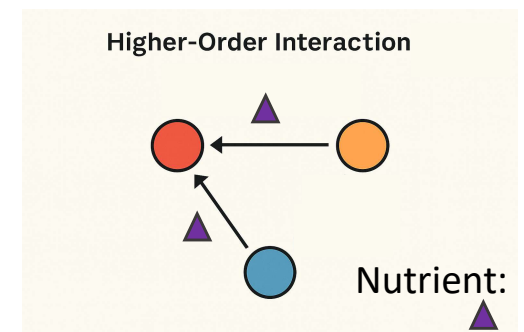
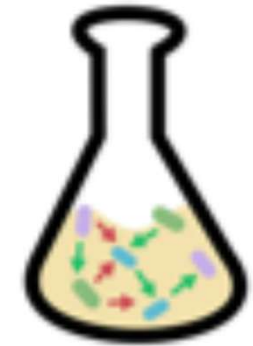
7. Evaluation of microbial network inference on biological data

- To date, no systematic evaluation of microbial network inference tools on biological data available
- Small-scale evaluations scattered throughout the literature, with mixed results (e.g. Lima-Mendez et al. Science 2015, Biswas et al. Lecture Notes in Bioinformatics 2015, Durán et al. Cell 2018)



The challenges of biological evaluation

- With complex ecosystems in situ, it is hard to know whether two species do not interact (confirming the negative is harder than confirming the positive)
- That means that we cannot assess network inference accuracy in situ (only sensitivity)
- In vitro, we can measure all pair-wise interactions comprehensively and thus can assess accuracy (i.e. sensitivity and specificity)
- But we don't know whether interactions in vitro also happen in situ
- And there may be higher-order interactions (roughly: an interaction between two species that is modified by the presence of additional species)

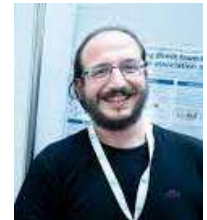


Building the basis for biological evaluation: μ GrowthDB

- Goal: make it easy to store, upload, visualize, compare and share data of microbial monocultures and defined communities

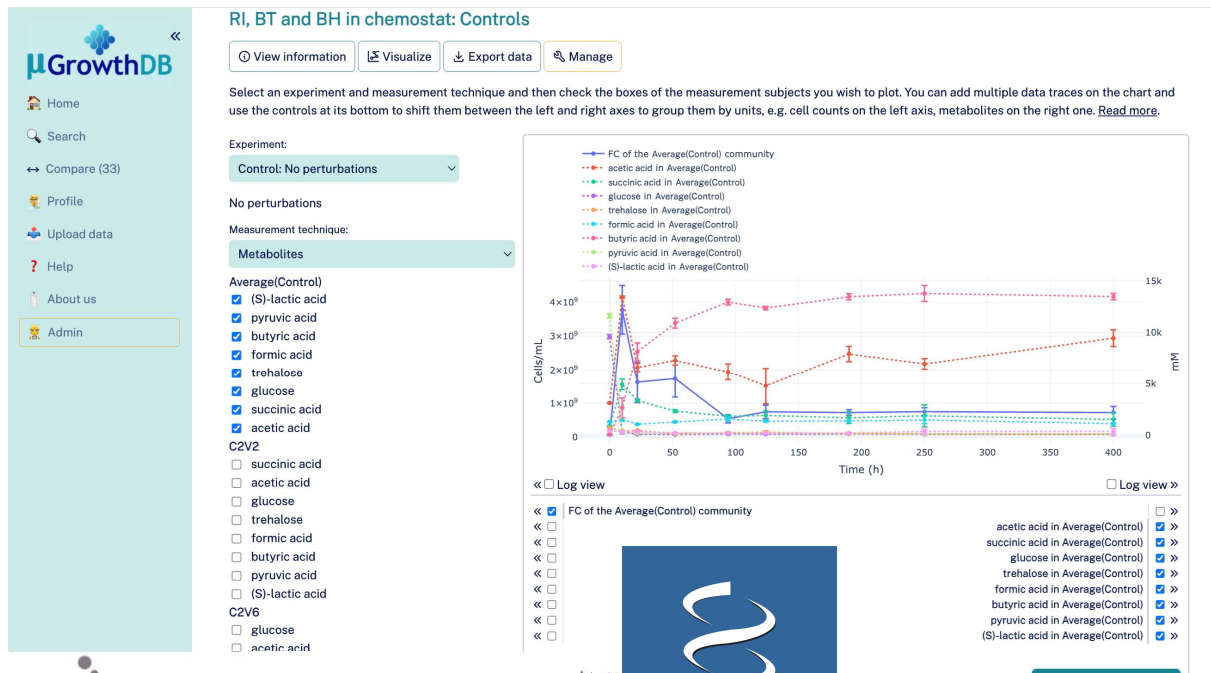


Andrey
Radev



Haris
Zafeiro-
poulos

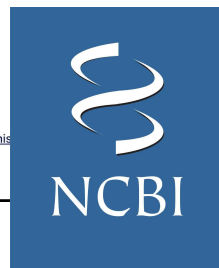
<https://mgrowthdb.gbiomed.kuleuven.be>



You too can improve
 μ GrowthDB by
uploading data



MediaDive



Today's Key Lessons

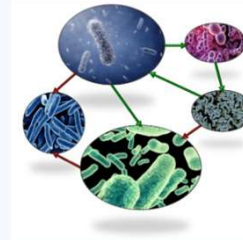


- **Correlation != causation**
- Use existing biological knowledge in (meta-omic) data analysis
- Do not use tools without first evaluating them, ideally on biological data with known ground truth
- **Do not trust predictions without experimental validation**

Acknowledgements

KU LEUVEN

Lab of Microbial Systems Biology



Sam Röttjers
(now at Biome
Makers)



Haris
Zafeiro-
poulos



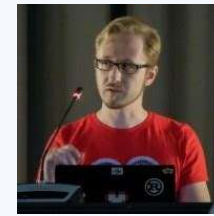
Daniel
Garza
(now at
INRAE)



Anna
Greffe
(Postdoc)



Ermis Delo-
poulos
(Master
student)



Andrey
Radev
(Master
student)

**UNIVERSITY
OF
CALIFORNIA**

**Biocomputing,
and Informatics**

John
Morris



ULB



**Theoretical
Chronobiology**
Didier Gonze



**University of
Copenhagen
Globe Institute**

Antton
Alberdi



**UNIVERSITY
OF TURKU**

**Turku Data
Science Group**

Leo Lahti



ETH zürich

**Institute of
Food, Nutrition
and Health**



Annelies
Geirnaert



Andi
Erega

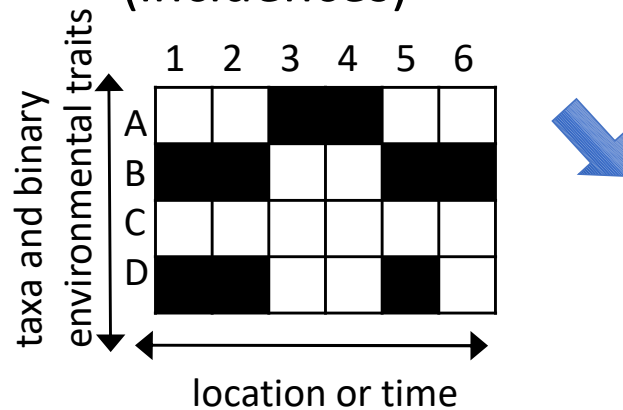


Thank you!

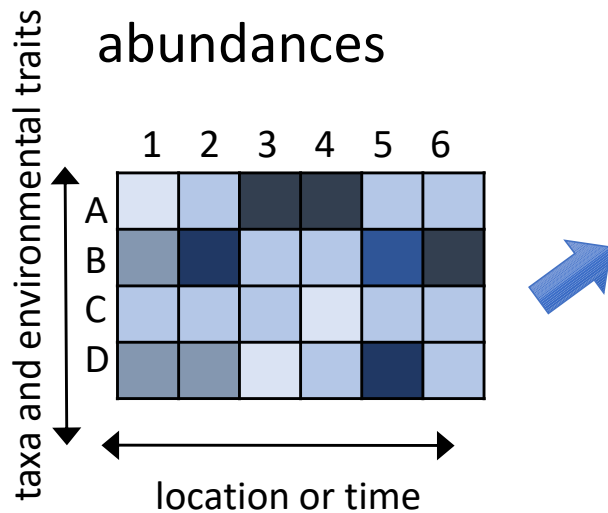
Appendix: Similarity-based network inference

INPUT

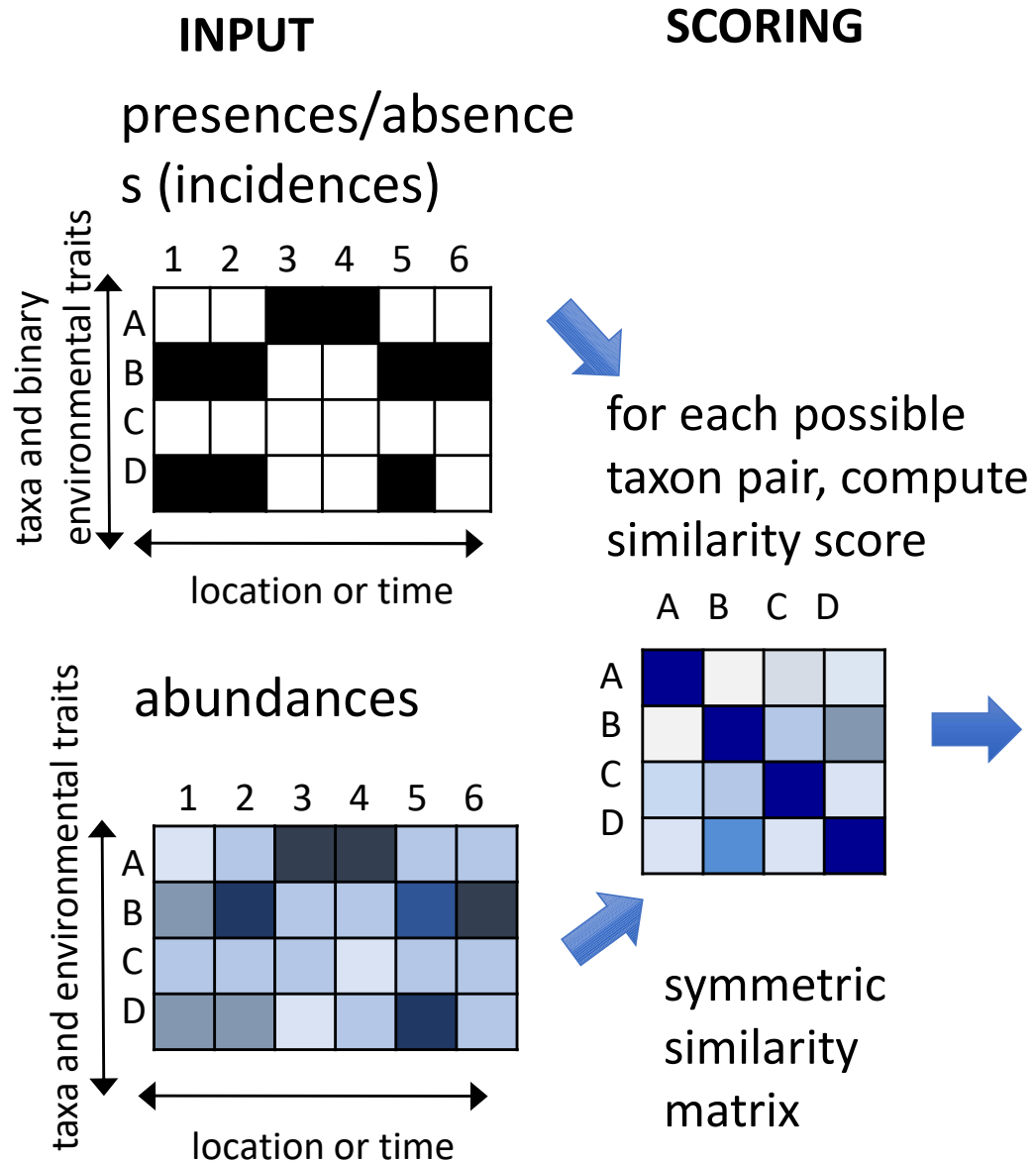
presences/absences
(incidences)



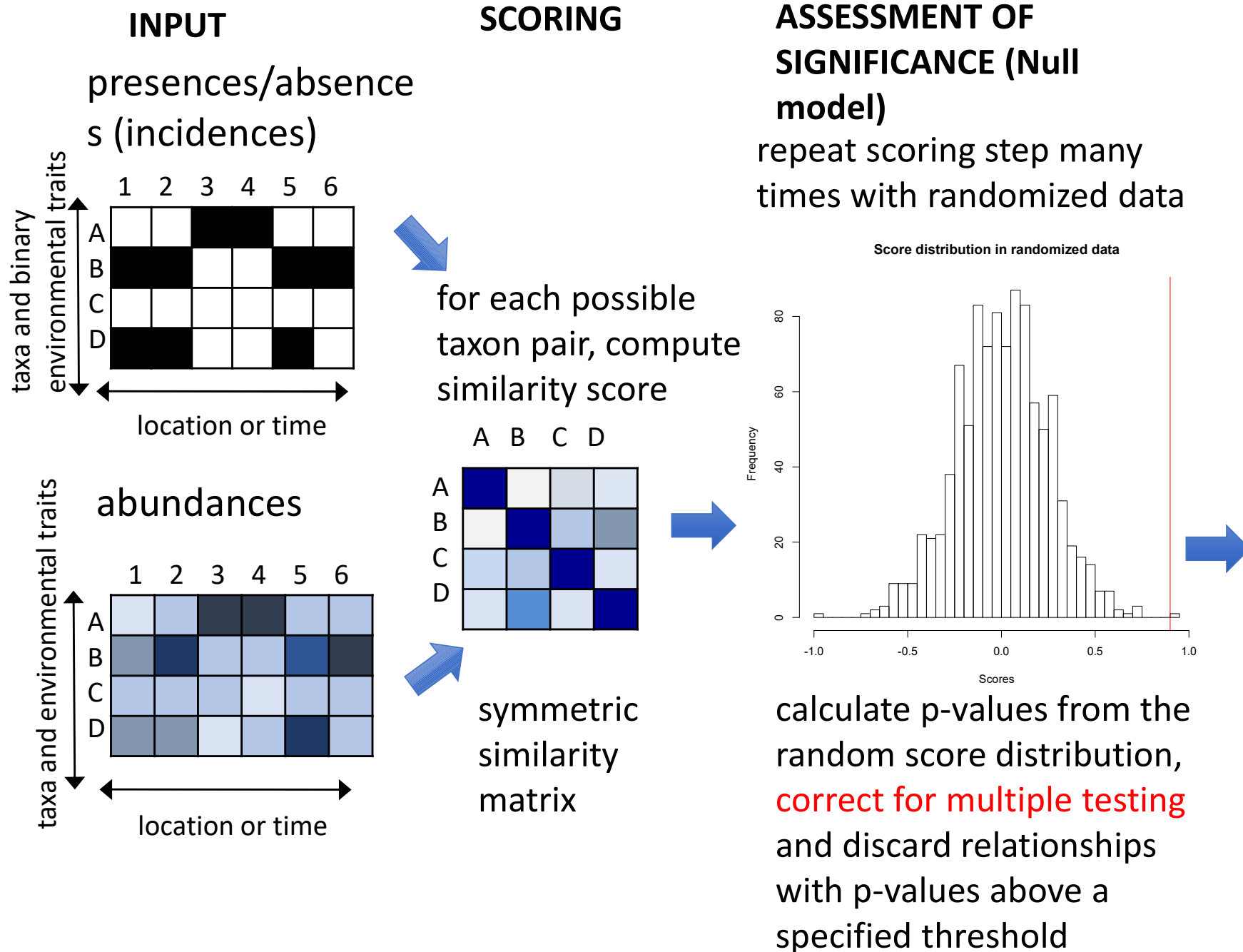
abundances



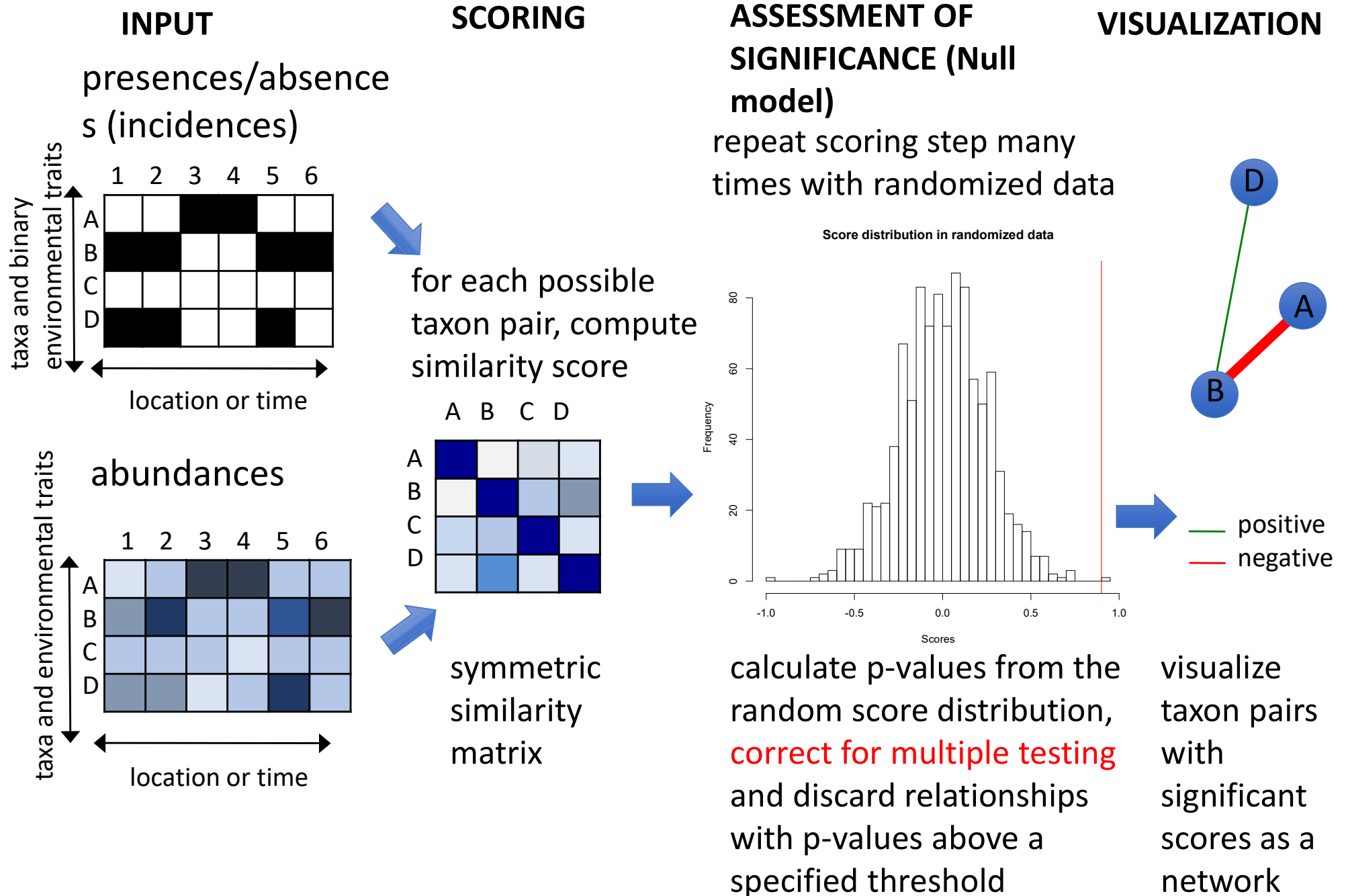
Appendix: Similarity-based network inference



Appendix: Similarity-based network inference



Appendix: Similarity-based network inference



Zero treatment in HARMONIES

- HARMONIES is a recent microbial network inference tool
- It models count data with a mixture distribution: an **indicator function I for sampling zeros (zero inflation)** and a **negative binomial distribution (NB)** representing counts and structural zeros
- π_i determines mixing **weights** in sample i ; s_i describes **sequencing depth**, a_{ij} is the **true taxon count**

$$y_{ij} \sim \pi_i I(y_{ij} = 0) + (1 - \pi_i) NB(\overset{\text{mean}}{\lambda_{ij}}, \overset{\text{dispersion}}{\phi_j})$$

observed
count

"there but not seen"
(zero inflation part)

"there or absent" (simulating
read counts and true absences)

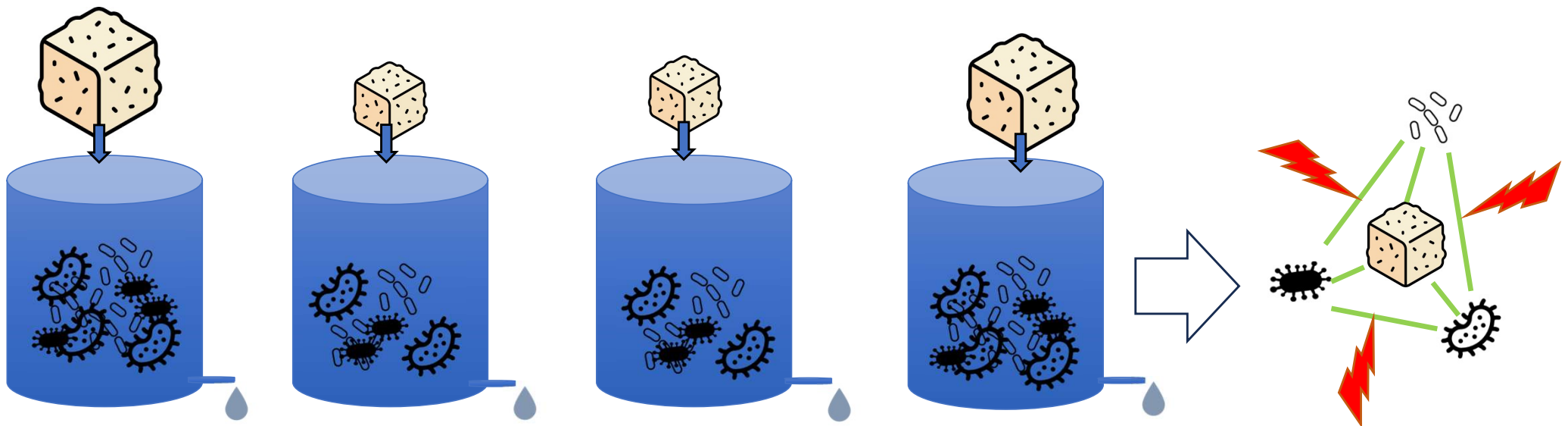
$$\lambda_{ij} = s_i \overset{\text{true count}}{a_{ij}}$$

(Logarithms of a_{ij} values
are assumed to be
normally distributed)

Appendix: Total count as a confounder in network inference

- If change in total cell count is not driven by interactions it can introduce false positive edges between taxa

Example: continuous flow reactors with different nutrient concentrations in the inflow (size of sugar cube) that make species grow proportionally better or worse



Appendix: SPIEC-EASI

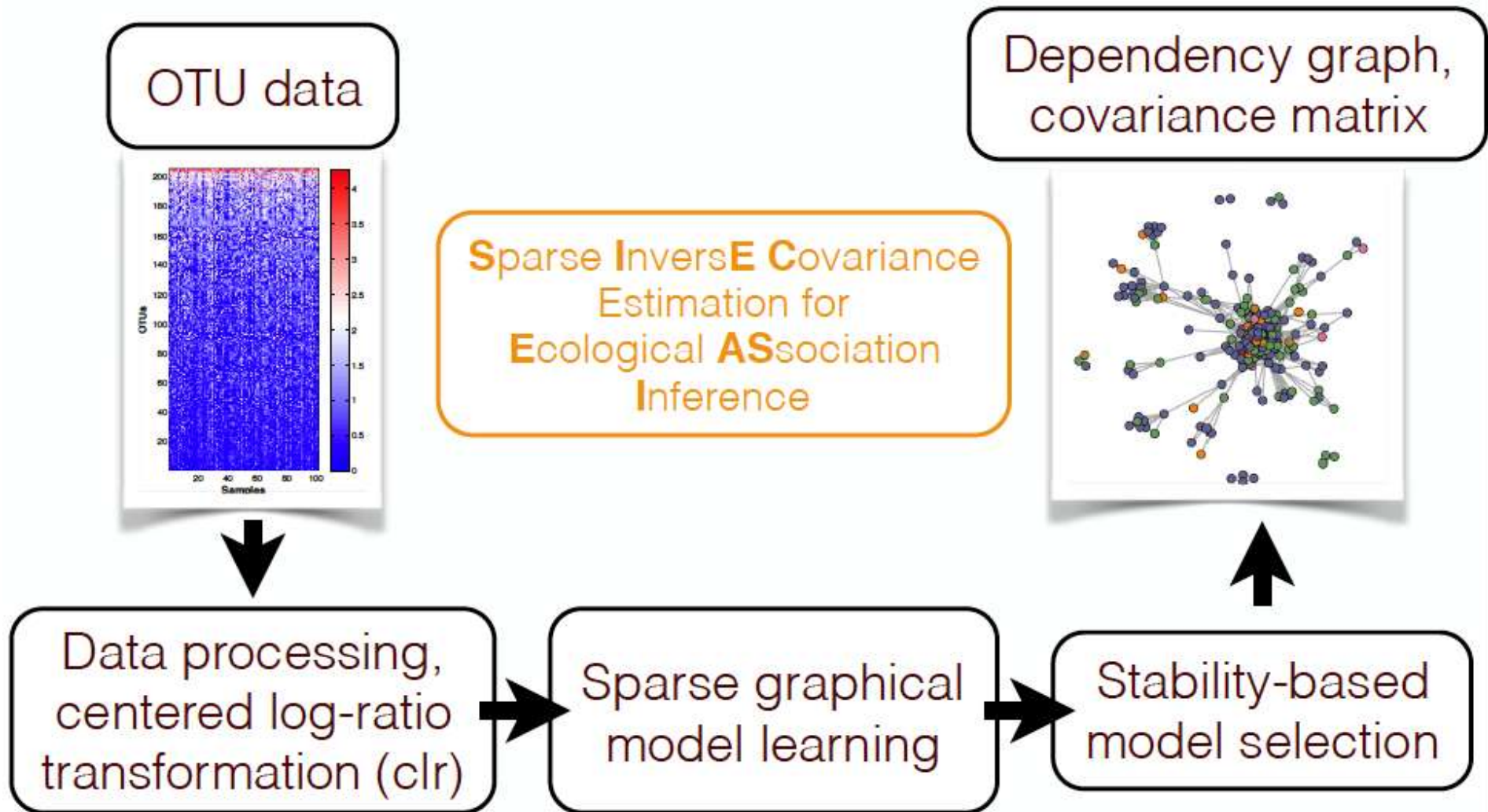
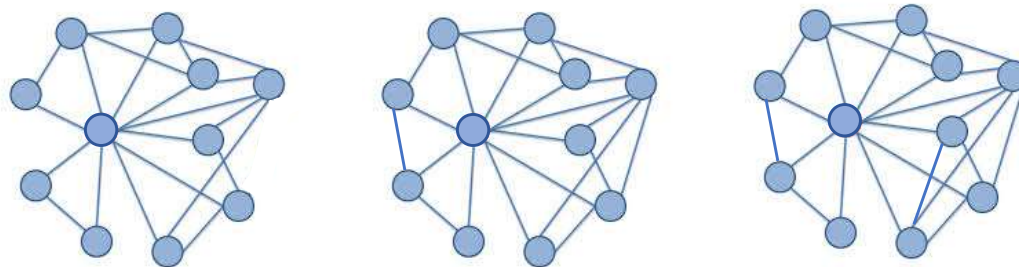


Image taken from: https://stamps.mbl.edu/images/f/f0/STAMPS_Network_1.pdf
(Christian Mueller)

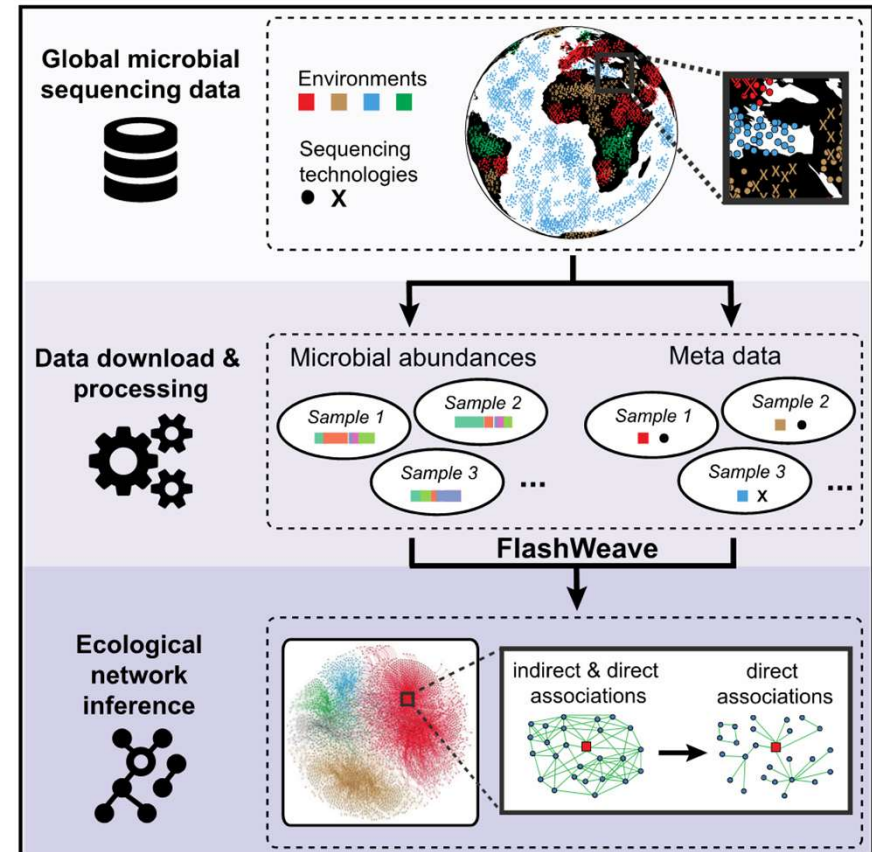
Appendix: SPIEC-EASI: Stability-based model selection

- **StARS: Stability Approach to Regularization Selection**
- Bootstrap technique: Repeat network construction a number of times with 80% of the samples (bootstrap iteration number = rep.num parameter)
- Purpose: select **penalty parameter λ** such that the number of edges present across bootstrap iterations is maximized
- Stability means here: stable with respect to small changes in the data



Appendix: FlashWeave

- SPIEC-EASI's main weakness: does not take environmental data into account
- "FlashWeave = SPIEC-EASI + metadata": clr-transforms data and exploits conditional independence to reduce indirect edge number, taking metadata into account (algorithm: si-HITON-PC)
- Implemented in Julia



Tackmann, Rodrigues and van Mering (2019): "Rapid Inference of Direct Interactions in Large-Scale Ecological Networks from Heterogeneous Microbial Sequencing Data" Cell Systems 2019.08.002.

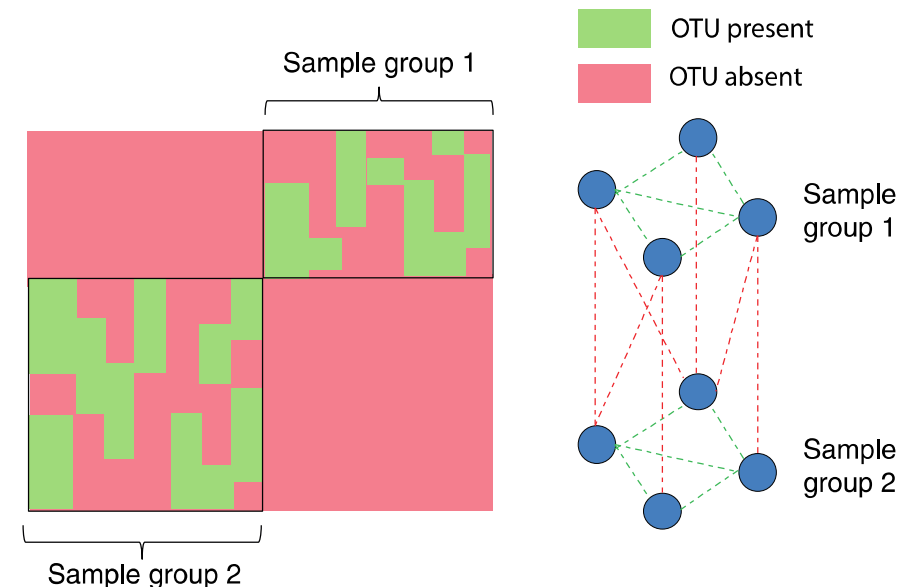
Appendix: FlashWeave modes

Sensitive vs fast mode

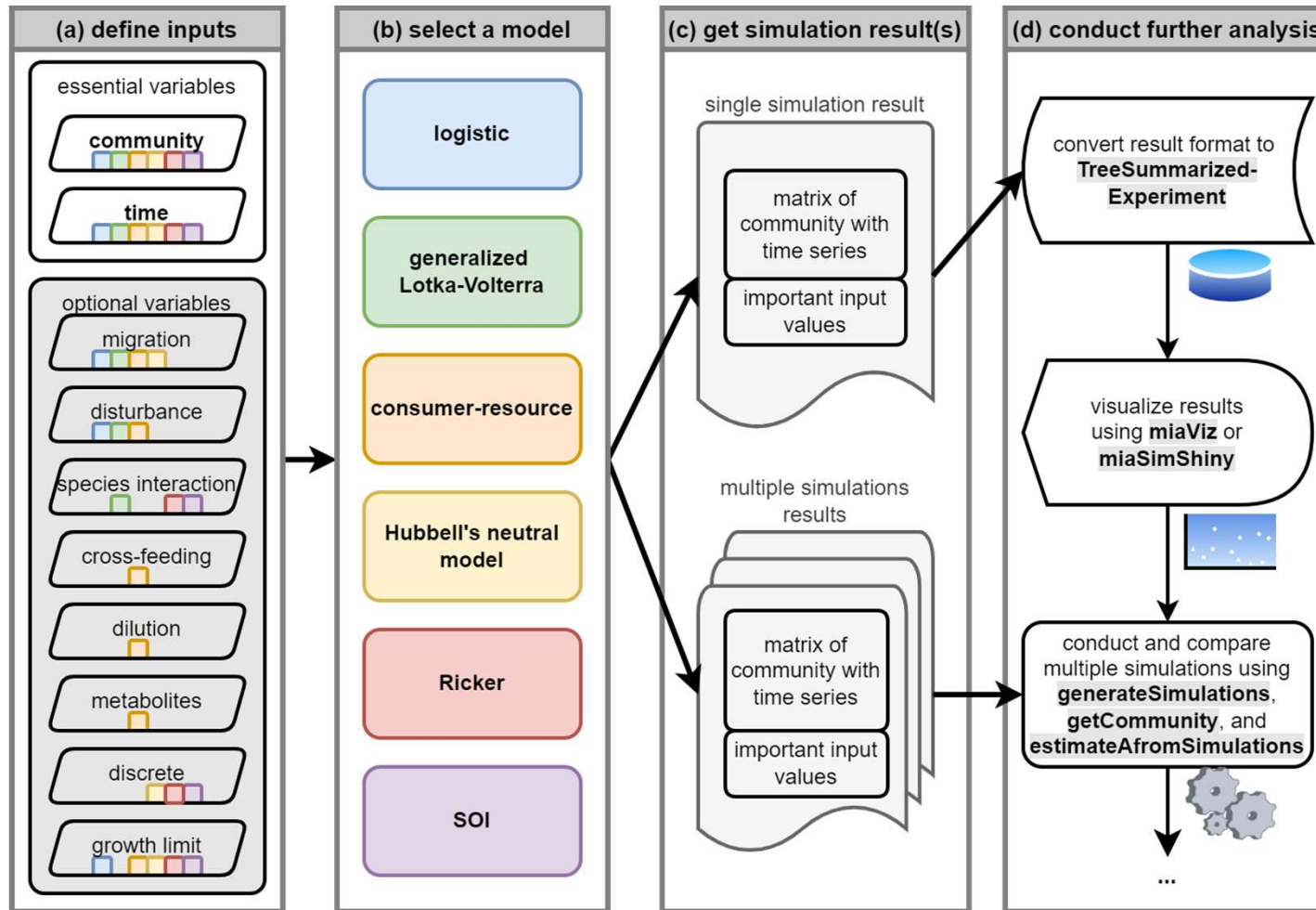
- Implementation of conditional independence:
 - Sensitive mode: partial correlations on abundances, assumes multivariate normal distribution (weak assumption)
 - Fast mode: mutual information on presence/absences

HE mode

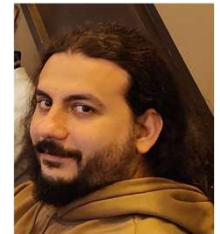
- FlashWeave can optionally ignore zeros ('structural zeros') to deal with heterogeneous samples



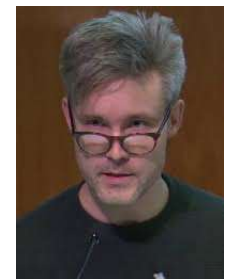
miaSim for in silico microbiome data generation



Yu Gao



Daniel Garza



Leo Lahti

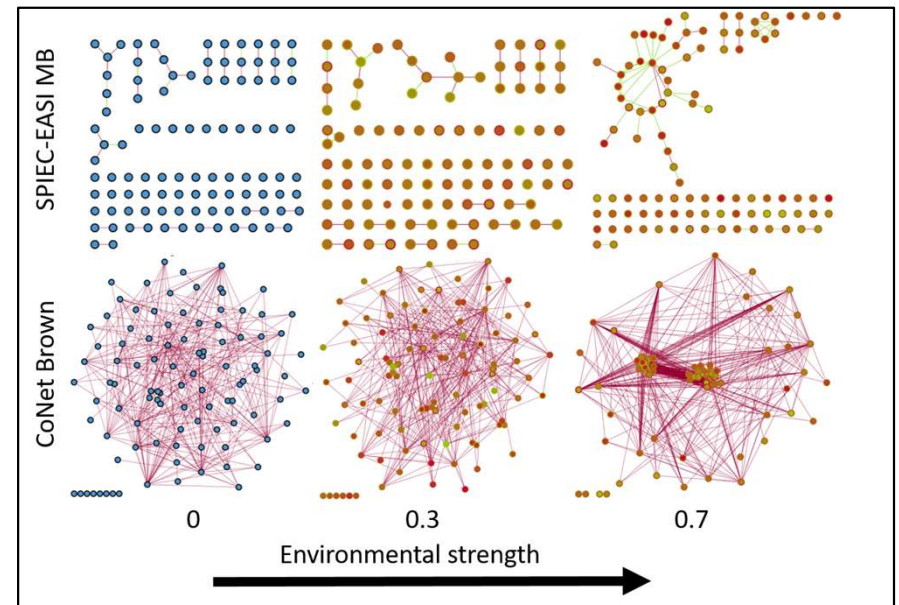
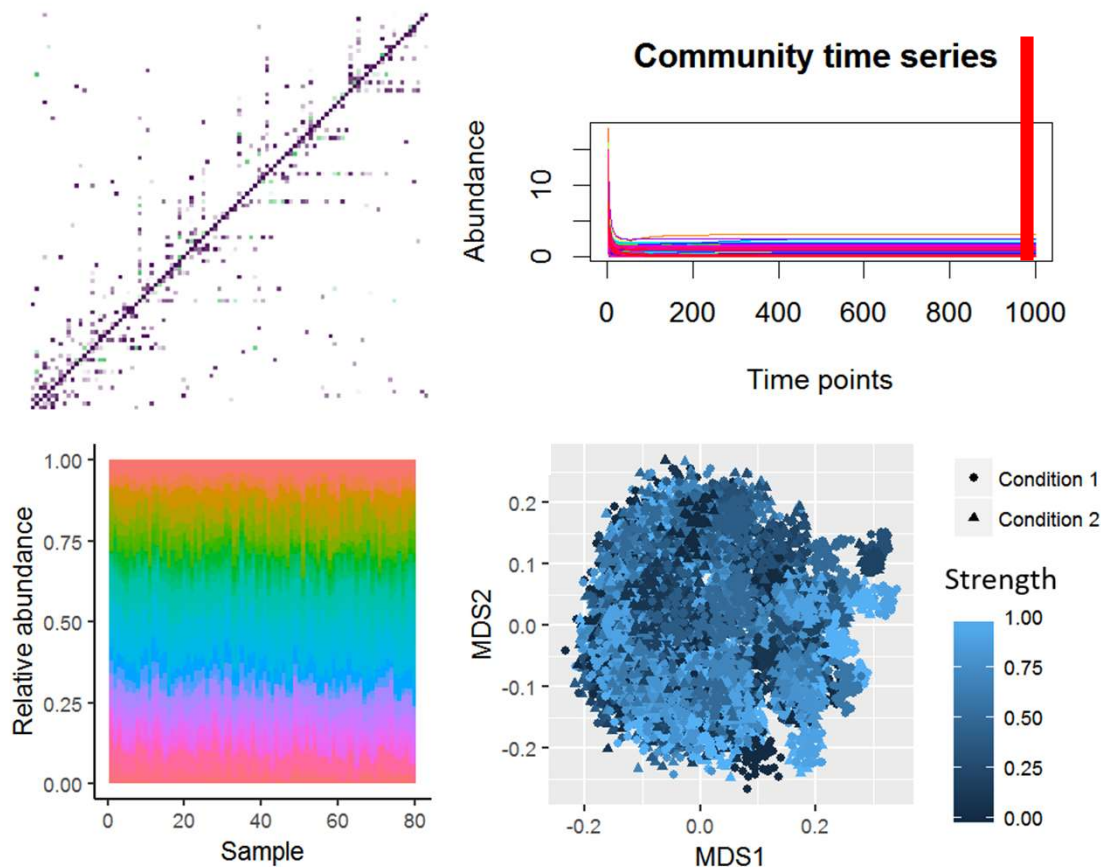
<http://bioconductor.org/packages/release/bioc/html/miaSim.html>

Yu Gao et al. "miaSim: an R/Bioconductor package to easily simulate microbial community dynamics." *Methods in Ecology and Evolution* 14, 1967-1980.

Appendix: Data generation for manta

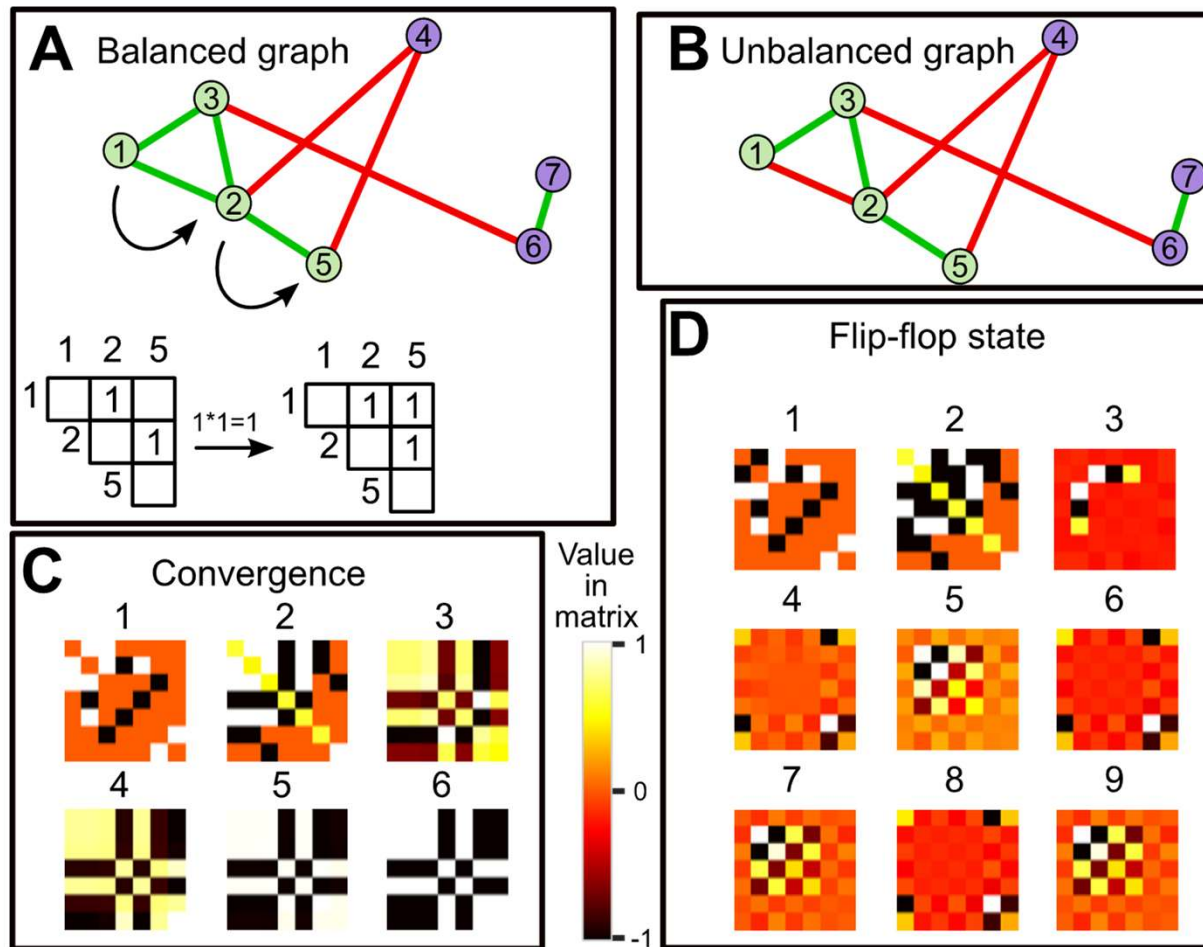
- Modular and scale-free interaction matrix generated with Klemm-Eguiluz algorithm
- Simulations with generalized Lotka-Volterra including environmental effects
- Cross-sectional microbiome abundances generated

Sam Röttjers



With increasing environmental impact in the simulations, clusters form in the networks.

Appendix: Manta internals



B, D) Unbalanced graphs: no convergence (flip-flop state)

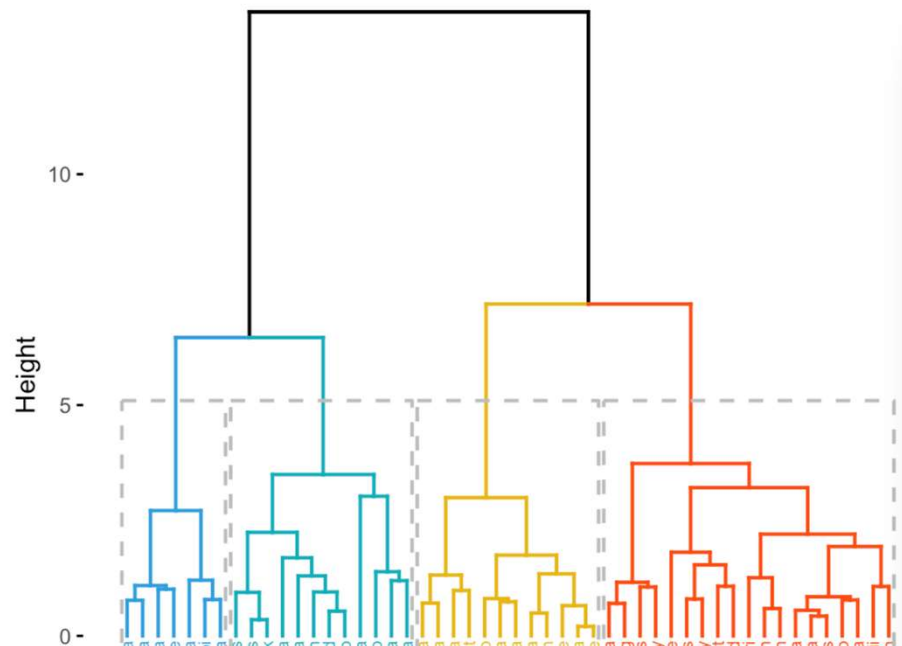
Trick: use sub-sets of the network to generate scoring matrix. For balanced subsets, follow procedure for balanced graphs. For unbalanced subsets, carry out only one iteration. When merging, only use nodes with signs consistent across most subsets.

A, C) Balanced graphs: Multiplication and transformation (retaining signs) of weighted adjacency matrix MCL-style until convergence (scoring matrix)

=> Scoring matrix

Appendix: Manta internals

- Clusters derived from scoring matrix through agglomerative clustering using Ward's minimum variance method
- Optimal cluster number determined with sparsity score S



$S = 1/E * (\text{number of negative edges not in clusters} + \text{number of positive edges in clusters} - \text{number of negative edges in clusters} - \text{number of positive edges not in clusters})$

E: total edge number

S ranges from -1 to 1, with -1 being the worst score

Appendix: Manta internals

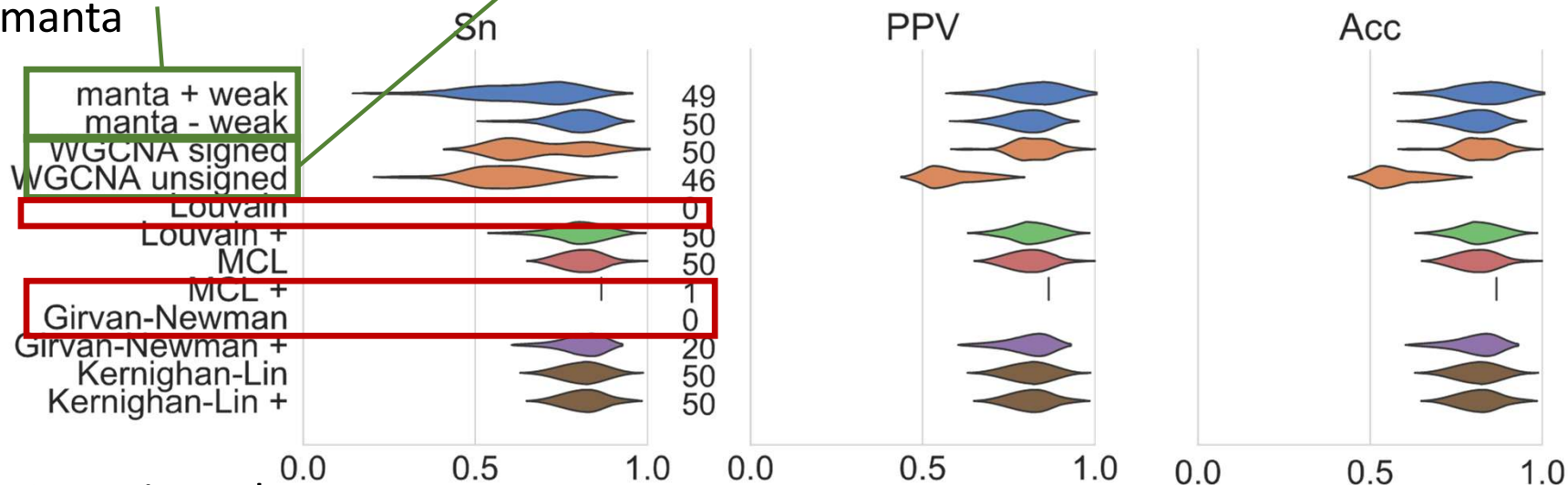
- Weak cluster assignments:
 - Find nodes that fluctuate strongly in flip-flop iterations: oscillators (some diagonal values of scoring matrix)
 - Compute weight of shortest path from each node to closest oscillator
 - Negative weight or weight below user-defined threshold: node has a weak cluster assignment
- Treatment of small clusters: clusters in size below threshold are removed from scoring matrix and cluster membership of their nodes is assigned based on average shortest path weight to cluster members

Appendix: Manta evaluation on biclusters

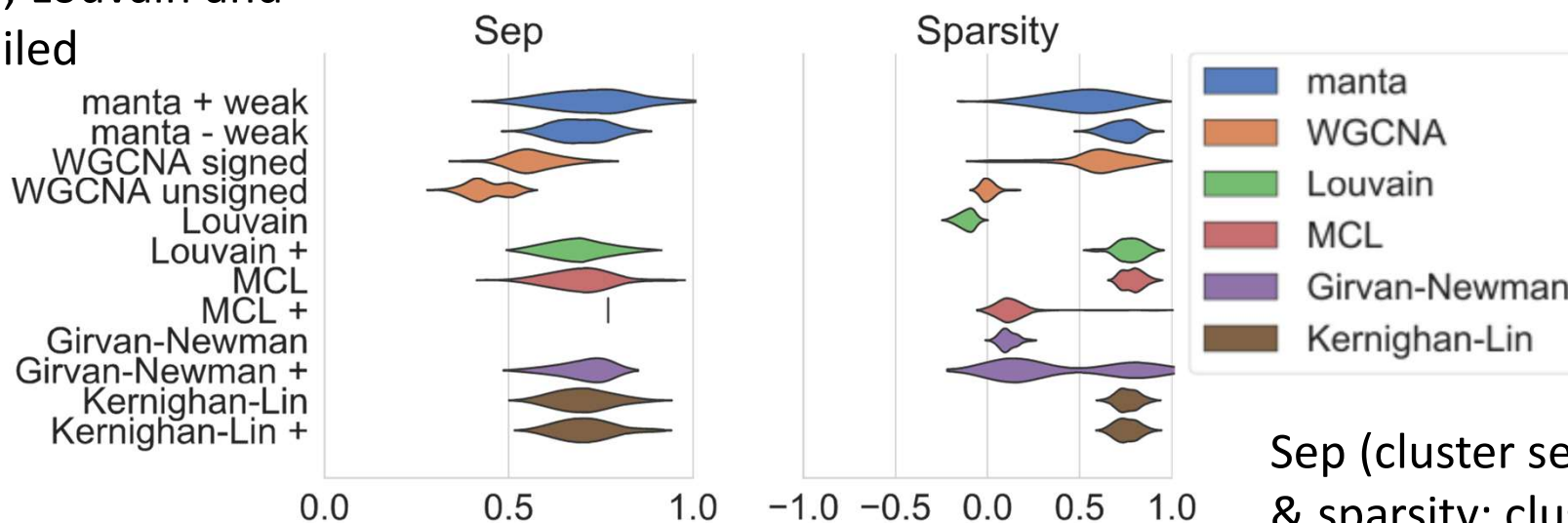


Weak node assignments
help manta

Keeping the edge signs improves WGCNA



MCL+, Louvain and
GN failed

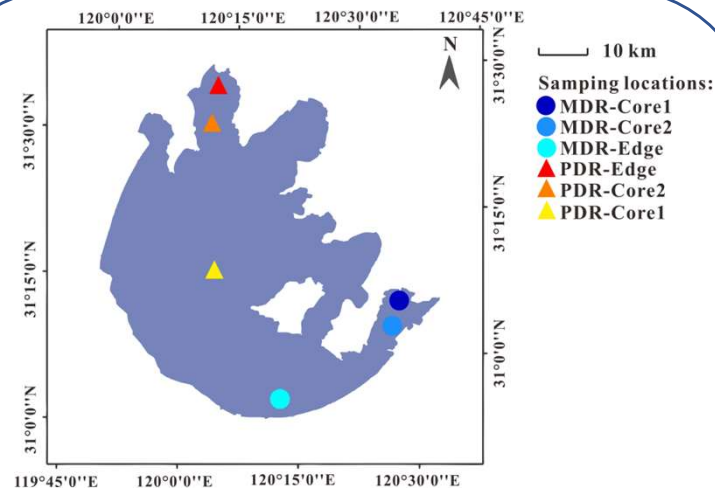


Sep (cluster separation)
& sparsity: cluster quality
scores

Appendix: Manta in action: Lake Taihu



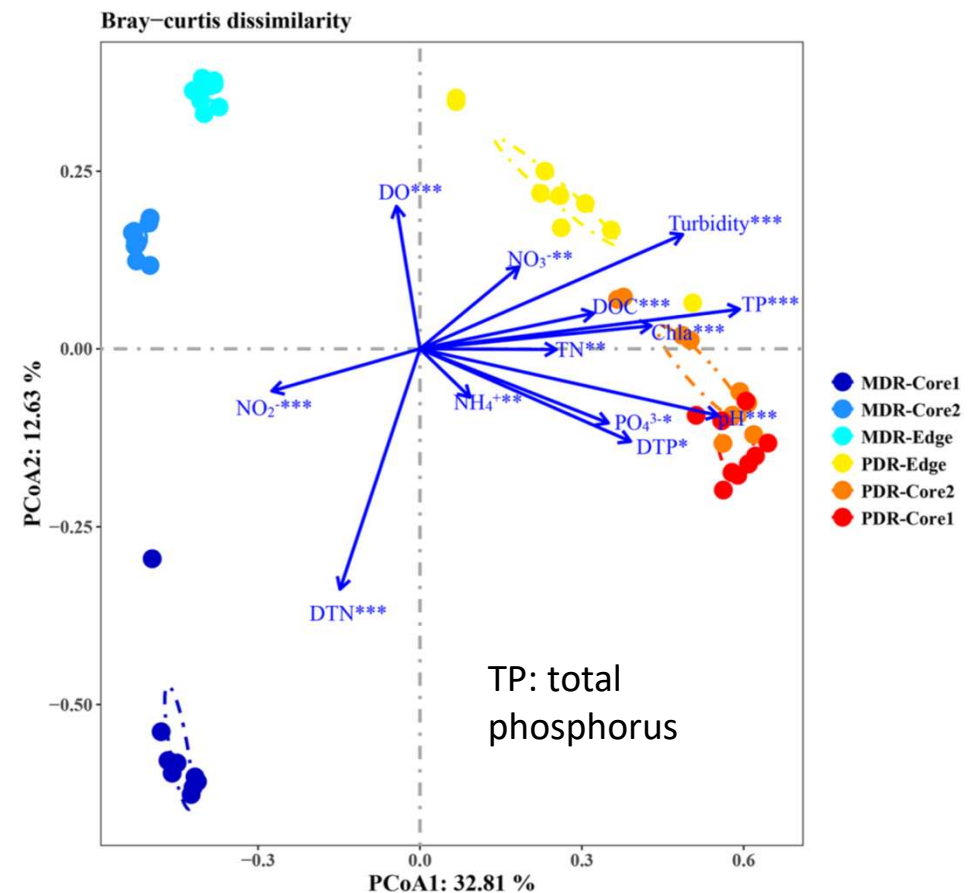
Eutrophication: two regimes driven by nutrient concentration



Macrophyte-dominated state (MDR)



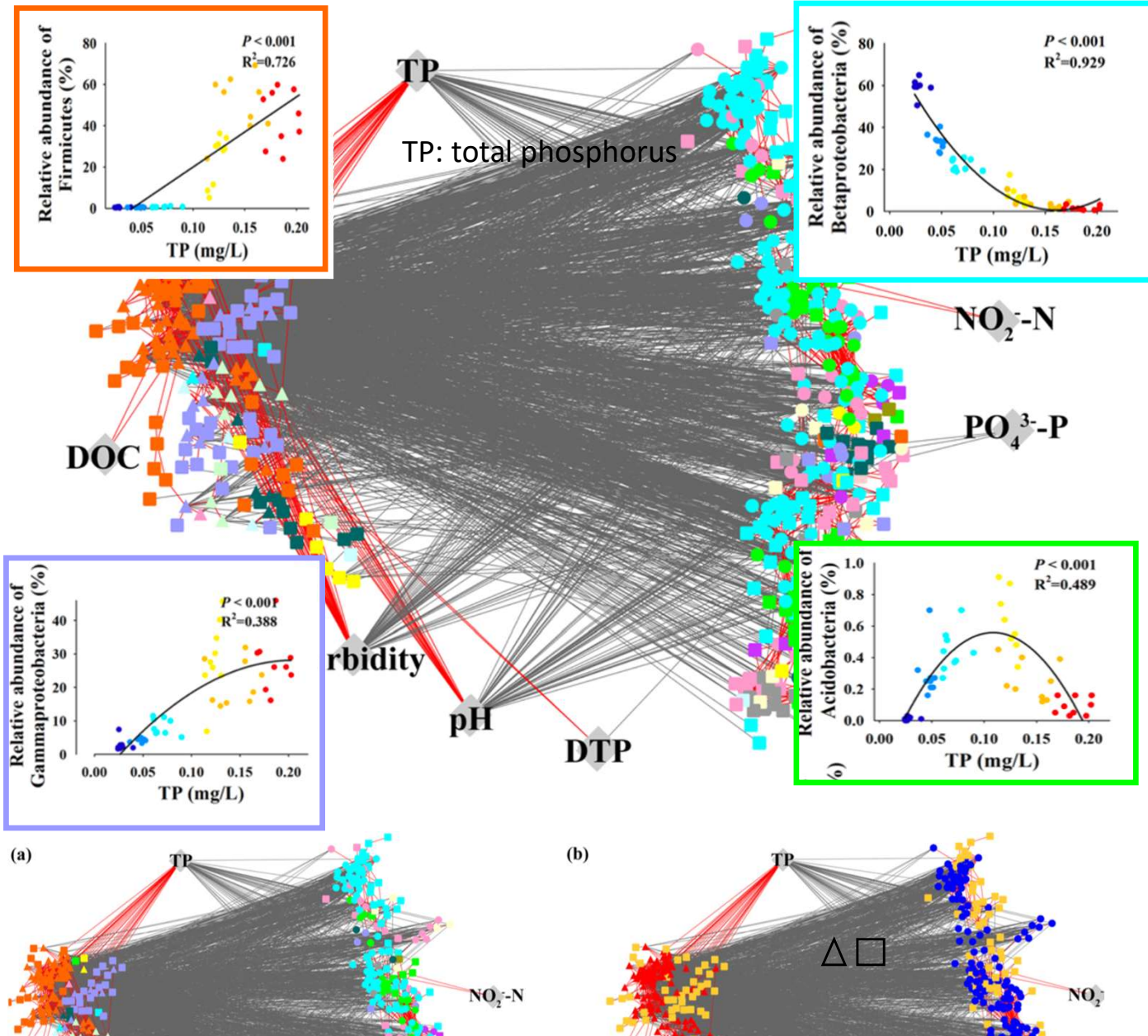
Phytoplankton-dominated state (PDR)



Appendix: Manta in action: Lake Taihu

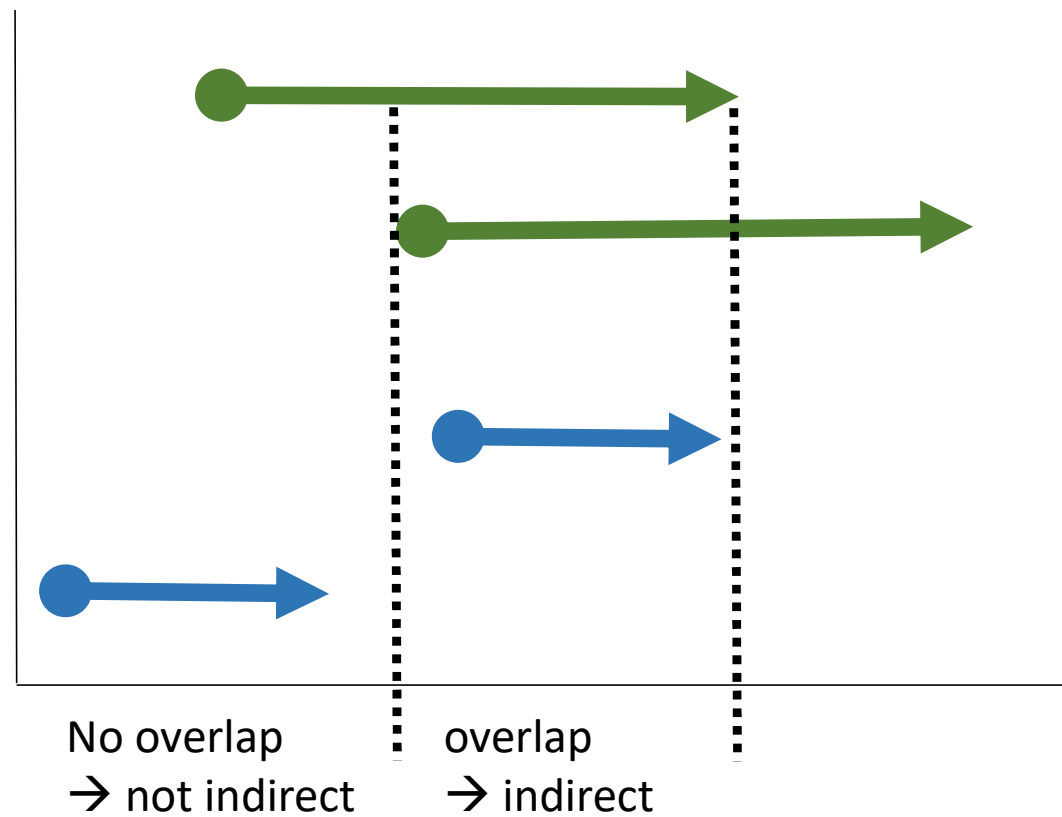


- Network constructed with CoNet on 54 samples and clustered with manta
- Cluster 1 mostly consists of *Betaproteobacteria*, which decline with nutrient levels
- Cluster 3 is dominated by *Firmicutes*, which tend to increase with nutrient levels
- Cluster 2: contains phyla with non-linear responses: *Gammaproteobacteria* (saturation) and *Actinobacteria* (optimum)

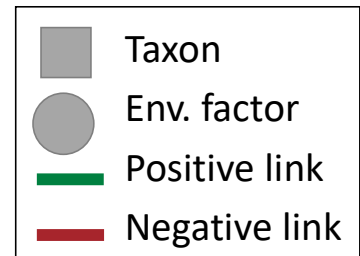


Appendix: Indirect edge removal - overlap

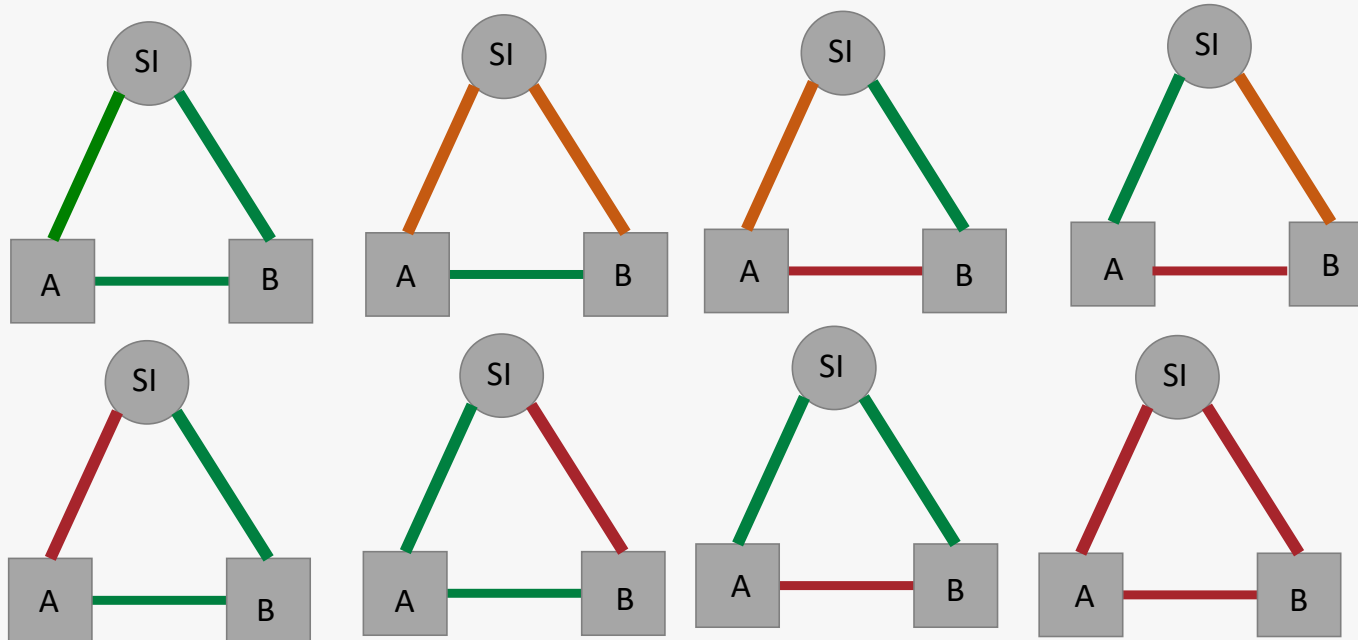
Overlap uses the start and length of co-occurrence in time



Appendix: Indirect edge removal – sign patterns



8 possible sign patterns for two taxa and one environmental factor in a triplet

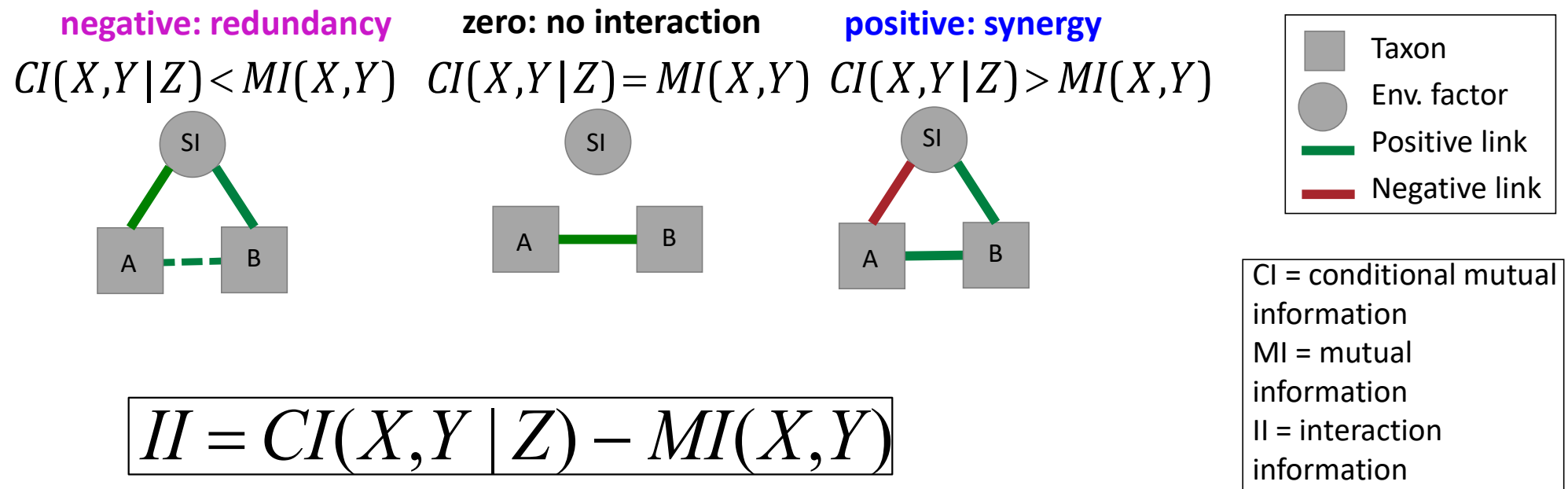


Sign patterns
indicating an
indirect taxon
relationship

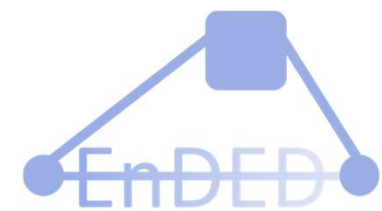
Sign patterns
indicating a **direct**
taxon relationship

Appendix: Indirect edge removal – interaction information

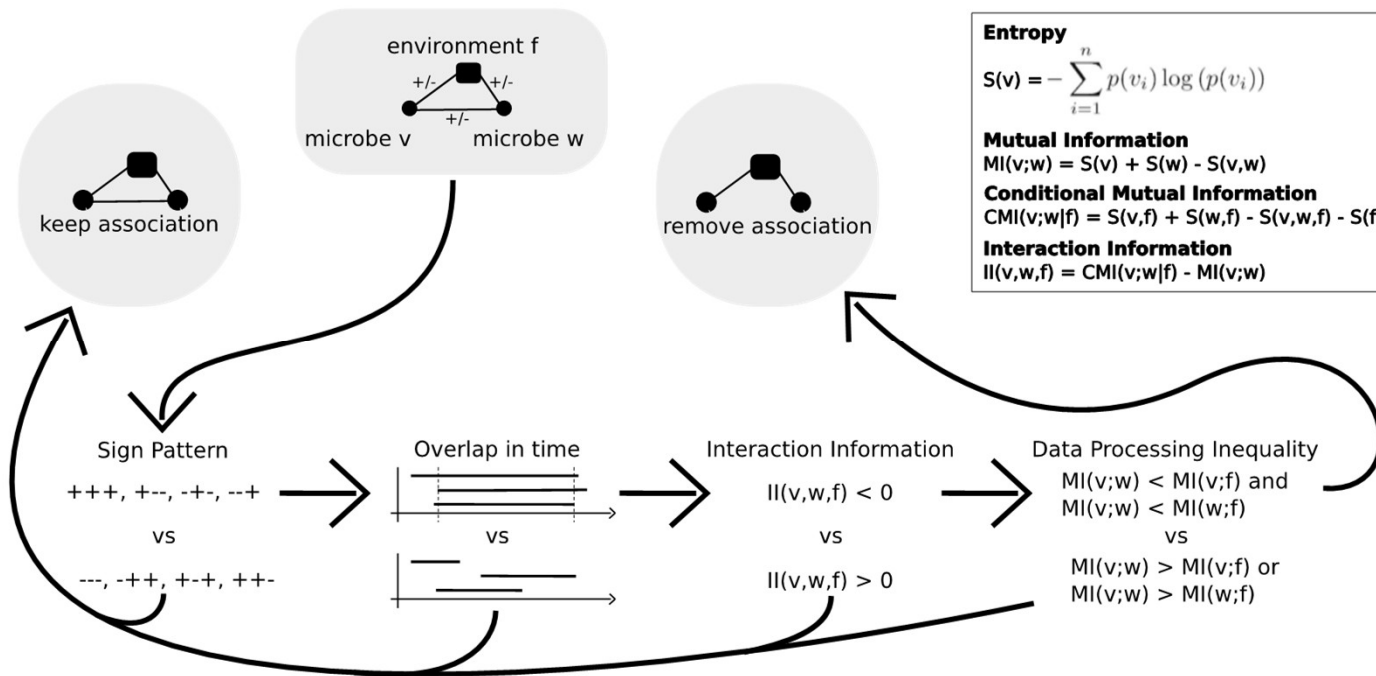
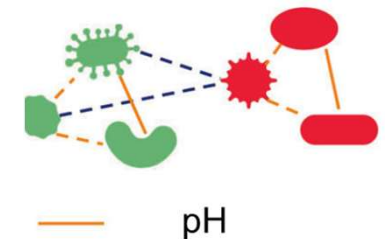
- Interaction information indicates whether a triplet contains an indirect edge
- It is an assumption that the indirect edge is the taxon-taxon edge (this is a good assumption for environmental factors that cannot be quickly influenced by taxa, such as temperature)



Appendix: EnDED: shrink the hairball



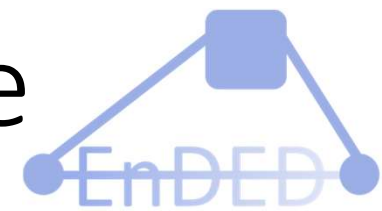
- Problem: edges in microbial networks are often driven by environmental factors
- EnDED combines several methods to remove indirect edges



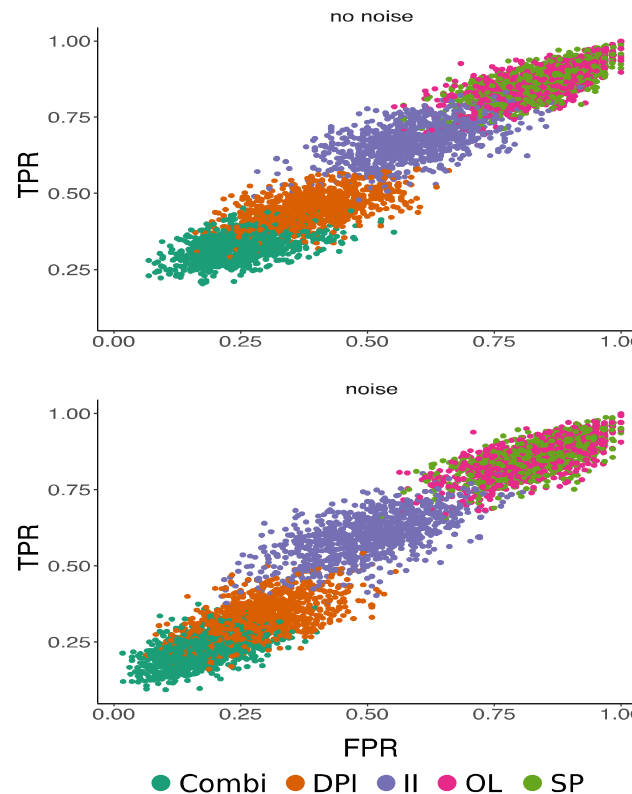
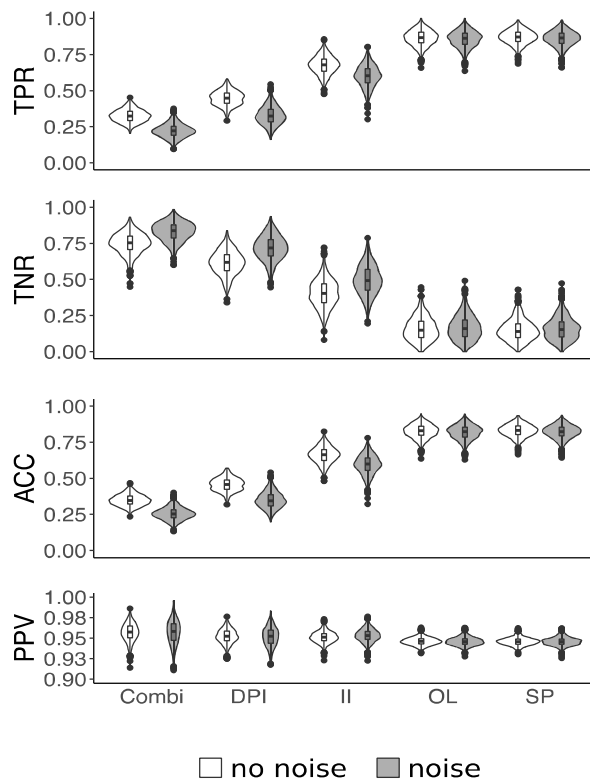
Ina Deutschmann

Deutschmann et al. (2021) "Disentangling environmental effects in microbial association networks" Microbiome 9, 232.

Appendix: EnDED performance



- Accuracy assessed on simulated data (extended Lotka Volterra model)
- Method combination: lowest accuracy but highest positive predictive value (removes fewer true edges at cost of keeping false ones)



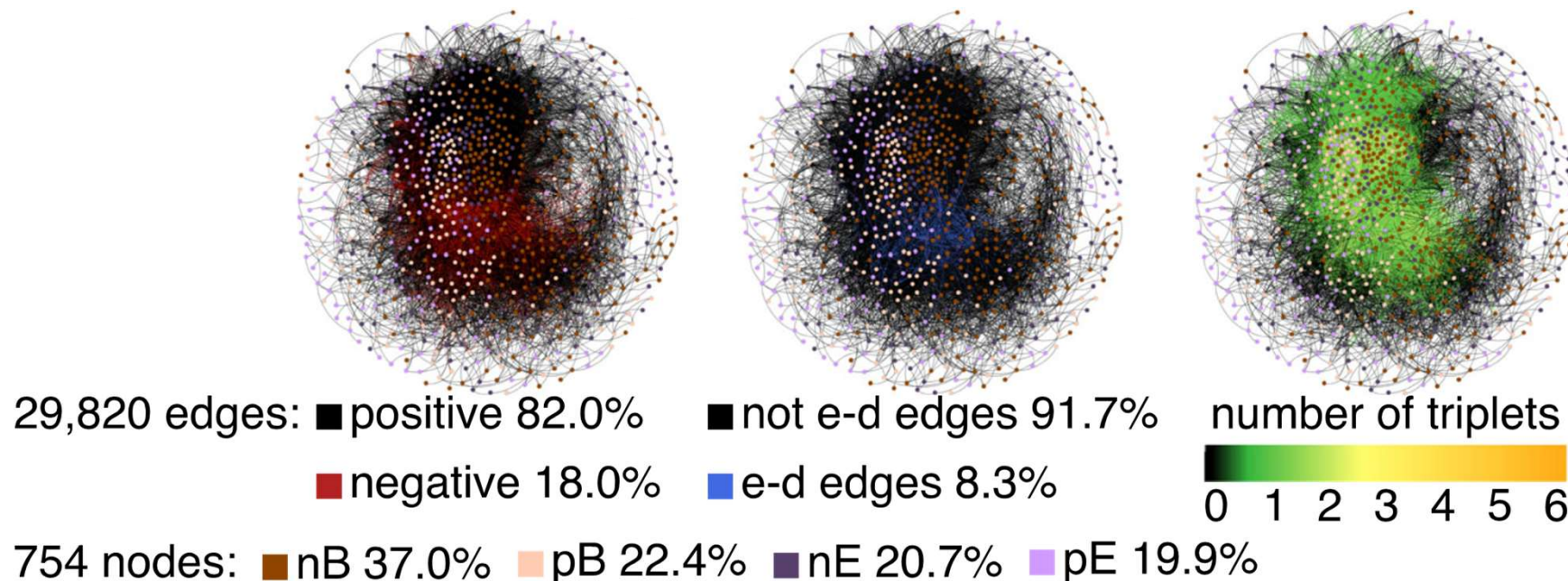
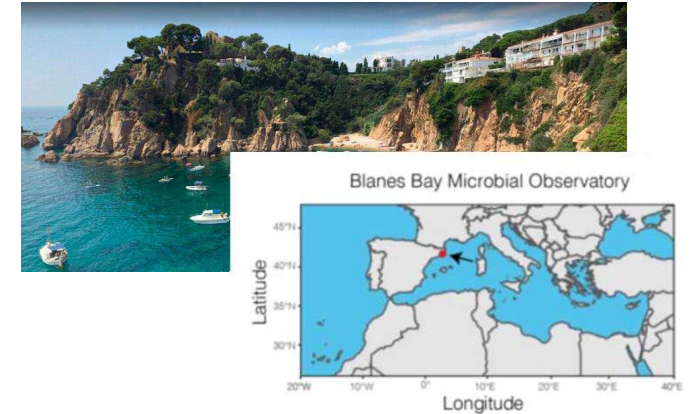
DPI: data-processing inequality (edge with smallest mutual information in triplet is removed)
II: Interaction information (indicates redundancy in triplets)
OL: overlap (time series)
SP: Sign pattern
Combi: Combination

True positive: false edge correctly removed
False negative: false edge not removed

False positive: correct edge falsely removed
True negative: correct edge not removed

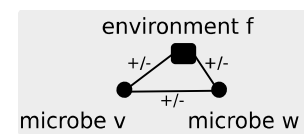
Appendix: EnDED in action

- Blanes Bay Microbial Observatory (BBMO) data: Mediterranean Sea sampled monthly from Jan 2004 to Dec 2013
- Environmental factors measured
- 18S & 16S V4 region sequenced
- Network constructed with eLSA
- EnDED (combi) removed 8.3% of the edges



5 out of 29
known
interactions
lost

Triplet:

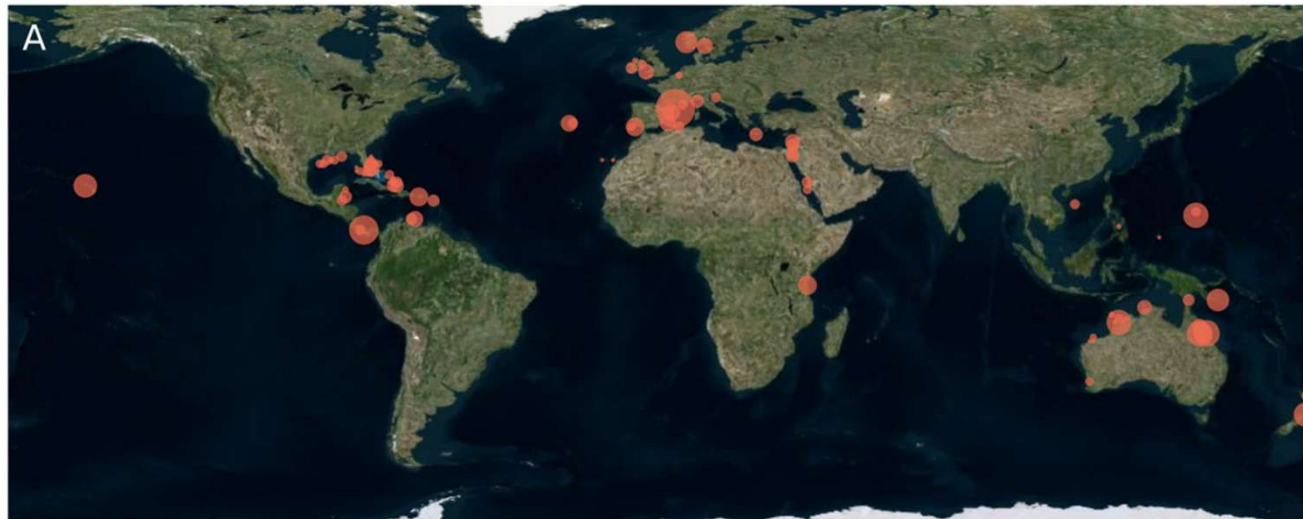


Taxon size fraction: nano and pico, kingdom: B=Bacteria, E=Eukaryotes

Appendix: Anuran in action - sponge microbiome



- Sponge microbiome project: microbiota of 268 different sponge host species collected around the globe
- > 3000 samples from ten sponge orders sequenced



Sample number

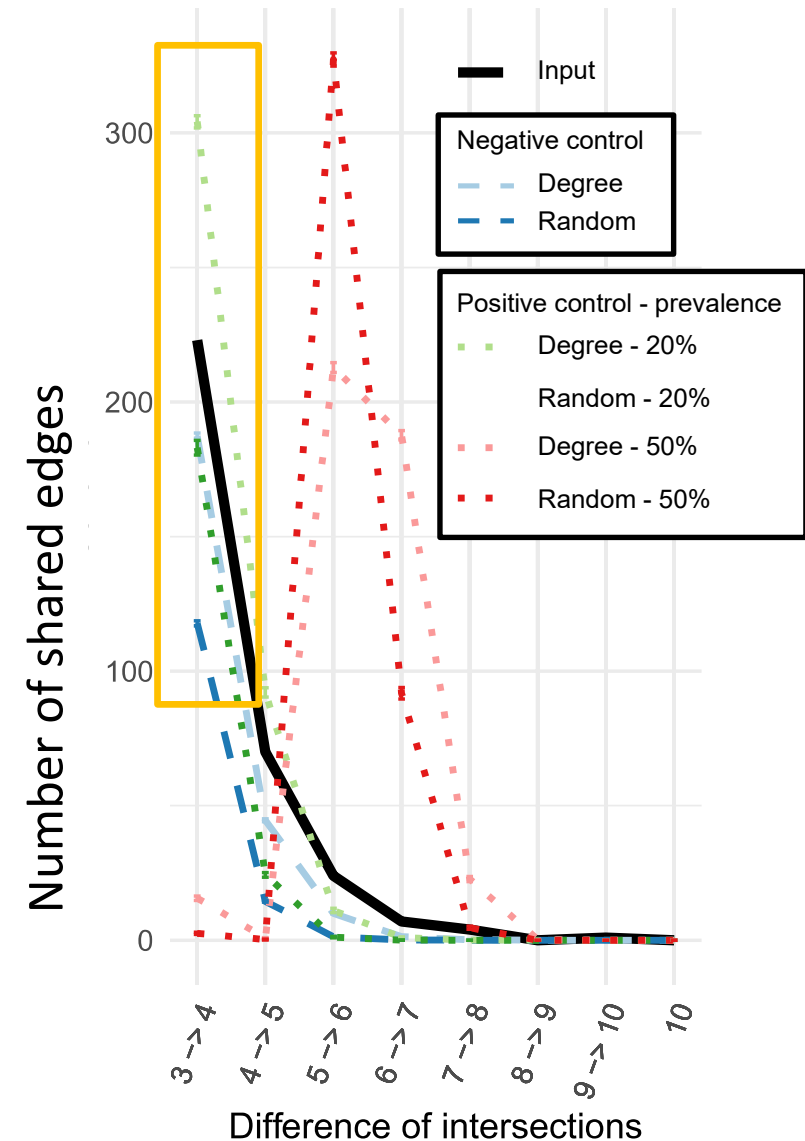
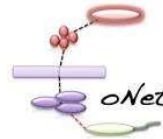


Moitinho-Silva, L., Nielsen, S., Amir, A., Gonzalez, A., Ackermann, G. L., Cerrano, C., ... & Steinert, G. (2017). The sponge microbiome project. *GigaScience*, 6(10), gix077.

Appendix: Anuran in action - sponge microbiome



- Ten sponge-order-specific networks constructed with CoNet
- Number of shared edges is significant for edges conserved in three networks -> core network

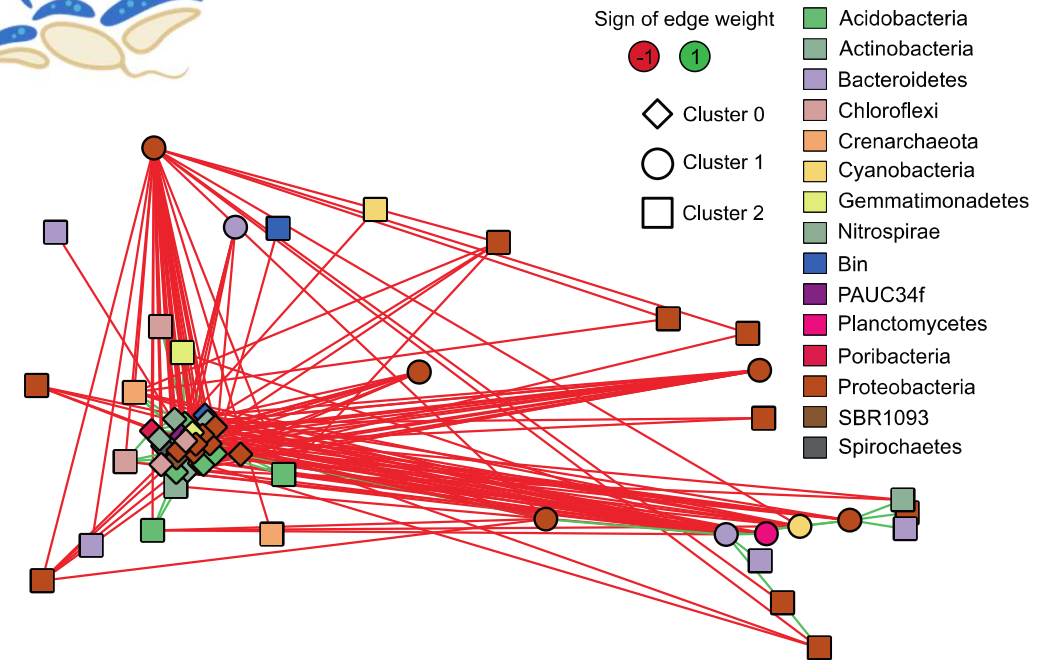


= Fraction of networks in which shared edges occur

Appendix: Anuran in action - sponge microbiome



- Core network clustered with manta
- Clusters are enriched in known indicator taxa for high versus low microbial abundance sponges (HMA vs LMA)
- Sponge orders are either HMA or LMA, so no 100% core network expected



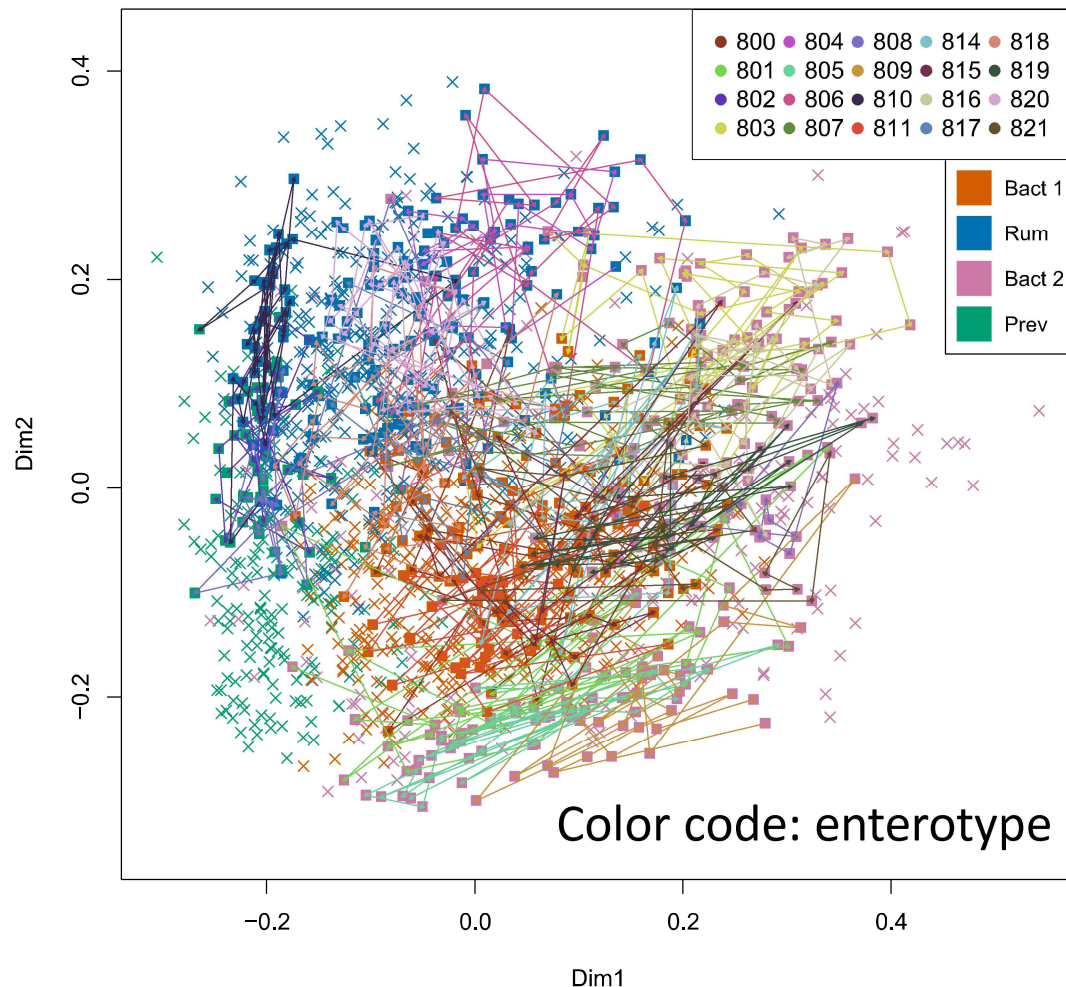
Cluster 0: enriched in HMA phyla

Cluster 1: only LMA phyla

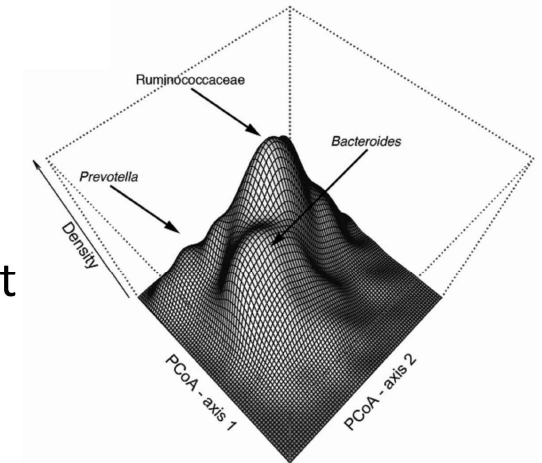
Moitinho-Silva, L., Steinert, G., Nielsen, S., Hardoim, C. C., Wu, Y. C., McCormack, G. P., ... & Hentschel, U. (2017). Predicting the HMA-LMA status in marine sponges by machine learning. *Frontiers in microbiology*, 8, 752.

Appendix: Anuran in action - gut microbiome

- Fecal samples collected for 20 women over six weeks (713 samples) and sequenced
- 20 microbial networks constructed with fastLSA



PCoA of time series data collected from 20 women, mapped onto the Flemish Gut Flora Data set (1104 samples)

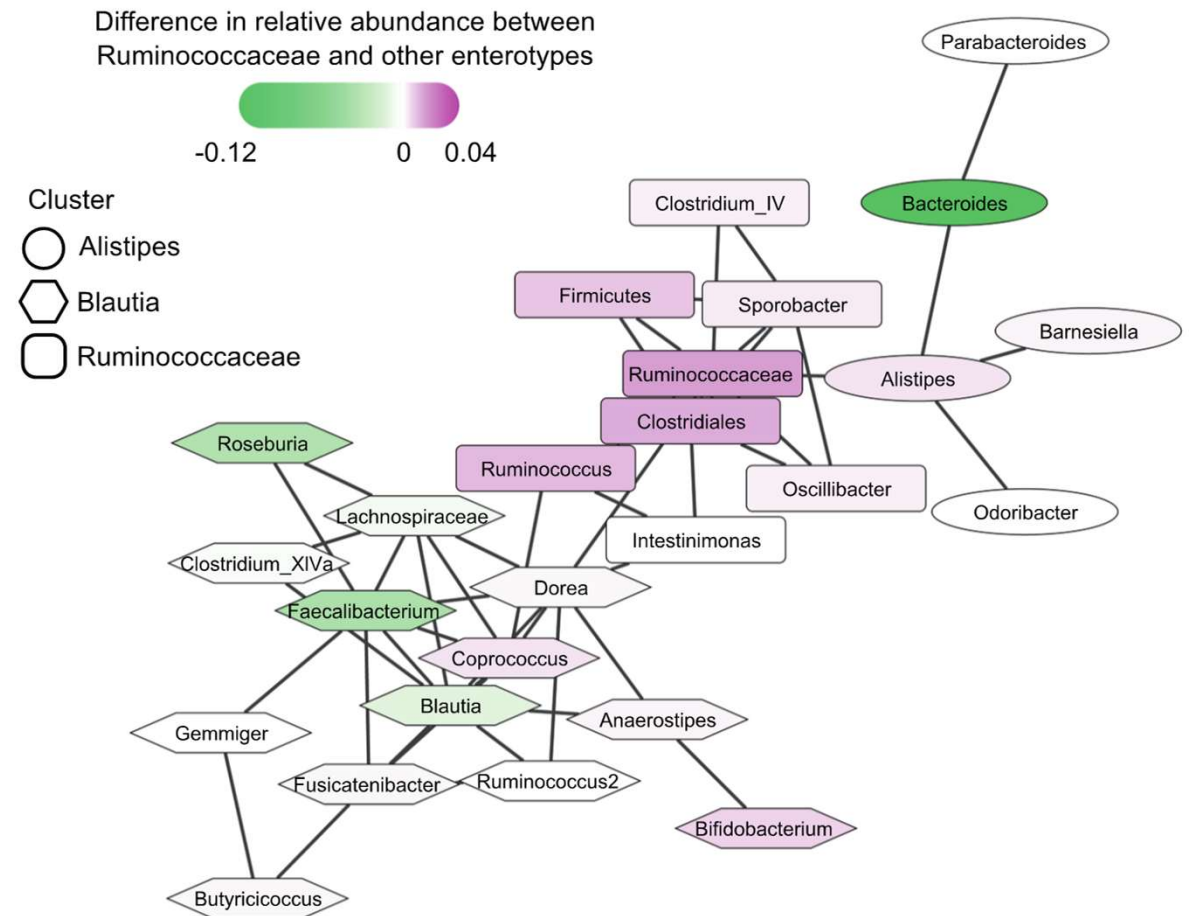
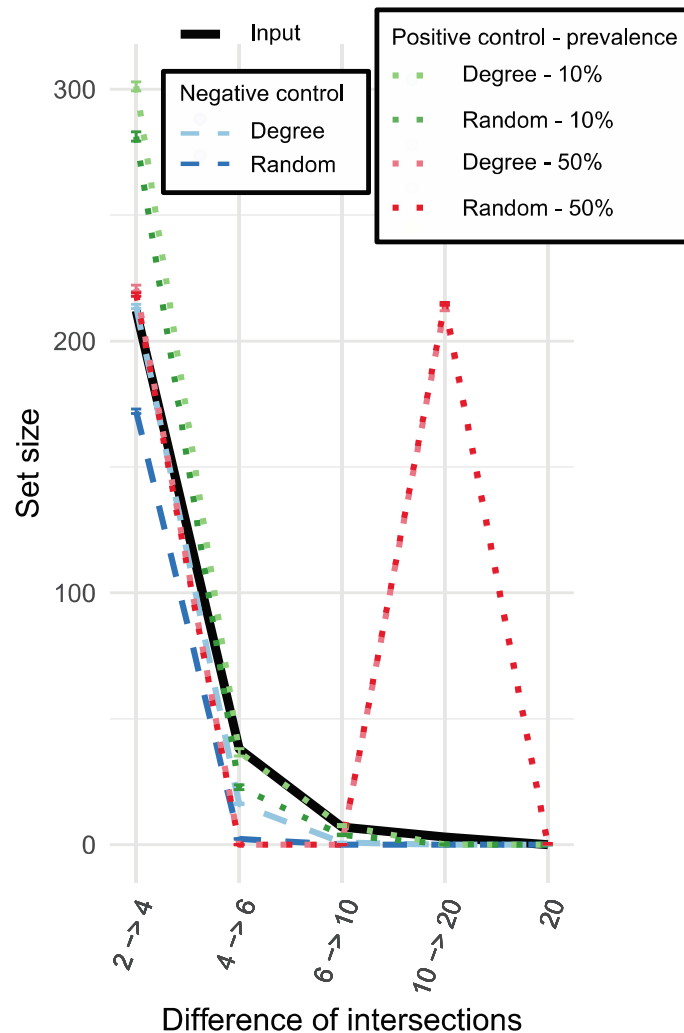


Enterotypes: gut microbial compositions dominated by different genera (Rum = Ruminococcus, Bact = Bacteroides, Prev = Prevotella)

Appendix: Anuran in action - gut microbiome



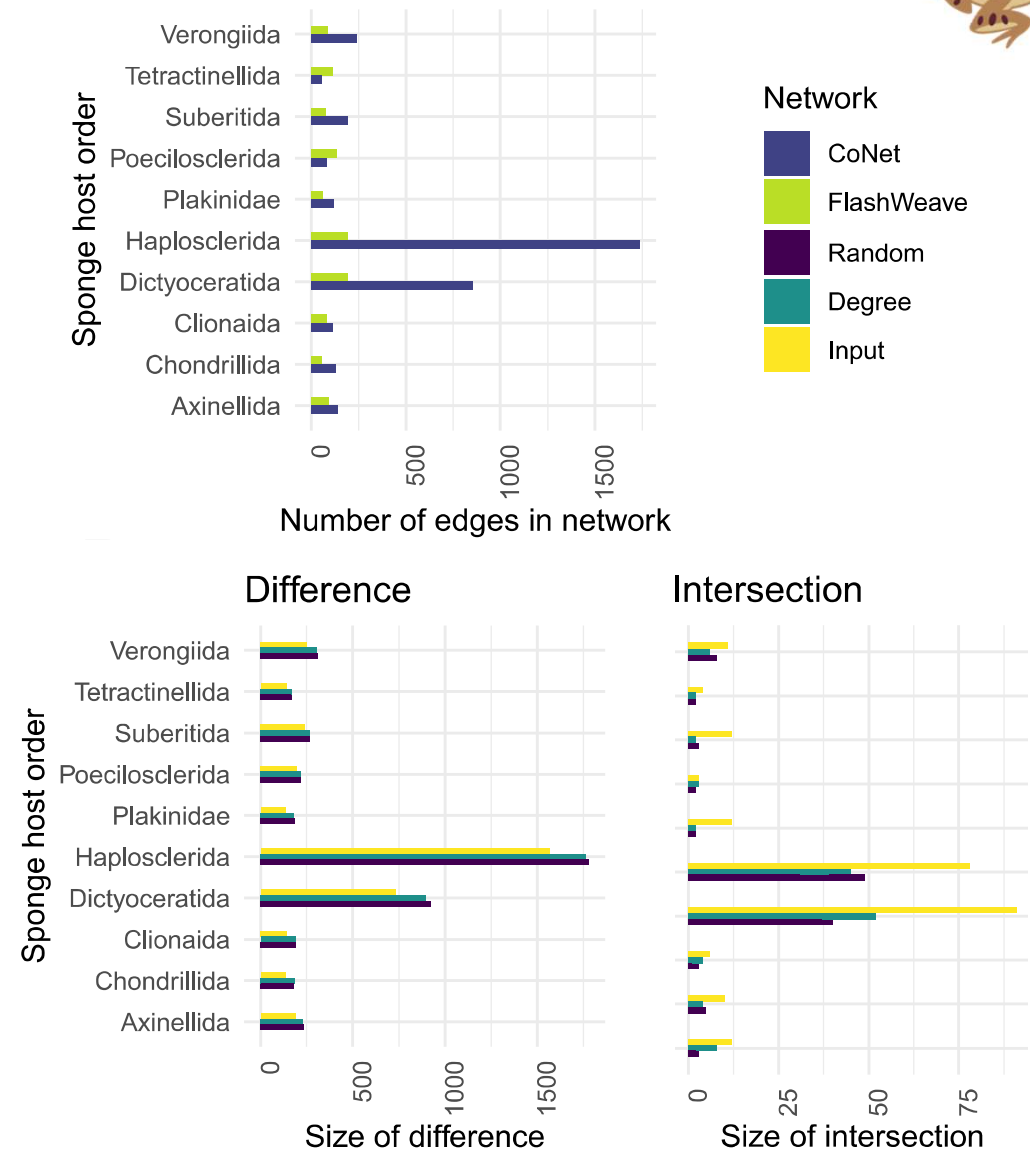
- Significant core network for edges in four or five networks
- Network clusters correspond to enterotypes



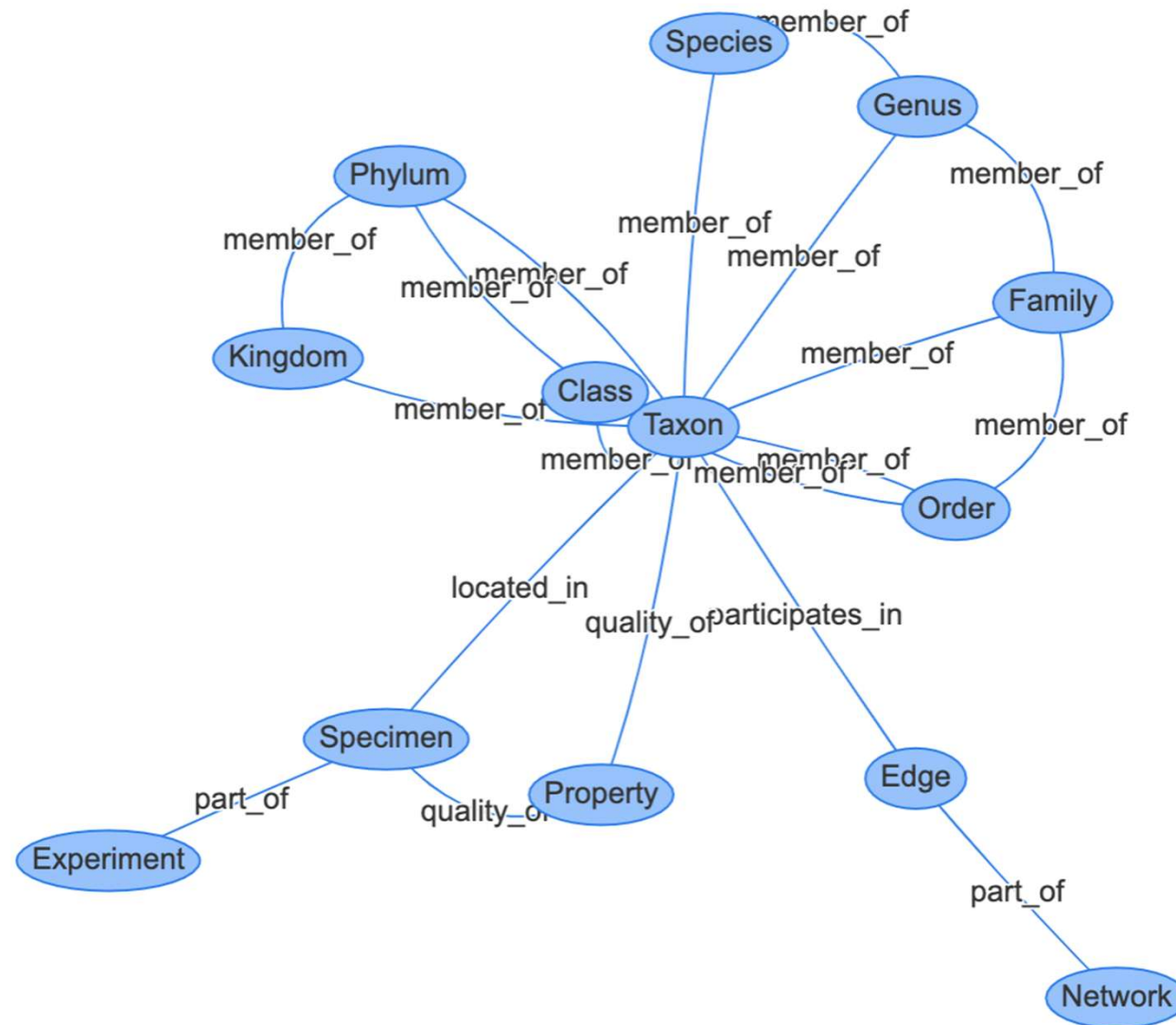
Appendix: Anuran in action - tool comparison



- Sponge-order specific networks constructed with CoNet and FlashWeave
- CoNet networks are systematically larger
- Tool-specific network intersection is highly significant
- Tools pick up the same associations, but CoNet reports many additional edges (does not filter indirect edges)



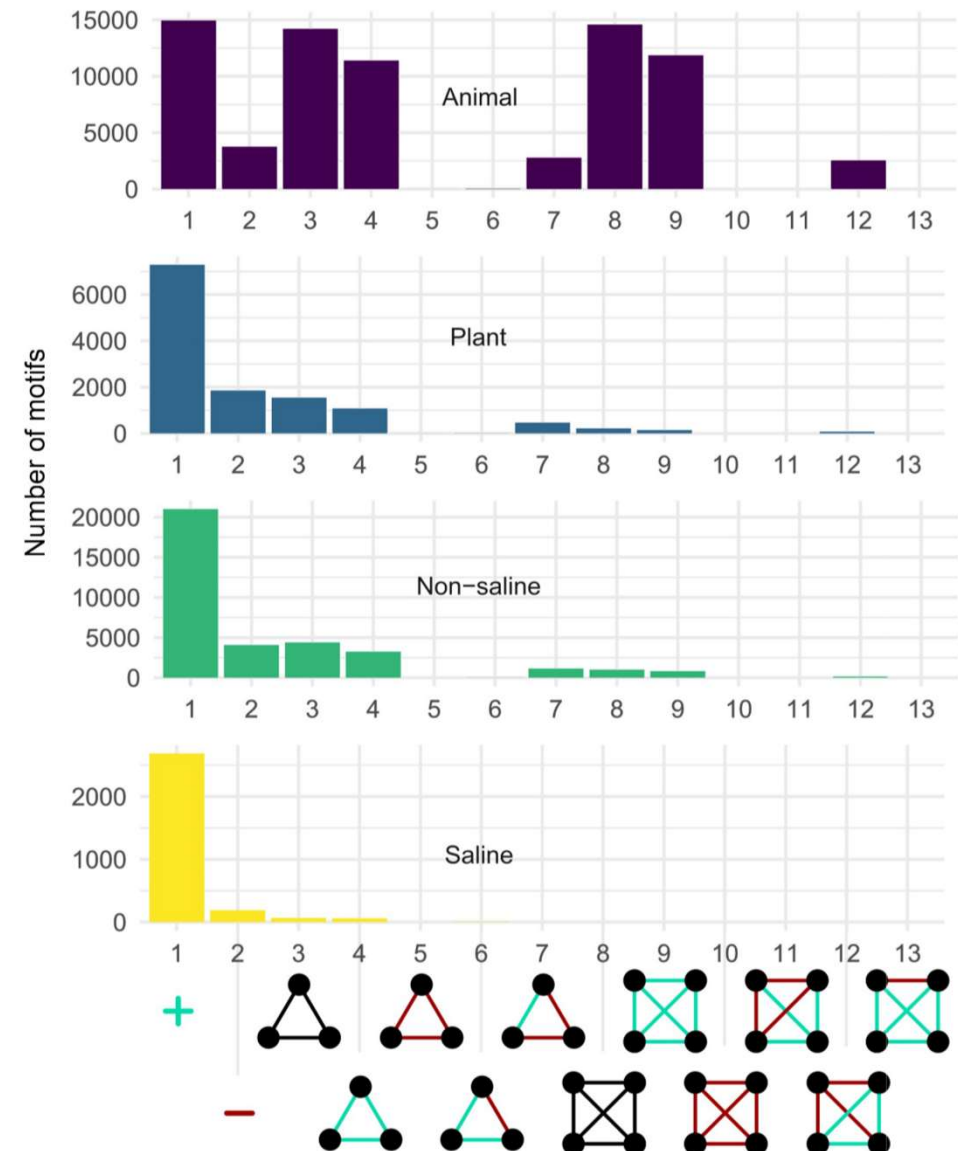
Appendix: Mako's data scheme



Appendix: Mako in action



- Networks constructed for 60 microbiome data sets taken from QIITA (tool: FlashWeave)
- Networks queried for specific motifs
- More positive than negative edges found
- Animal-derived microbial data sets enriched in positive 4-node clique



Appendix: Mako in action

- Networks constructed for 60 microbiome data sets taken from QIITA (tool: FlashWeave)
- Search for edges representing taxon combinations that can produce a target metabolite
- Example propionate: most frequent taxon combination found that could produce propionate is Bacteroides and Lactobacillus

